

Familiarity in International Equity Investment: Concentrated Value, Pervasive Allocation Bias

Abstract

Does familiarity convey information or produce bias? Using 2000–2024 data from 1,663 active U.S. international equity funds and managers' foreign education as a familiarity proxy, we examine both country allocation and stock selection. Funds overweight education-linked countries, yet these sub-portfolios do not outperform. Within familiar countries, the most active quartile of stocks earns a monthly five-factor alpha of 41 basis points, while the remaining holdings earn no significant alpha. The overweight persists after removing top-quartile stocks. Familiarity's value is concentrated, its allocation footprint pervasive: information and bias coexist at margins that country-level averages obscure.

Keywords: Familiarity, Home Bias, Mutual Funds, International Portfolio Choice, Active Management, Stock Selection

JEL Codes: G11, G14, G15, G23, G41

1 Introduction

When investors overweight familiar markets, are they acting on information or displaying bias? A large literature shows that investors tilt their portfolios toward assets that are geographically, socially, or otherwise familiar, both within domestic markets and across borders (Coval and Moskowitz, 1999, 2001; Grinblatt and Keloharju, 2001; Huberman, 2001; Portes and Rey, 2005; Jagannathan et al., 2022). Familiarity may have informational value if closer ties help investors acquire, process, or interpret local information; consistent with this, domestic investors in several international markets trade at better prices or earn higher profits than foreigners (Choe et al., 2005; Dvořák, 2005). Yet the same force may also distort portfolio choice: familiarity may foster overconfidence, leading investors to overweight those markets even when the additional exposure does not improve performance. Distinguishing these forces is therefore central to evaluating whether familiar-market tilts reflect informed active management or inefficient portfolio concentration.

We find that neither force dominates outright: the two operate at different margins of the same portfolio. Familiarity is associated with two distinct patterns inside international mutual fund portfolios. Its allocation footprint is broad: funds systematically overweight countries connected to their managers' educational backgrounds. Its performance footprint is narrow: the broad familiar-country tilt does not, on average, generate abnormal returns; the value associated with familiarity appears instead in selected positions, especially the stocks within familiar countries where fund teams express their strongest active views. Combining existing theories yields distinct predictions across these margins. Theories of investor recognition and familiarity-based preference predict breadth: investors hold and overweight the assets they know, whether or not knowing them confers an advantage (Merton, 1987; Huberman, 2001). Theories of information choice predict concentration: because learning capacity is limited, an informational edge is more likely to be developed intensively in a few securities than spread evenly across a familiar market (Van Nieuwerburgh and Veldkamp, 2009; Schumacher, 2018). Distinguishing these two margins, which country-level averages

blend together, helps reconcile the view that familiarity reflects bias with the view that it conveys information.

We study these margins using actively managed U.S. international equity mutual funds and managers' post-secondary education in non-U.S. countries as a source of familiarity. Educational experience can provide knowledge of local institutions, language, business norms, and informal networks, while also generating familiarity-based preferences for investing in those countries. We compile quarterly holdings for U.S. actively managed international equity funds over 2000–2024 from Morningstar and link them to manager biographies; the resulting estimation sample comprises 1,663 funds and identifies 2,976 managers with post-secondary education across 52 countries. Throughout the paper, a fund's benchmark is the market index that Morningstar assigns to it, most commonly the Morningstar Global ex-US index. We observe each benchmark's security-level holdings, which provide the passive comparison for the fund's country and stock weights. The separation of margins is empirically meaningful because funds are much more concentrated than their benchmarks: the median fund holds equities in about 19 countries and 80 stocks, against roughly 50 countries and 5,000 stocks in the Global ex-US benchmark.

We first find that funds overweight countries in which their current managers received post-secondary education by 0.97 percentage points of excess weight, defined as the fund's country weight net of its benchmark weight, or 39.4% relative to benchmark allocations. This tilt is economically large: the average country position is about 5% of the portfolio. In the continuous specification, a fully education-linked management team is associated with an 80.7% higher benchmark-scaled allocation, more than four times the 18.8% increase that Pool et al. (2012) estimate for a fully home-state-linked team in U.S. domestic funds. Because the settings and familiarity measures differ, we view this comparison as a benchmark for economic magnitude.

A central identification concern is that education ties may line up with country tilts through fund-level choices rather than through the managers themselves: a fund that intends

to overweight country c may select managers educated in c , and large markets both carry high benchmark weights and supply many foreign-educated managers. Our baseline specification includes country $\times$ time fixed effects, which absorb shocks common to a country at a given time, and fund fixed effects, which absorb persistent fund characteristics. Identification therefore comes from cross-fund differences within a country-time after removing fund-level averages. An event study around manager turnover provides complementary timing evidence: benchmark-relative holdings show no pre-trend before a fund gains its first education-tied manager, build gradually afterward, and unwind gradually after the last one leaves.

Yet the broad tilt is not matched by broad performance. Our preferred specification includes country $\times$ time fixed effects, so that a fund’s holdings in a country are compared with other funds’ holdings in the same country at the same time. In this specification, education-country sub-portfolios earn less than one basis point more per month than non-education-country sub-portfolios, a difference statistically indistinguishable from zero. The only estimate approaching significance is a marginally significant three-month differential in the least-saturated specification, which does not survive country $\times$ time fixed effects and has no counterpart in the continuous education-share measure. Funds hold substantially more of their managers’ education countries, but on average they earn nothing extra for doing so.

The average, however, is not where information-choice theories predict an edge to reside. We therefore ask whether familiarity is associated with value at a narrower margin: managers’ most active stock positions within familiar markets. To identify these positions in an international setting, we construct the I-measure, a within-country activeness measure that captures a fund team’s ex-ante conviction in an individual stock.¹ Intuitively, within each country, the I-measure asks which stocks a fund holds most aggressively relative to the benchmark or market portfolio. Within each fund-country-quarter cell, we rank held stocks by the I-measure and classify the top quartile as high-I positions, the fund’s top picks within

¹The I-measure extends the activeness measure in Antón et al. (2021) to the international context. Section 3 explains why their measure does not map cleanly to the international setting; Appendix C shows that a within-country, universe-adjusted version of it recovers the pattern we report.

the country.

The value associated with familiarity is highly concentrated in exactly these positions. High-I positions in managers' education countries earn a Fama-French five-factor α of 41.0 basis points per month, or 5.03% annualized, while the remaining education-country holdings earn an insignificant -3.8 basis points; the difference is significant at the 1% level. The non-education-country sample provides a natural falsification comparison: applying the same sort there produces no significant alpha, so activeness alone does not generate the result. Within education countries, the premium is confined to the top quartile rather than rising gradually with activeness. It remains large under alternative sorting measures, net of estimated bid-ask spread costs, and when the United Kingdom is excluded.

The final step connects the two margins. If the education-linked country tilt merely carried these profitable positions, removing them should eliminate it. It does not: after excluding the top-quartile holdings, funds still overweight their education countries by 0.58 percentage points of excess weight, or 32.1% relative to benchmark weights. This separation between a broad allocation footprint and a concentrated performance footprint is the paper's central contribution. The evidence is consistent with education-linked information or skill whose value is far narrower in scope than the associated allocation tilt. The residual exposure earns no abnormal return, consistent with familiarity-based allocation bias operating alongside a concentrated information or skill channel.

This decomposition speaks to two strands of the familiarity literature that have largely developed separately. One strand emphasizes familiarity as an information channel: social networks, geographic proximity, professional expertise, and home-country links help investors acquire or interpret value-relevant information (Cohen et al., 2008; Baik et al., 2010; Bernile et al., 2015; Ben-David et al., 2019), and in delegated portfolios, funds of hedge funds with a stronger local bias earn higher risk-adjusted returns (Sialm et al., 2020). The other strand emphasizes familiarity as a distortion: investors overweight familiar assets even when these positions deliver no superior returns (Huberman, 2001; Pool et al., 2012). The two strands

thus read the familiar-asset tilt in opposite ways. Our evidence shows why both readings persist: each is correct at a different margin of the same portfolio.

The paper also contributes to the active-management literature. Mutual funds rarely outperform on average after fees (Jensen, 1968; Berk and Green, 2004; Barras et al., 2010), but skill is visible where managers deviate most: funds’ most active positions predict returns (Jiang et al., 2014; Antón et al., 2021), and funds concentrated in a few industries outperform through stock selection (Kacperczyk et al., 2005). In this literature, deviation from the benchmark serves as the marker of the skilled manager. International familiarity splits that marker in two. The allocation tilt toward managers’ familiar markets, the broadest deviation in our data, carries no performance information, while position-level deviation within those same markets isolates economically large stock-selection skill. The concentration-reveals-skill inference is therefore margin-specific: it holds at the position margin, fails at the allocation margin, and the two can decouple inside the same portfolio.

Our paper is most closely related to Jagannathan et al. (2022) (JJK), who study U.S. international mutual funds over 1991–2014 and find that home-linked managers overweight their home countries by roughly 14 percentage points and outperform in them, with a long–short alpha of 31 basis points per month. Neither magnitude survives in our data. On allocation, replicating their design on their own fund list recovers their order of magnitude, but the tilt falls to 3.2 percentage points once the specification absorbs both confounds their designs control one at a time: managers educated in a country tend to run funds already mandated to tilt toward it, and large markets both carry high benchmark weights and supply more home-linked managers. On performance, the same replication yields an alpha indistinguishable from zero, in line with our country-level estimates. More importantly, JJK stop at the country level; our within-country decomposition shows why that average is too coarse—it blends a broad tilt that earns nothing with a narrow set of high-I positions that earns substantial alpha. Appendix D compares the samples, definitions, and estimation strategies in detail.

The remainder of the paper proceeds as follows. Section 2 describes the data and the construction of manager education ties. Section 3 presents the empirical design and the construction of the I-measure. Section 4 reports the main results on country allocation and stock-level performance. Section 5 presents robustness checks. Section 6 discusses the broader pattern of concentrated value alongside pervasive bias, and Section 7 concludes.

2 Data

2.1 Sources

The analysis draws on several data sources: U.S. mutual funds’ international equity holdings, fund benchmark holdings, fund managers’ educational backgrounds, and international equity price data. The sample covers the period from 2000 to 2024 across all data sources. We begin in 2000 because the inputs required by our benchmark-relative measures are not consistently available earlier: each fund’s assigned Morningstar benchmark holdings and the developed- and emerging-market factor and risk-free series—which enter the excess-weight, universe-adjusted, and abnormal-return measures—begin only in 1999, and both fund and manager-education coverage are thin before 2000. The appendix exercises that extend back to 1991 to replicate Jagannathan et al. (2022) therefore drop the benchmark and rely on raw portfolio weights and returns (Appendix D).

Funds and benchmarks’ holdings: We use the Morningstar universe² of U.S. actively managed international equity funds as our primary sample. To avoid double-counting, we restrict to one share class per fund using Morningstar’s “oldest share class” designation, and refer to each such share class as a “fund” throughout the paper; holdings are measured from this share class, while the fund-level size, fee, and return characteristics in Table 2

²According to Morningstar’s definition, stock funds that have invested 40% or more of their equity holdings in foreign stocks (on average over the past three years) are classified as international equity funds. The holding data are free of survivorship bias—the sample includes funds that liquidated, merged, or were closed during 2000–2024. Morningstar data selection criteria: U.S. Category Group = International Equity; Oldest Share Class = Yes; Index Fund = No; the “Only surviving investments” filter is unchecked.

aggregate across all of a fund’s share classes. We collect equity holdings data at a quarterly frequency for all funds, with monthly holdings available for a subset. Table 2 summarizes key characteristics of the sample. Morningstar holdings data are cross-validated and have been employed in numerous prior studies. For each international fund, we also collect the monthly holdings of its corresponding benchmark, as assigned by Morningstar. The sample funds are mapped to 17 different Morningstar benchmarks. Table 3 presents the benchmark distribution across the sample.

Fund and Benchmark Characteristics: We collect both total returns and gross returns for each benchmark, as well as for each fund at the share class level. In addition, we obtain fund-level characteristics such as net expense ratios and net asset values from Morningstar.

Fund manager education: Morningstar provides a biographical paragraph for each manager that includes information on post-secondary education. We collect these biographies from Morningstar’s website, use text search to identify educational institutions, and map each institution to its country using a large-language-model classification that is subsequently reviewed by hand. We then merge the manager education data with fund holdings using manager names and tenure dates. Our estimation sample comprises the 1,663 funds that have at least one manager matched to a biography with identifiable post-secondary education, an assigned Morningstar benchmark, and at least one fund-country-quarter cell with a non-missing fund or benchmark country weight. We refer to these 1,663 funds as the sample throughout the paper.

International equity price and return: We obtain daily international equity price, return, and market capitalization data from Datastream for the period 2000–2024. All prices and returns are expressed in U.S. dollars; we use Datastream’s USD-denominated total-return series, so currency movements are absorbed in the reported returns rather than treated separately. These data are merged with mutual fund holdings using equity ISIN identifiers. We assign each equity’s country of origin based on the location of the company’s primary

business activity, as recorded by Datastream.³ This classification reflects our view that fund managers’ information advantages or confidence in a company are more likely to stem from where the firm operates rather than where it is listed. The distinction matters for how country exposure is measured: Demirci et al. (2022) show that a fund’s international exposure is substantially understated by the domicile of the stocks it holds, because domestically listed multinationals generate a large share of their sales abroad, and that recognizing this indirect exposure makes measured home bias considerably less severe. Assigning each equity to the location of its primary business activity moves our country weights toward economic rather than purely legal exposure, though it does not resolve the issue entirely, since a firm’s operations may themselves span several countries.

Fama-French Factors: We obtain the Fama and French (2015) five factors from the Dartmouth Data Library separately for emerging markets and developed markets over 2000–2024, yielding ten monthly factors in total. We use the 3-month U.S. Treasury rate as the risk-free rate.

2.2 Summary Statistics

Table 2 summarizes the sample of 1,663 U.S.-domiciled international equity funds over 2000–2024. Statistics are reported at three frequencies: fund-quarter for portfolio and manager-education characteristics (Panel A), fund-year for size, fees, turnover, and performance (Panel B), and once per fund for manager tenure and team size (Panel C). The average fund manages \$1.6 billion in net assets, though the distribution is sharply right-skewed (median \$0.23 billion), and is larger than the average U.S. domestic fund in Pool et al. (2012), who also use Morningstar data. Funds charge an average net expense ratio of 1.22%, comparable to the average across all U.S. actively managed mutual funds.⁴ Portfolios turn over about 64% of holdings per year and are run by teams of roughly three managers whose

³We follow Datastream’s classification of countries, which may include both sovereign countries and broader geographic regions; the classification does not reflect our own views.

⁴<https://www.morningstar.com/business/insights/blog/funds/us-fund-fee-study>

average tenure is 5.6 years, close to the 4.75-year average Pool et al. (2012) report for U.S. domestic funds.

Central to our analysis is the international composition of fund management teams. Across the sample we identify 2,976 distinct managers with education ties to 52 countries (Table 4). U.S. training dominates—75.2% of managers hold a U.S. degree—but foreign education is widespread: 739 managers (24.8%) have no record of any U.S.-based education, and the United Kingdom (19.4%), Canada (2.8%), India (2.3%), and China (2.1%) are the leading non-U.S. origins. The pattern is more pronounced at the fund level (Table 5): 91.7% of funds employ at least one U.S.-educated manager, but 39.5% employ a U.K.-educated manager, and Canada (11.9%), India (11.2%), and China (8.2%) each appear in a meaningful share of funds. On average, a fund’s management team carries education ties to 1.58 countries, of which 0.72 lie outside the United States. This dispersion in manager backgrounds is the source of variation we exploit throughout the paper.

The sample is dominated by funds with broad global mandates. Table 3 shows that most funds are assigned to global or global ex-US benchmarks—the Morningstar Global ex-US index is the single most common assignment—and that funds tracking broader global indices tend to be the largest, while only a small share are mapped to single-country benchmarks. Relative to these benchmarks, the funds hold strikingly concentrated portfolios. Table 6 reports that the typical fund holds stocks in about 19 countries and roughly 80 stocks (median), against the roughly 50 countries and more than 5,000 stocks in the Global ex-US benchmark. Sample-fund positions are correspondingly larger—the average country weight is about 4.8% of the portfolio and the average stock weight about 0.5%, versus 2.0% and 0.02%, respectively, in the benchmark. This concentration is what makes individual position sizes informative about managers’ selection decisions, and it motivates the within-country activeness measure we develop in Section 3.1.2.

Finally, Figure 4 compares the net-of-fee performance of the sample funds with that of their benchmarks. Fund returns aggregate reported share-class total returns weighted by

each share class’s net asset value, and benchmark returns weight each Morningstar index by the frequency of its assignment across the sample. The funds track their benchmarks closely but underperform slightly on average, consistent with the well-documented difficulty active managers face in outperforming after fees—and setting up the paper’s central question of whether the value of familiarity is instead concentrated in managers’ most active positions.

3 Empirical Methods

Before describing the variables, it is useful to summarize the empirical predictions that the analysis distinguishes. Two views of familiarity yield distinct empirical signatures. Under a pure-bias view, education-country tilts are not associated with abnormal returns at any level of the portfolio. Under an information-with-limited-scale view, the return premium concentrates in managers’ most active top picks, while broader allocation tilts need not generate corresponding returns. The variables and specifications introduced below are designed to separate these views by allowing allocation and returns to be examined at different levels of the portfolio.

3.1 Variable Construction

We use two main variables in the empirical analysis: a fund-level measure of manager education ties to a country, and an equity-level measure of within-country investment activeness. The remainder of this section describes each in turn.

3.1.1 Education Measure

The education measure captures whether a fund has any currently employed manager with post-secondary education ties to a given country. For each manager, we observe the start and end dates of every fund tenure; when multiple managers are employed concurrently, however, we only observe the fund’s aggregate portfolio rather than manager-level holdings.

We therefore aggregate education ties to the fund-quarter level.

Multi-manager funds are not expanded into separate manager-level portfolios: each fund’s observed portfolio enters each analysis only once, regardless of the number of managers on the team. In the country-level analyses, the fund contributes a single fund-country-quarter observation, while in the stock-level portfolio analysis each fund-security-quarter holding is included once before aggregation. Multiple managers affect only the fund-level education indicator and team-share measure; they do not duplicate holdings or portfolio weights.

Concretely, define $\text{Education}_j(t)$ as the set of countries associated with the post-secondary education of any manager actively employed by fund j at time t . Our primary education variable is the indicator

$$\mathbb{1}(\text{EC})_{jc}(t) = \mathbb{1}(c \in \text{Education}_j(t)),$$

which equals one if any current manager of fund j at time t has an education tie to country c , and zero otherwise. As a continuous alternative, we also define $\text{Frac}_{EC,jc}(t)$ as the fraction of fund j ’s current managers at time t whose education-country list includes country c .

3.1.2 Activeness Measure

Antón et al. (2021) introduce a measure of equity-level activeness that identifies fund managers’ top stock picks in U.S. domestic mutual fund portfolios and demonstrate its predictive power for cross-sectional returns. Their measure is designed for a single-country setting, and adapting it to international portfolios requires two extensions.^{5,6} Stock-selection ability is

⁵Appendix C reports the Antón et al. (2021) measure applied to our data. Two design features make our adaptation useful in an international setting: (i) we condition on the destination country, since our analysis of stock selection is most natural within country, and (ii) we adjust for the substantial differences in investment universes between funds and benchmarks. The original Antón et al. (2021) measure is not designed to address either of these features, which we view as complementary extensions for the international context.

⁶The active-share measure of Cremers and Petajisto (2009) is a fund-level statistic that aggregates benchmark-relative deviations across all of a fund’s holdings into a single number. It captures a different, complementary dimension of fund activeness; our setting calls for a within-country, equity-level measure that identifies individual top-activeness positions.

plausibly country-specific in international portfolios, shaped by informational advantages or familiarity with local market conditions that themselves vary across managers and destinations. We therefore develop a measure designed for the within-country, international setting.

Building on their framework, we propose the **I-measure**, designed to proxy for a fund team’s ex-ante conviction in an individual equity position in an international portfolio. The “I” refers to “informational activeness within the fund’s held-stock reference set,” which is the construct the measure is designed to capture. The I-measure addresses a simple question: within the set of stocks a fund currently holds in country c , how much heavier is the fund’s weight on stock i than what would be implied by a market-capitalization-weighted portfolio of the same stocks? A positive value indicates an active overweight on stock i relative to a passive “market-cap tracker” restricted to the fund’s current holdings. A negative value indicates an active underweight, and zero indicates that the fund holds the selected stocks in proportion to their market capitalizations. Because the baseline reference set conditions on stocks the fund already owns, the measure captures position intensity on the intensive margin rather than the decision to include or exclude a stock; the U-measure below broadens the reference set using the fund’s prior holdings.

We arrive at the I-measure in three steps, each addressing a problem with the measure that precedes it. Throughout, equity i belongs to country c , and we denote by $\omega_{jc}^i(t)$ and $B_{jc}^i(t)$ the portfolio weights of equity i in fund j and its assigned benchmark at time t , respectively—where the country is not emphasized we write $\omega_j^i(t)$ and $B_j^i(t)$ for the same objects—and by $V_c^i(t)$ the market capitalization of equity i at time t . Let $\Omega_{jc}(t)$ be the set of equities fund j holds in country c at time t (the fund’s held-stock reference set in c), and let $\Omega_{jc}^B(t)$ be the analogous set for the benchmark.

Step 1 (within-country scaling). A natural starting point is the raw excess weight $\omega_j^i(t) - B_j^i(t)$. Comparing such excess weights across countries, however, is problematic in an international setting because countries differ enormously in equity-market size, and

aggregate country allocations primarily reflect benchmark mandates rather than individual stock selection. To isolate within-country stock selection from broader country-level tilts, we scale each weight by the fund’s (or benchmark’s) total weight in the country. This gives a country-scaled excess weight,

$$A_{jc}^i(t) = \frac{\omega_{jc}^i(t)}{\omega_{jc}(t)} - \frac{B_{jc}^i(t)}{B_{jc}(t)}, \quad (1)$$

where $\omega_{jc}(t) \equiv \sum_{i' \in \Omega_{jc}(t)} \omega_{jc}^{i'}(t)$ is the fund’s total weight in country c and the benchmark’s total weight is $B_{jc}(t) \equiv \sum_{i' \in \Omega_{jc}^B(t)} B_{jc}^{i'}(t)$. Dividing through by $\omega_{jc}(t)$ and $B_{jc}(t)$ renormalizes both sides to sum to one within country c , so the country-allocation margin is removed by construction and what remains is the fund’s allocation across stocks inside the country.

Step 2 (universe restriction). Equation 1 still suffers from a mechanical problem: benchmarks span far broader investment universes than funds. As documented in Table 6, the median sample fund holds roughly 80 equities globally across about 19 countries, whereas its assigned benchmark holds around 1,300 securities at the median (about 1,900 on average); the Morningstar Global ex-US benchmark alone holds over 5,100 securities across nearly 50 countries. Within a given country, the typical fund therefore holds only about four stocks (roughly 80 equities spread across about 19 countries), while the Global ex-US benchmark holds over 100 (about 5,100 equities across nearly 50 countries). As a result, any single equity in A_{jc}^i has a fund-side weight that is mechanically much larger than its benchmark-side weight, and this distorts the measure in two ways. The held stocks appear overweight on average—their excess weights must sum to the benchmark weight of everything the fund omits—so the measure reflects how narrow the fund’s universe is rather than how strongly the fund favors any particular position within it. And among the stocks it does hold, the measure ranks by the fund’s own weight, labeling the largest absolute positions the “most active”—which biases attention toward large-cap stocks regardless of how deliberately they were chosen. The illustration below shows both distortions, and shows how restricting the

reference set removes them.

To correct this asymmetry, we restrict the comparison to the fund’s held-stock reference set: instead of comparing the fund’s within-country weights with the benchmark’s weights over its own, far broader universe, we renormalize the benchmark’s weights over exactly the equities the fund holds. This yields the B-measure,

$$B\text{-measure}_{jc}^i(t) = \begin{cases} \frac{\omega_{jc}^i(t)}{\omega_{jc}(t)} - \frac{B_{jc}^i(t)}{\sum_{i' \in \Omega_{jc}(t)} B_{jc}^{i'}(t)} & \text{if } \sum_{i' \in \Omega_{jc}(t)} B_{jc}^{i'}(t) \neq 0, \\ \frac{\omega_{jc}^i(t)}{\omega_{jc}(t)} & \text{if } \sum_{i' \in \Omega_{jc}(t)} B_{jc}^{i'}(t) = 0. \end{cases} \quad (2)$$

In the first case, the reference term is the weight a passive investor would place on equity i when holding exactly the fund’s stocks in country c and weighting them as the assigned benchmark does; both terms sum to one over the same set of stocks, so the mechanical overweighting of held positions disappears, and the measure asks only which of the fund’s positions are aggressive relative to its others. The second case is required because the benchmark reference can be empty: when the assigned benchmark holds none of the equities the fund owns in country c , the B-measure collapses to the fund’s within-country weight, so positions can still be ranked within the cell—by their raw within-country weights—even where the benchmark provides no reference.

Step 3 (a benchmark-free reference). The B-measure resolves the universe asymmetry, but its reference weights still depend on the identity of the assigned benchmark—a dependence that is stronger still in the raw and country-scaled excess weights, where the benchmark also supplies the entire comparison universe. This residual dependence is unnecessary: once the reference set is the fund’s own holdings, the benchmark’s only remaining role is to dictate relative weights among stocks the fund already owns, so two funds holding identical portfolios could be scored differently merely because Morningstar assigns them different indices. We therefore replace the renormalized benchmark weights with the held stocks’ market capitalizations, which are defined for every position—including holdings out-

side the assigned index—and are free of benchmark selection. Conceptually, we ask how the fund’s position sizes depart from a passive proportional-tracking allocation that holds the same equities in proportion to their market capitalizations.⁷ This yields the I-measure,

$$I_{jc}^i(t) = \frac{\omega_{jc}^i(t)}{\omega_{jc}(t)} - \frac{V_c^i(t)}{\sum_{i' \in \Omega_{jc}(t)} V_c^{i'}(t)}. \quad (3)$$

A further advantage of the market-cap reference is that no fallback case is needed: market capitalizations are strictly positive, so, unlike the benchmark reference in Equation 2, the reference term is defined in every fund-country cell. In practice, the switch from benchmark weights to market capitalization moves the reference little: Morningstar’s benchmarks are predominantly market-capitalization-weighted indices that cover, on average, 97% of the market capitalization within their respective categories—the only exceptions being four “Target Market Exposure” (TME) benchmarks, whose mild individual-stock caps leave the indices close to market-cap-weighted in practice—so the B-measure is numerically close to the I-measure. We adopt the benchmark-free I-measure as the baseline throughout and re-estimate the main portfolio sorts under the B-measure as a robustness check (Section 5, Appendix B.1).

Numerical illustration. Consider one country in which the benchmark holds three stocks: A, carrying 9% of the benchmark’s country weight; B, carrying 1%; and C, carrying the remaining 90%. Read C as a stand-in for the many stocks the benchmark holds in a country but the fund does not own—in our sample the assigned benchmarks hold on the order of 70 stocks in a typical country, and the Global ex-US benchmark over 100, against the fund’s four. The fund holds only A and B, at 60% and 40% of its own country exposure. Because the benchmarks are essentially capitalization-weighted, these benchmark weights also describe relative size: A is the large-cap of the pair, accounting for $9/(9 + 1) = 90\%$ of the market capitalization inside the fund’s universe $\{A, B\}$, against 10% for B.

⁷Concretely, the proportional-tracking allocation puts weight $V_c^i(t)/\sum_{i' \in \Omega_{jc}(t)} V_c^{i'}(t)$ on each $i \in \Omega_{jc}(t)$. If stock A has nine times the market capitalization of stock B inside the fund’s reference set, the proportional tracker also holds A and B in a 9:1 ratio.

The country-scaled excess weight $A_{jc}^i(t)$ of Equation 1 compares each of the fund's weights with the benchmark's weight on the same stock: $60 - 9 = +51$ percentage points for A and $40 - 1 = +39$ for B. Two problems surface at once. First, both positions look strongly overweight. This is largely mechanical: the held stocks carry all of the fund's country weight but only a tenth of the benchmark's, so their excess weights must sum to +90—precisely the benchmark weight of the stocks the fund does not own. Individual positions can still come out negative, but the measure is shifted upward for every held stock by this missing-universe term. What the measure registers is the narrowness of the fund's universe, not the intensity of its convictions. Second, the ranking inverts: A is labeled the more active position, when A is merely the large-cap whose benchmark weight was diluted by the presence of C.

Restricting the reference to the two stocks the fund actually holds removes both problems. Renormalizing the benchmark's weights over $\{A, B\}$ gives reference weights of $9/(9 + 1) = 90\%$ on A and 10% on B, so the B-measure is $60 - 90 = -30$ for A and $40 - 10 = +30$ for B; and because the benchmark is capitalization-weighted, the I-measure's market-cap tracker holds the same $90\%/10\%$ and delivers identical values while depending on no benchmark at all. The two excess weights now sum to zero, as they must once both sides are measured over the same universe, so the measure speaks only to which of the fund's positions is aggressive relative to its others. The ranking also reverses, identifying B as the active position—the economically sensible answer. A passive investor holding these same two stocks would put 90% in A, so the fund's 60% is in fact an underweight, while its 40% in B is four times the 10% such an investor would hold. B is where the manager has departed from passive, and A's apparent activeness under the country-scaled measure was an artifact of the stocks missing from the fund's book rather than a view about A. Had the fund held A and B in the same 9:1 ratio as their capitalizations, both would receive an I-measure of zero, indicating no active deviation.

3.1.3 An Additional Activeness Measure for Robustness: The U-Measure

The I-measure defines the fund’s investment universe as the set of equities the fund holds at time t . As a robustness check, we widen this universe to include all equities the fund has held in country c at any point in the prior eight quarters (two years). We call the resulting measure the U-measure, defined as

$$U_{jc}^i(t) = \frac{\omega_{jc}^i(t)}{\sum_{i' \in \Omega_{jc}(t-8,t)} \omega_{jc}^{i'}(t)} - \frac{V_c^i(t)}{\sum_{i' \in \Omega_{jc}(t-8,t)} V_c^{i'}(t)}. \quad (4)$$

The only difference from the I-measure is that the contemporaneous universe $\Omega_{jc}(t)$ is replaced by the rolling two-year universe $\Omega_{jc}(t-8, t)$. Together with the B-measure of Step 2, this provides two robustness variants of the baseline measure—one changing the reference weights, the other widening the reference set—both examined in Section 5.

3.2 Regression Analysis

This subsection sets up the country-level regression analyses used in the paper. We first examine whether managers’ foreign education ties are associated with systematic deviations from benchmark country allocations, and then assess the performance implications of those deviations. Together, these specifications clarify how familiarity is associated with country-level allocation in international mutual funds and provide initial evidence on whether the resulting tilts are consistent with the bias view or the information view; the equity-level analysis follows in Section 4.3. We exclude U.S. equity holdings throughout, since the funds in our sample are U.S.-based international equity funds whose intended exposure is non-U.S.

3.2.1 Excess Holdings in Managers’ Education Countries

A large literature documents that mutual funds systematically deviate from their benchmark allocations. Pool et al. (2012), for example, show that a 100-percentage-point increase in the share of a fund’s managers from a given U.S. state raises the fund’s allocation to that state

relative to benchmark weights. We extend this idea to an international setting by asking whether managers' education backgrounds are associated with country-level portfolio tilts beyond what standard international diversification would imply.

We estimate

$$Y_{jc}(t) = \alpha + \beta \mathbb{1}(\text{EC})_{jc}(t) + \text{FE} + \epsilon_{jc}(t). \quad (5)$$

We consider two dependent variables for $Y_{jc}(t)$. The raw excess weight, $\omega_{jc}(t) - B_{jc}(t)$, is the fund's portfolio weight in country c minus the corresponding benchmark country weight, where $\omega_{jc}(t)$ is fund j 's portfolio weight in country c at time t and $B_{jc}(t)$ is the weight of country c in fund j 's Morningstar benchmark. The benchmark-scaled excess weight,

$$\frac{\omega_{jc}(t) - B_{jc}(t)}{B_{jc}(t)},$$

expresses the same deviation as a percentage of the benchmark country weight; combined with the continuous team-share specification below, this makes the magnitude directly comparable to domestic-fund estimates. We also re-estimate Equation 5 with $\mathbb{1}(\text{EC})_{jc}(t)$ replaced by the continuous fraction $\text{Frac}_{\text{EC},jc}(t)$, yielding

$$Y_{jc}(t) = \alpha + \beta \text{Frac}_{\text{EC},jc}(t) + \text{FE} + \epsilon_{jc}(t). \quad (6)$$

Throughout, the coefficient of interest is β : when the dependent variable is the raw excess weight, β measures the percentage-point increase in the fund's excess weight in country c associated with the indicator turning on (Equation 5) or with a 100-percentage-point increase in the manager-education share (Equation 6); when the dependent variable is the benchmark-scaled excess weight, β measures the corresponding proportional increase relative to the benchmark country weight. We estimate alternative specifications with combinations of country $\times$ time fixed effects, fund fixed effects, and fund $\times$ time fixed effects. Country $\times$ time fixed effects absorb time-varying shocks common to a given country. The most demanding

specifications replace the fund fixed effect with a fund $\times$ time fixed effect. Because any fund-level characteristic in a given period—size, flows, fees, turnover, team size—is constant across the countries the fund holds at that date, this fixed effect absorbs all time-varying fund-level heterogeneity nonparametrically, acting as an exhaustive set of fund-level controls; identification comes solely from whether the same fund, on the same date, tilts toward the countries in which its managers were educated. The event study below applies the same device through its fund $\times$ quarter fixed effect α_{jq} .

These panel regressions describe a robust association, but they cannot, on their own, distinguish two channels through which education and country exposure could move together: a fund may hire managers educated in countries it intends to overweight, or managers themselves may tilt the fund toward countries where they studied. Both channels involve education-linked familiarity, but they attribute the tilt to different decision makers. To separate them, we exploit the timing of changes in manager composition. Let $Fund_EC_{jqc}$ equal one if fund j has at least one manager with a non-U.S. education tie to country c in quarter q . An entry event occurs when $Fund_EC_{jqc}$ switches from zero to one—the fund gains its first manager with an education tie to country c —and an exit event when it switches from one to zero, so that all such managers have left. We follow each event for up to twelve quarters while the treatment status is unchanged, and require a six-quarter clean window before the event; Appendix B.4 details these filters and the resulting event counts. On the quarterly fund-country panel we then estimate

$$ExcessHold_{jqc} = \sum_{k \neq -1} \beta_k \mathbb{1}\{EventTime_{jqc} = k\} + \alpha_{jc} + \alpha_{cq} + \alpha_{jq} + \varepsilon_{jqc}, \quad (7)$$

where α_{jc} , α_{cq} , and α_{jq} denote fund-country, country-quarter, and fund-quarter fixed effects, and standard errors are clustered at the fund level. The specification compares the same fund-country pair before and after manager turnover while absorbing shocks common to the fund in a quarter and to the country in a quarter.

The two margins play complementary roles in identification. Entry events provide the primary evidence that allocations respond to the arrival of familiarity, and they do so most precisely. Because hiring is an endogenous decision, however, entry alone cannot separate the two channels: a fund may recruit a manager educated in country c precisely because it intends to overweight that country. Exit events discipline that interpretation. If the tilt merely reflected fund-level plans that happened to align with a new hire’s background, there would be no reason for it to recede when that manager later leaves. A tilt that builds after the first education-linked manager arrives and unwinds after the last one departs therefore ties the allocation to the managers themselves rather than to a fixed fund mandate.

A mandate that itself varies over time is harder to dismiss: a fund whose appetite for country c rises might hire a c -educated manager, and might part with that manager when the appetite fades. What speaks against this reading is the timing. A portfolio can be rebalanced within days, whereas hiring or replacing a manager takes quarters, so a mandate-driven account predicts that allocations move first and the roster follows. The event studies show the opposite ordering: benchmark-relative holdings are flat in the quarters preceding both entry and exit, and begin to move only once the manager has actually arrived or left. A time-varying mandate cannot be excluded outright, but it would have to shift the portfolio only after—and never before—the corresponding change in the management team. The results, reported alongside the allocation estimates in Section 4.1, display exactly this two-sided pattern.

3.2.2 Education-Country Return Predictability

Deviations from benchmark holdings can reflect either informational advantage or behavioral bias. If a fund’s overweighting of a manager’s education country reflects superior information, we should observe stronger sub-portfolio performance in that country. If the overweighting is driven primarily by familiarity-based preference, as under the pure-bias view, returns in education countries should not differ systematically from returns elsewhere. We test this by

estimating

$$R_{jc}(t+1) = \alpha + \beta X_{jc}(t) + \text{FE} + \epsilon_{jc}(t), \quad (8)$$

where the right-hand-side education variable $X_{jc}(t)$ is alternatively $\mathbb{1}(\text{EC})_{jc}(t)$ or $\text{Frac}_{EC,jc}(t)$. The dependent variable is fund j 's value-weighted return in country c over the holding period following portfolio formation at t : the subsequent month in Panel A of Table 11, and the return compounded over the subsequent three months in Panel B. Writing the horizon generically,

$$R_{jc}(t+1) \equiv \frac{\sum_{i \in \Omega_{jc}(t)} \omega_{jc}^i(t) R_c^i(t+1)}{\sum_{i \in \Omega_{jc}(t)} \omega_{jc}^i(t)},$$

where $\Omega_{jc}(t)$ is the set of equities fund j holds in country c at time t , $\omega_{jc}^i(t)$ is the corresponding portfolio weight, and $R_c^i(t+1)$ is equity i 's return from t to $t+1$.

As before, we include alternative combinations of fixed effects. The main specification includes country $\times$ time fixed effects, which absorb all time-series variation in aggregate country returns, plus fund fixed effects, which absorb time-invariant fund characteristics. Identification therefore comes from within-country, cross-fund variation in manager education ties. The coefficient β measures whether, for a given country at a given time, funds with managers educated in that country earn systematically higher returns than otherwise-comparable funds—that is, whether familiarity translates into country-level outperformance. With the indicator specification, β is the average return differential associated with the presence of any education-tied manager; with the continuous specification, it is the differential associated with a 100-percentage-point increase in the manager-education share.

3.3 Portfolio Construction and Performance Evaluation

To assess the informational content of education-based familiarity at the stock level, we test whether fund managers exhibit superior stock-picking ability in their education countries. Antón et al. (2021) show that, while U.S. mutual funds rarely outperform their benchmarks on average, their implied “best ideas” can still generate significant alpha—suggesting that

stock-picking skill may be obscured when performance is evaluated at the aggregate portfolio level. We extend this idea to the international setting and ask whether managers’ best ideas are systematically stronger in countries where they received their education.

Our baseline portfolio analysis uses the full 2000–2024 sample and the I-measure in Equation 3, whose passive reference weights stocks by market capitalization. Alternative activeness measures, sample restrictions, and familiarity definitions are presented as robustness exercises rather than alternative headline specifications.

For each fund j , country c , and quarter t , we sort the equities held by the fund in country c into four groups, or quartiles, based on the I-measure within that fund-country-quarter cell. Denote by $\Omega_{jc}^k(t)$ ($k = 1, \dots, 4$) the set of equities in the k -th quartile, with quartile 4 having the highest ex-ante I-measure. Pooling across all funds, countries, and quarters, we form a top portfolio from the union of quartile-4 equities and a non-top portfolio from the union of quartiles 1–3. We construct these two portfolios separately for education and non-education countries, yielding four sorted portfolios: top–education, top–non-education, non-top–education, and non-top–non-education.

We evaluate the performance of each portfolio against the Morningstar Global Markets ex-US NR USD index (the most commonly assigned benchmark in our sample), which tracks large-, mid-, and small-cap equities across developed and emerging markets outside the United States and covers approximately 97% of the investable universe by market capitalization. We choose this benchmark because our constructed portfolios consist exclusively of non-U.S. equities.

Portfolios are formed and rebalanced quarterly using quarter-end holdings; we then compute monthly value-weighted returns over the next three months. To isolate within-country stock selection from differences in country allocation, at each quarterly formation date we reweight the portfolio’s country exposures to match the benchmark weights across the countries represented in that portfolio. These target weights are then allowed to drift with realized security returns during the three-month holding period; we do not reimpose contemporane-

ous benchmark country weights each month. Letting $\omega_c^i(t)$ denote the original weight of equity $i \in \Omega_c(t)$ at quarterly formation date t , the target reweighted weight is

$$\hat{\omega}_c^i(t) = \omega_c^i(t) \cdot \frac{B_c(t) / \sum_{c' \in \Gamma(t)} B_{c'}(t)}{\omega_c(t) / \sum_{c' \in \Gamma(t)} \omega_{c'}(t)}, \quad (9)$$

where $B_c(t)$ is the benchmark's total weight in country c , $\omega_c(t)$ is the portfolio's total weight in country c , and $\Gamma(t)$ is the set of countries in which the portfolio has non-zero exposure at formation date t . This procedure aligns country exposures—and therefore approximately aligns currency exposures—with the benchmark at formation, although modest differences can emerge as weights drift during the holding period. The portfolio's monthly return is then

$$R_{P_n}(t) = \frac{\sum_{c \in \Gamma_{P_n}} \sum_{i \in \Omega_c(t-1)} \hat{\omega}_c^i(t-1) R_c^i(t)}{\sum_{c \in \Gamma_{P_n}} \sum_{i \in \Omega_c(t-1)} \hat{\omega}_c^i(t-1)}, \quad (10)$$

where $R_c^i(t)$ is the monthly return of equity i in country c , $\hat{\omega}_c^i(t-1)$ is the position's weight at the end of month $t-1$ —the reweighted formation weight of Equation 9, subsequently drifting with realized returns within the quarter—and Γ_{P_n} is the set of countries held by portfolio P_n .

To evaluate each portfolio's risk-adjusted performance, we estimate an international Fama-French five-factor specification augmented by the portfolio's benchmark return. We use the five factors (market excess return, SMB, HML, CMA, and RMW) constructed separately for developed and emerging markets, denoted $F_i^{\text{dev}}(t)$ and $F_i^{\text{em}}(t)$ for $i = 1, \dots, 5$. Rather than combine the two regional series or constrain their coefficients to be equal, we allow the portfolio to load separately on each developed- and emerging-market factor. The regression specification is

$$R_{P_n}(t) - R_f(t) = \alpha_n + \beta_n [R_B(t) - R_f(t)] + \sum_{i=1}^5 \gamma_{n,i}^{\text{dev}} F_i^{\text{dev}}(t) + \sum_{i=1}^5 \gamma_{n,i}^{\text{em}} F_i^{\text{em}}(t) + \epsilon_n(t), \quad (11)$$

where $R_f(t)$ is the monthly 3-month U.S. Treasury rate and $R_B(t)$ is the monthly benchmark

return. Thus, the regression includes the benchmark excess return and ten regional factor returns—five for developed markets and five for emerging markets. The estimated α_n measures the portfolio’s risk-adjusted abnormal return relative to these regional Fama-French factors and the Morningstar Global Markets ex-US index. For brevity, we refer to it below as the international five-factor α .

Equation 11 is estimated separately for each of the four sorted portfolios, so the comparisons we report—top versus non-top within a country group, and education versus non-education within an activeness group—are tests of the equality of intercepts across two of these regressions. Because the four portfolios are formed from overlapping holdings and are observed over the same months, their residuals are contemporaneously correlated, and treating the estimates as independent would understate the standard error of the difference. We therefore stack the relevant pair of equations and test the equality of their intercepts with a Wald statistic, using a joint covariance matrix estimated by the sandwich (robust) method that permits arbitrary correlation between the two equations’ residuals. All p -values reported for differences between portfolio alphas, in the figures and in the text, come from this test.

Our main portfolio analysis sorts on the I-measure; we report robustness checks in Section 5 using the B-measure and the U-measure. Figure B14 (Appendix B.11) also reports portfolio sorting results based on the country-scaled excess weight $A_{jc}^i(t)$ in Equation 1, which omits the universe-restriction step and produces less precise sorting—underscoring the value of restricting the comparison to the fund’s investable universe. Beyond risk-adjusted alphas, we also plot cumulative returns of the four portfolios against the benchmark to visualize relative performance over time.

4 Results

4.1 Manager Education and Excess Holdings

We document robust evidence that mutual funds systematically overweight countries in which their current managers received their education. This over-allocation manifests as significantly larger excess portfolio weights on equities from those countries. The main regression results are reported in Table 1.

VARIABLES	(1) Excess ω	(2) Excess ω	(3) Scaled Excess ω	(4) Scaled Excess ω	(5) Excess ω	(6) Excess ω	(7) Scaled Excess ω	(8) Scaled Excess ω
1(EC)	0.974*** (0.219)		39.39*** (5.857)		0.955*** (0.218)		40.22*** (5.917)	
Frac _{EC}		2.465*** (0.435)		80.72*** (11.29)		2.417*** (0.427)		81.61*** (11.41)
Constant	-0.0462*** (0.00339)	-0.0488*** (0.00312)	-19.42*** (0.0842)	-19.38*** (0.0737)	-0.0474*** (0.00335)	-0.0498*** (0.00303)	-19.43*** (0.0844)	-19.38*** (0.0735)
Observations	5,613,030	5,613,030	5,244,238	5,244,238	5,611,449	5,611,449	5,242,603	5,242,603
R-squared	0.145	0.146	0.119	0.120	0.191	0.192	0.139	0.139
Fund FE	Y	Y	Y	Y	—	—	—	—
Fund×Time FE	N	N	N	N	Y	Y	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y	Y	Y
Sub-Sample	N	N	N	N	N	N	N	N
Funds Number	1663	1663	1663	1663	1663	1663	1663	1663

Table 1: Education Links and Excess Country Weight

Notes: This table reports panel regressions of country-level excess weight on manager education ties to the destination country. The unit of observation is a fund-country-quarter. Columns (1)–(2) and (5)–(6) use raw excess weight (fund country weight minus benchmark country weight, in percentage points) as the dependent variable, and columns (3)–(4) and (7)–(8) scale the same difference by the benchmark country weight (multiplied by 100). 1(EC) equals one if any current manager of the fund received post-secondary education in the country in question; Frac_{EC} is the share of the fund’s current managers educated in that country. Columns (1)–(4) include fund and country×time fixed effects. Columns (5)–(8) repeat the same specifications but replace the fund fixed effect with a fund×time fixed effect (which subsumes it, indicated by “—”), absorbing all time-varying fund-level characteristics; singleton fund-date cells are dropped, slightly reducing the sample. Robust standard errors clustered at the fund level are in parentheses. The “Sub-Sample” row indicates whether the estimation is restricted to funds assigned to a global Morningstar benchmark; all columns here use the full sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In column (1) of Table 1, the dependent variable is the raw country-level deviation of a fund’s portfolio weight from the benchmark allocation. The primary explanatory variable, 1(EC), equals one if any of the fund’s current managers received post-secondary education

in country c . The coefficient indicates that the presence of a manager with educational ties to a country raises the fund’s excess weight in that country by 0.97 percentage points on average. This magnitude is economically meaningful: the median fund holds positions in 19 countries, so an average country position is around 5% of the portfolio. The effect is robust across alternative specifications, including models with different sets of fixed effects and across fund subsamples; as reported in Table 7, point estimates of over-allocation range from 0.96 to 1.07 percentage points.

Column (3) replaces the dependent variable with the benchmark-scaled excess weight, defined as the raw deviation divided by the benchmark’s country weight. The coefficient on $\mathbb{1}(\text{EC})$ implies that funds overweight education countries by 39.39% relative to benchmark allocations. For example, if the benchmark assigns 10% weight to a country, a fund with at least one manager educated there allocates, on average, an additional 3.94 percentage points beyond benchmark—relative to an otherwise similar fund without such ties. This benchmark-scaled effect remains robust across alternative specifications and subsamples, as reported in Table 8.

Equation 6 replaces the indicator $\mathbb{1}(\text{EC})$ with Frac_{EC} , the fraction of the fund’s current managers educated in the focal country. Column (2) reports the corresponding raw-weight regression: the coefficient of 2.47 implies that a 100-percentage-point increase in the manager-education share raises the fund’s excess country weight by 2.47 percentage points. To illustrate, if a fund with one foreign-educated manager hires a second who was educated in country c , the manager-education share in c rises from 0 to 50%, implying an expected $0.5 \times 2.47 = 1.23$ percentage-point increase in the fund’s excess weight in c . Robustness across alternative specifications (Table 9) yields coefficients between 2.42 and 2.77.

Direct comparisons of raw weight deviations across domestic and international settings can be misleading because of differences in portfolio granularity: U.S. domestic funds allocate across 50 states, while international funds typically invest across a much smaller set of countries with very heterogeneous benchmark weights. We therefore place greater emphasis on the

benchmark-scaled measure, which is more comparable across contexts. Column (4) of Table 1 reports the benchmark-scaled specification with Frac_{EC} : a 100-percentage-point increase in the manager-education share is associated with an 80.72% increase in the benchmark-scaled portfolio weight. Robustness checks are reported in Table 10.

Columns (5)–(8) of Table 1 replace the fund fixed effect with a fund $\times$ time fixed effect—the nonparametric counterpart of an exhaustive set of time-varying fund-level controls, since any fund characteristic in a given quarter is constant across the countries the fund holds at that date. The education coefficient is then identified solely by comparing, within the same fund on the same date, education countries with the fund’s other markets. The estimates are essentially unchanged: 0.96 percentage points for the indicator and 2.42 for the team share in the raw-weight specifications (columns (5)–(6)), and 40.22% and 81.61% in the benchmark-scaled specifications (columns (7)–(8)). The over-allocation is therefore not an artifact of time-varying fund-level heterogeneity but reflects where, within a portfolio at a point in time, a fund chooses to deviate.

For comparison, the corresponding domestic estimate in Pool et al. (2012) is 18.8%. Our international estimate of 80.72% is more than four times larger, consistent with familiarity exerting a stronger influence on portfolio allocation when information frictions and market segmentation are greater.

The panel estimates establish the association; the event study of Equation 7 establishes that the tilt follows the managers. Figure 1 plots the dynamic coefficients around entry and exit events. When a fund gains its first manager with an education tie to country c , its benchmark-relative holding in that country does not jump in a single quarter but builds over the following several quarters, reaching roughly 0.5–0.7 percentage points of excess weight and remaining elevated through the end of the twelve-quarter window. The pre-event pattern supports the design: the coefficients in the six quarters before entry are small, flat, and statistically indistinguishable from zero, showing no build-up toward the event. The post-entry increase is therefore not the continuation of a pre-existing trend—exposure

begins to accumulate only once the education-linked manager arrives.

The entry estimates are the sharper of the two panels statistically; the exit estimates, though noisier, carry the complementary identification argument. After the last education-linked manager departs, benchmark-relative holdings decline and remain negative for ten or more quarters, with the largest declines two to five quarters after exit. As argued in Section 3.2.1, this unwinding is the pattern that endogenous hiring alone cannot generate: a tilt reflecting a fixed fund mandate would have no reason to recede when the manager who embodied the tie leaves. The holdings also do not move in anticipation of the departure—the coefficients through quarter -1 are flat and statistically indistinguishable from zero—so the decline follows the manager rather than preceding them, as it would if the fund had first decided to retreat from the country and the manager’s exit merely followed that decision. The reversal is partial rather than complete: holdings fall by roughly 0.2–0.5 percentage points after exit, against the 0.5–0.7 percentage points accumulated after entry. This is what one would expect if the managers who remain retain part of the position their departing colleague built, and a full reversal is not required for the timing evidence to bind—what identifies the tilt to the manager is that exposure recedes at all when the tie is severed. The benchmark-weighted version of the outcome delivers the same two-sided pattern (Appendix B.4, Figure B6).

4.2 Education Country Sub-portfolio Return

Table 11 reports country-level return regressions that compare the performance of a fund’s sub-portfolios in education countries with those in non-education countries at one- and three-month horizons. Panel A uses the return in the month following portfolio formation. Education-country sub-portfolios outperform non-education-country sub-portfolios by 0.7 basis points in our preferred country $\times$ time fixed-effects specification, but the estimate is not statistically significant; the corresponding continuous education-share estimate is -2.7

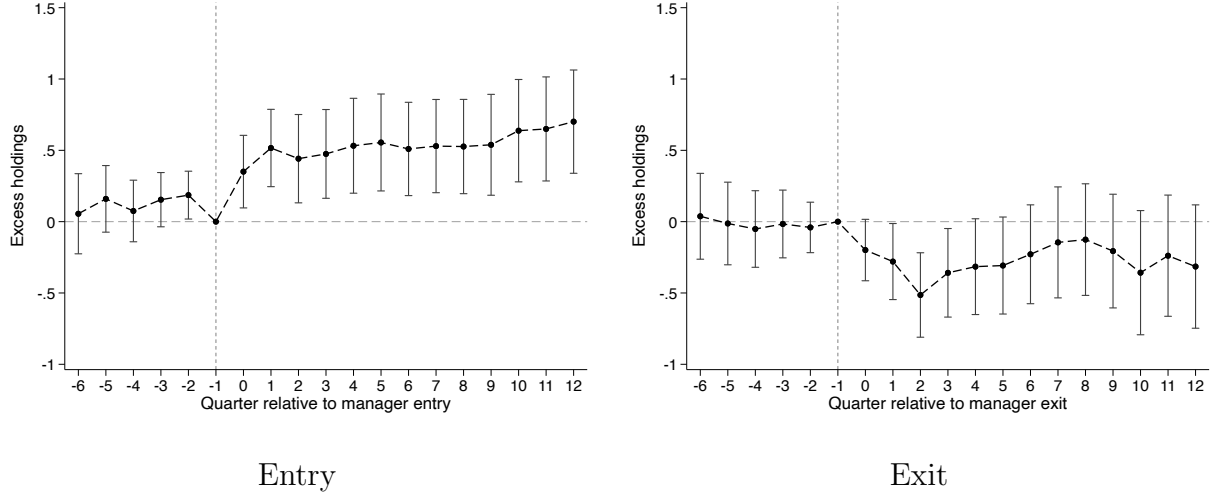


Figure 1: Excess Holdings Around Manager Entry and Exit

Notes: This figure plots event-study coefficients from Equation 7 for raw excess holdings (the fund's country weight minus the benchmark country weight, in percentage points) around changes in $Fund_EC_{jq}$, an indicator equal to one if fund j has at least one manager with a non-U.S. education tie to country c in quarter q . Entry events are quarters in which the indicator switches from zero to one, capturing the arrival of the fund's first education-linked manager for that country; exit events are switches from one to zero, capturing the departure of the last such manager. Event windows run up to twelve quarters and end if the treatment status reverses; events failing a six-quarter clean-lead washout and overlapping windows are removed (974 entry and 777 exit events remain; see Appendix B.4). Both panels share the same vertical scale. Regressions include fund-country, country-quarter, and fund-quarter fixed effects, with quarter -1 as the reference period. Capped bars denote 95% confidence intervals; standard errors are clustered at the fund level.

Alt text: Two event-study panels plotting coefficients with 95% confidence intervals for excess country holdings, in percentage points, against event-time quarters. In the entry panel, coefficients are flat near zero in the six quarters before a fund gains its first education-linked manager and rise gradually to roughly 0.5–0.7 percentage points afterward; in the exit panel, coefficients turn negative after the last education-linked manager departs and remain below zero.

basis points and is likewise insignificant.

Panel B uses the cumulative return compounded over the next three months. In the least-saturated indicator specification, the estimated cumulative differential is 12.81 basis points with a standard error of 6.81, marginally significant at the 10% level. This cumulative estimate corresponds to approximately 4.3 basis points per month. The evidence is not robust to the preferred fixed effects: with country $\times$ time fixed effects, the cumulative three-month estimate falls to 8.97 basis points with a standard error of 6.57 and is statistically insignificant. The continuous education-share estimates are close to zero and insignificant in both three-month specifications.

Panel A has fewer than half as many observations as Table 1. The difference reflects sample construction: the excess-weight regression fills in zero weights for fund-country-quarter cells where the fund holds no equity, while the return regression is restricted to cells with positive holdings. Panel B is smaller still because it uses quarter-end formation dates, producing non-overlapping three-month return horizons.

Together, these results provide no robust evidence that, despite substantially overweighting their managers' education countries, funds earn superior country-level returns in those markets. The isolated marginal estimate in the least-saturated three-month specification does not survive country $\times$ time fixed effects and has no counterpart in the continuous-measure regressions. The broad allocation tilt is therefore not matched by a reliable broad country-level performance advantage. We next examine whether education-linked information or skill appears at the stock-selection margin instead.

4.3 Portfolio Analysis

The previous subsection shows that holdings in managers' education countries do not generate higher country-level returns. This contrasts with Jagannathan et al. (2022), who report positive abnormal returns on funds' home-country holdings; as Appendix D discusses, this difference is tied to sample coverage. We now turn to within-country, stock-level selection

and ask whether the value of familiarity instead manifests in managers’ most active equity picks.

Figure 2 reports the monthly Fama-French five-factor α (in basis points) for four portfolios constructed by sorting the fund-country-quarter sample on the I-measure: (i) non-education country, non-top quartile; (ii) non-education country, top quartile; (iii) education country, non-top quartile; and (iv) education country, top quartile. We refer to the top-quartile and non-top-quartile portfolios as the “top portfolio” and “non-top portfolio” for brevity. The figure plots all four alphas together with three p -value comparisons: black brackets near the markers report top vs. non-top differences within each country group, the wider top bracket reports the education vs. non-education comparison among top portfolios, and the wider bottom bracket reports the analogous comparison among non-top portfolios.

The top portfolio in education countries—the equities held most actively by funds in countries linked to the current management team’s education—earns a monthly α of 41.0 basis points, or roughly 5.03% annualized. This exceeds the 2.8–4.5% annual α that Antón et al. (2021) report for “best ideas” portfolios in the U.S. domestic market and is consistent with education-linked information or skill concentrated in a narrow subset of high-I positions.

By contrast, the non-top portfolio within education countries earns a monthly α of -3.8 basis points, indistinguishable from zero. The top–non-top differential within education countries is therefore both economically large and statistically significant at the 1% level (p -value = 0.001), underscoring that the value of familiarity is concentrated in the most active positions rather than diffused across all education-country holdings. This concentration is not an artifact of the five-factor risk model: Appendix B.6 repeats the sort adjusting for the assigned benchmark only, where the education-country top portfolio still earns a large monthly α of 33.9 basis points—if anything smaller than the five-factor estimate, so the factor adjustment is not manufacturing the result.

In non-education countries, the top and non-top portfolios are statistically indistinguish-

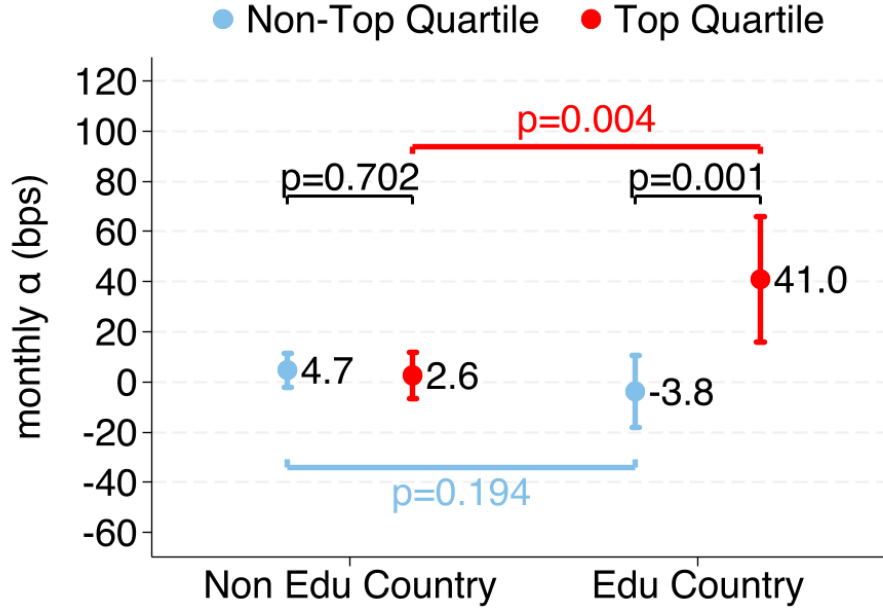


Figure 2: Portfolio α from I-Measure Sorting

Notes: This figure provides portfolio α from I-measure sorting with the Fama-French five-factor model. The two black brackets near the markers report the p -value of the top vs non-top quartile difference within each fund type (non-education country, education country). The wider red bracket on top reports the p -value of the education vs non-education country difference within the top quartile, and the wider blue bracket on the bottom reports the corresponding difference within the non-top quartile. All reported p -values are from Wald tests of the equality of intercepts across the relevant pair of portfolio regressions, computed from a joint covariance matrix that allows the two portfolios' residuals to be contemporaneously correlated (Section 3). The bars provide 95% confidence intervals.

Alt text: Dot plot of monthly five-factor alphas in basis points with 95% confidence intervals for four portfolios. In non-education countries, the non-top and top portfolios earn 4.7 and 2.6 basis points; in education countries, the non-top portfolio earns -3.8 and the top portfolio 41.0 basis points. Brackets report p -values of 0.702 and 0.001 for the top versus non-top comparisons within each group, 0.004 for the top-portfolio comparison across groups, and 0.194 for the non-top comparison.

able: the top portfolio earns a monthly α of 2.6 basis points and the non-top portfolio earns 4.7 basis points, corresponding to annualized alphas of 0.31% and 0.56% respectively. Higher activeness in non-education countries therefore does not translate into superior performance, consistent with limited informational advantage in those markets. Comparing across country groups, the top portfolio in education countries significantly outperforms its non-education counterpart (p -value = 0.004), while the two non-top portfolios are indistinguishable from each other.

Figure 5 decomposes the non-top portfolio further, reporting the α of each of the four I-measure quartiles separately for education and non-education countries. In non-education countries, alphas do not differ meaningfully across quartiles, indicating that variation in activeness alone does not generate performance. In education countries, by contrast, the first three quartiles all earn alphas indistinguishable from zero (-4.0 , -11.4 , and 2.0 basis points for quartiles 1–3), and α then jumps to 41.0 basis points in quartile 4, which significantly outperforms the bottom quartile. The premium is thus a discrete feature of the most active quartile rather than a gradient across activeness. The contrast reinforces the interpretation that the return premium is concentrated in fund teams’ most active positions in their education countries.

Figure 3 translates these alphas into long-run cumulative returns. The left panel plots cumulative returns of the education-country top and non-top portfolios alongside the assigned Morningstar and global factor benchmarks, normalized to one at January 2000. Over the 25-year window, the education-country top portfolio grows roughly tenfold, while the non-top portfolio grows about fourfold, closely tracking the benchmarks. The right panel reports the analogous picture for non-education countries: both top and non-top portfolios closely track the benchmarks. The stark divergence of the education-country top portfolio underscores funds’ ability to identify a small set of high-performing stocks in countries where their managers received their education.

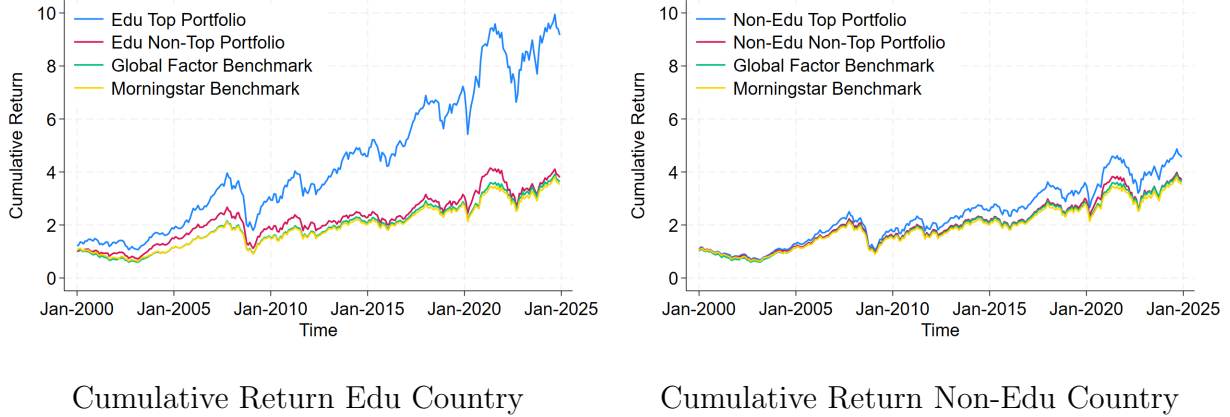


Figure 3: Portfolio Cumulative Return

Notes: This figure provides portfolio cumulative return by quartile using I-measure. The four lines on the left graph are cumulative returns for the Morningstar benchmark portfolio, global factor benchmark portfolio, the education country’s top quartile portfolio, and the education country’s non-top portfolio. The right plots are the non-education country counterparts. The starting period (Jan 2000) is normalized to 1. The education country’s top quartile portfolio performs the best, while all other portfolios closely track the benchmark.

Alt text: Two line charts of cumulative returns over 2000–2024, normalized to one in January 2000. In the left panel, the education-country top portfolio grows roughly tenfold, well above the non-top portfolio and the Morningstar and factor benchmarks, which grow about fourfold; in the right panel, the non-education-country top and non-top portfolios track the benchmarks closely.

Table 12 reports portfolio performance metrics at monthly frequency. The education-country top portfolio delivers the strongest performance among the four sorted portfolios, with an annualized Sharpe ratio of 0.438 and an information ratio of 0.546 against the Glb. ex-US benchmark, comfortably exceeding the benchmark’s own Sharpe ratio of 0.253. The remaining three sorted portfolios deliver Sharpe ratios close to the benchmark’s and information ratios near zero (0.055–0.287). Table 13 reports the corresponding annual return distributions, which show the same ordering: the education-country top portfolio averages 11.18% per year against 6.83% for the benchmark, with comparable return volatility.

The magnitude remains economically meaningful when compared with fund fees. The average annual net expense ratio in our sample is 1.22%, or roughly 10.2 basis points per month. Deducting an amount of that size from each synthetic portfolio would leave the education-country top portfolio with a substantial mean excess return, while the other three portfolios would fall below the Glb. ex-US benchmark. This comparison is illustrative—

fund expenses cannot be allocated mechanically across portfolio sleeves—but it highlights the economic importance of the concentrated education-linked performance advantage.

Next, we investigate the persistence of the education-linked performance advantage. Prior research, such as McLean and Pontiff (2016), shows that the return predictability of factors and estimated alphas often declines over time, particularly after an anomaly is published. More broadly, return predictability may weaken as markets become more integrated and efficient. We therefore repeat the I-measure portfolio sort over several subsamples. Figure B11 (Appendix B.8) presents results for the post-recession period (July 2009–December 2024), the last 10 years (January 2015–December 2024), and the most recent five years (January 2020–December 2024).

The education-country top portfolio continues to generate economically large point estimates across all subsamples. During the post-recession period, it earns a monthly alpha of 21.9 basis points and significantly outperforms the education-country non-top portfolio (p -value = 0.006). Over the past 10 years, the alpha is 24.3 basis points (p -value = 0.052); over the most recent five years, it is 26.0 basis points, although the difference from the non-top portfolio becomes marginal as the sample shortens (p -value = 0.076). Statistical precision naturally declines in shorter windows, but the point estimates provide no evidence of a sharp recent decay. This persistence is consistent with an education-linked advantage that continues to inform stock selection, although the subsample tests do not by themselves distinguish private information from other forms of skill.

Figure B14 (Appendix B.11) presents the portfolio sorting result based on the country-scaled excess weight $A_{jc}^i(t)$ of Equation 1, which compares the fund against the full benchmark instead of restricting the comparison to the fund’s held-stock reference set. While the overall pattern remains directionally consistent, the performance sort is noticeably less sharp than under the I-measure. This contrast reflects the substantial difference between the breadth of fund holdings and benchmark holdings and underscores the importance of comparing position intensity over a common reference set.

4.4 Portfolio Performance After Bid-Ask Spread Adjustment

A potential concern with the portfolio-sorting results is that the top portfolios may require substantial trading, so that their gross outperformance may not survive plausible implementation costs. To address this, we re-estimate the baseline portfolio analysis after subtracting security-level bid-ask spread costs at each quarterly rebalance; the construction is detailed in Appendix B.7. The main conclusion is robust: net of estimated spread costs, the education-country top portfolio earns a monthly five-factor α of 33.7 basis points and the education-country non-top portfolio -11.4 basis points, with the top–non-top differential significant at the 1% level (p -value < 0.001), while the two non-education-country portfolios remain indistinguishable (p -value $= 0.590$). The education-country top portfolio is the most actively traded (133% average one-way annual turnover, 0.85% average annual spread cost), yet it is the only portfolio delivering large positive alpha after the adjustment, so the result is not compensation for mechanically higher turnover (Figures B9–B10 and Table B2, Appendix B.7).

4.5 Characteristics of Top Picks

4.5.1 Do High-I Positions Overlap Across Fund Teams?

A natural question is whether fund teams converge on the same high-I equity positions within a country or whether their selections are largely fund-specific. We focus on the education-country top-quartile portfolio (sorted by the I-measure) and examine the fraction of non-overlapping high-I positions across fund teams within each country-time pair.

Because the top-quartile portfolio is constructed at the fund-country level, the number of unique equities in a given country varies with the number of education-affiliated fund teams represented there. The median country-time cell contains only seven unique equities, while the United Kingdom, with the largest number of education-affiliated teams, has a median of 164. Mechanically, country-time cells with more funds tend to exhibit more overlap, so we

group cells by their equity count to account for this dependence. Figure 6 reports six groups (1; 2–5; 6–10; 11–25; 26–50; more than 50 equities), with green bars showing the number of country-time observations in each group and the red line plotting the average fraction of equities held by only one fund team.

The non-overlap rate remains consistently high across all groups, mostly above 80% (see Figure 6). Even in country-time cells with more than 50 unique equities, 62.9% of those equities are held by only one fund team. Education-country high-I positions are therefore largely fund-specific, consistent with heterogeneous information or skill rather than a single common signal shared across all education-affiliated funds.

4.5.2 Market Cap Comparison of Equities in Each Quartile

We next ask whether the I-measure mechanically sorts on firm size and whether education-country and non-education-country portfolios differ systematically in the size of the firms they target. Figure 7 reports average log market capitalization across the four I-measure quartiles, separately for education and non-education countries. Across quartiles 2–4, log market caps lie between 21.2 and 22.8 (approximately \$1.6–\$8.0 billion). Education-country holdings are larger on average in every quartile, but the gap is essentially constant across quartiles—about one log point (0.9 in quartile 4 versus 1.0–1.2 in quartiles 1–3)—and the 95% confidence intervals overlap throughout. Because the size difference does not widen in the top quartile, where the return premium is concentrated, the differences in stock-picking behavior between the two groups do not appear to reflect systematic differences in the size of firms targeted.

Within each education group, average market capitalization is also stable across I-measure quartiles. In both education and non-education countries, quartile 1 contains slightly larger-cap stocks, which follows from how the measure is built: a mega-cap accounts for a large share of within-country capitalization, so it takes an unusually large fund weight for such a stock to register as active. Across the remaining quartiles, market capitalizations are similar,

and the 95% confidence intervals confirm the absence of meaningful differences. Firm size is therefore unlikely to be a primary driver of the performance differences documented above.

4.5.3 Position Sizing in Education-Linked and Other High-I Holdings

We next compare position sizing across education-linked and other holdings. Because the I-measure is constructed partly from within-country portfolio weights, high-I positions are mechanically expected to be relatively large after controlling for the market-capitalization reference. The economically relevant question is whether this relation is unusually strong in education countries. We therefore compare average excess portfolio weight, $\omega_j^i(t) - B_j^i(t)$, across I-measure quartiles, separately for education and non-education countries. We focus on raw excess weights because they directly determine a fund's excess returns.

Figure 8 reports the comparison. Across the bottom three quartiles, average excess weights are modestly higher for education-country than non-education-country positions, with cross-group differences of around 0.1–0.3 percentage points. By contrast, in the top quartile the gap widens sharply: funds with a manager educated in the country assign an average excess weight of 1.0%, compared with 0.4% for high-I positions in non-education countries. This 0.6-percentage-point gap is economically large—it exceeds the average raw weight on a single equity in the sample (about 0.5%)—and shows that the relation between the I-measure and position size is substantially stronger in education countries. The figure also shows the 5th–95th percentile range of excess weights, illustrating substantial dispersion across fund teams.

To formally evaluate these patterns while controlling for country-specific and time-varying factors that may confound the raw comparison, we estimate the following regression:

$$\begin{aligned} \omega_j^i(t) - B_j^i(t) = & \alpha + \beta \mathbb{1}(\text{Top})_j^i(t) + \chi \mathbb{1}(\text{EC})_{jc}(t) \\ & + \gamma \mathbb{1}(\text{Top})_j^i(t) \times \mathbb{1}(\text{EC})_{jc}(t) + \theta X_{i,j,t} + \text{FE} + \epsilon_j^i(t) \end{aligned} \tag{12}$$

Here $\mathbb{1}(\text{Top})_j^i(t)$ equals one if equity i is in the top quartile of fund j 's I-measure,

$\mathbb{1}(\text{EC})_{jc}(t)$ equals one if any manager of fund j was educated in equity i 's country c , and $X_{i,j,t}$ contains controls for equity-level market capitalization and the number of securities the fund holds within the country. The coefficient of interest is γ , the interaction between education-country status and high-I status. We estimate alternative specifications with various combinations of country and time fixed effects.

Table 14 reports the results, which mirror the pattern in Figure 8 and are robust across specifications. The interaction coefficient ranges from 0.30 to 0.32, indicating that the additional excess weight associated with education-country status is concentrated in the top quartile of holdings rather than spread across all positions.

4.6 Education-Linked Overweighting Outside the Outperforming Positions

The previous results show that the return premium in education countries is concentrated in the top-quartile holdings; the rest of the education-country portfolio earns returns indistinguishable from the benchmark. A natural follow-up is whether funds continue to overweight education countries after these outperforming positions are removed. To investigate this question, we reconstruct each fund's country-level allocation after dropping the top-quartile holdings in both education and non-education countries and re-estimate Equations 5 and 6 on the resulting non-top weights.

Formally, let $\Omega_{jc}(t)$ denote the set of equities fund j holds in country c at time t , and let $\Omega_{jc}^k(t)$ ($k = 1, \dots, 4$) be the four I-measure quartile sub-portfolios. Equations 5 and 6 use the full country weight $\omega_{jc}(t) = \sum_{i \in \Omega_{jc}(t)} \omega_{jc}^i(t)$. We replace it with the bottom-three-quartile country weight,

$$\omega_{jc}^{NT}(t) = \sum_{i \in \bigcup_{k=1}^3 \Omega_{jc}^k(t)} \omega_{jc}^i(t), \quad (13)$$

and re-estimate Equations 5 and 6 with the dependent variable

$$Y_{jc}^{NT}(t) = \omega_{jc}^{NT}(t) - B_{jc}(t) \quad \text{or} \quad Y_{jc}^{NT}(t) = \frac{\omega_{jc}^{NT}(t) - B_{jc}(t)}{B_{jc}(t)}, \quad (14)$$

in place of the corresponding full-portfolio outcomes. In both cases the benchmark term is the fund’s full benchmark country weight $B_{jc}(t)$: the I-measure is defined only over the equities the fund holds, so the benchmark has no top-quartile counterpart to remove.

This asymmetry matters for how the estimates should be read. Because the fund side is reduced while the benchmark side is not—we still subtract, and divide by, the whole benchmark country weight—the coefficients here are mechanically smaller than their full-portfolio counterparts, and comparing the two levels directly understates how far the tilt extends. The bottom three quartiles hold about 73.6% of the fund’s country weight on average, so an education tilt of unchanged intensity would produce a coefficient roughly 73.6% as large. The question is therefore not whether the non-top coefficient is smaller—it must be—but whether it falls by more than this mechanical factor.

Recall from the portfolio analysis that the bottom-three-quartile education-country portfolios earn alphas indistinguishable from the non-education-country portfolios and the benchmark. If education-country over-allocation were confined to the positions associated with superior performance, we should observe no over-allocation in the bottom three quartiles. Table 15 shows otherwise: funds substantially overweight education countries even in this non-outperforming subset.

It does not. Column (2) of Table 15 reports a coefficient of 1.658 on Frac_{EC} : a 100-percentage-point increase in the manager-education share raises the bottom-three-quartile country weight by 1.658 percentage points. Rescaling by $1/0.736$ gives 2.25, close to the 2.465 estimated on the full country weight in Table 1. The other specifications behave the same way. Across the four fund-fixed-effect columns the non-top coefficient is between 59% and 84% of its full-portfolio counterpart—0.577 against 0.974 and 1.658 against 2.465 in the raw-

weight specifications, 32.06 against 39.39 and 67.90 against 80.72 in the benchmark-scaled ones—bracketing the 73.6% of country weight these positions represent. The fund $\times$ time specifications in columns (5)–(8) deliver very similar estimates (0.593 and 1.697 percentage points in the raw-weight specifications; 33.05% and 68.65% scaled). The intensity of education-linked overweighting outside the top quartile is therefore close to that in the portfolio as a whole: it is the smaller base, not a weaker tilt, that accounts for the smaller coefficient.

Together, these results show that education-linked over-allocation extends well beyond the most active positions. Even after removing the holdings associated with superior performance, funds continue to overweight their education countries by a substantial margin. Portfolio allocations therefore display both a concentrated education-linked performance advantage and a broader tilt toward familiar markets that does not translate into measured outperformance. This allocation-to-performance wedge is consistent with familiarity-based allocation bias extending beyond the positions in which education-linked familiarity appears to pay.

5 Robustness Checks

This section reports a series of robustness checks supporting the main portfolio result in Section 4.3: the I-measure top-quartile portfolio in managers’ education countries generates significant five-factor α , while no such premium appears in other portfolios. The checks below vary the activeness measure (B-measure and U-measure), the sample (excluding the United Kingdom), the definition of an education link (undergraduate degrees only), and the performance-attribution model (adding momentum and removing the benchmark control), and we provide additional figures in Appendix B. Across all tests, the qualitative pattern is unchanged.

5.1 Sorting with the Assigned-Benchmark B-Measure

The main I-measure defines the passive proportional-tracking reference using a market-capitalization-weighted portfolio of the equities the fund holds in country c . As a robustness check, we revert the reference to the fund’s Morningstar-assigned benchmark weights—the assigned-benchmark B-measure of Step 2 in Section 3.1.2. Because Morningstar benchmarks are themselves predominantly market-cap-weighted and cover, on average, roughly 97% of the market capitalization within their respective categories, the two references should yield similar rankings; this check verifies that the results are not driven by the use of market capitalization.

The B-measure, defined in Equation 2, replaces the market-cap reference with the assigned benchmark’s holdings on the fund’s universe: it captures a fund’s deviation from a benchmark-weighted portfolio of the equities it actually holds in country c .

Figure B1 (Appendix B.1) reports the B-measure sorting results, which are qualitatively and quantitatively consistent with the I-measure. In non-education countries, the monthly five-factor α is 5.5 bps for the non-top portfolio and 1.8 bps for the top portfolio (p -value = 0.520). In education countries, the monthly α is -4.0 bps for the non-top portfolio and 33.5 bps for the top portfolio, with the top–non-top differential significant at the 1% level (p -value = 0.002). The education-country top portfolio also significantly outperforms its non-education counterpart (p -value = 0.005).

5.2 Sorting with U-measure

The I-measure restricts the comparison universe to equities the fund is currently holding in country c . As a robustness check we widen this universe to include all equities the fund has held in country c at any point in the prior eight quarters (two years). This expanded definition is motivated by Kojen and Yogo (2019), who show that roughly 92% of equities held in a given quarter were also held by the same fund in the prior eight quarters; extending the lookback further has minimal effect.

We call the resulting measure the U-measure, defined in Equation 4; the only difference from the I-measure is that the contemporaneous universe $\Omega_{jc}(t)$ is replaced by the rolling two-year universe $\Omega_{jc}(t - 8, t)$.

Figure B2 (Appendix B.2) reports the U-measure sorting results, which are quantitatively and qualitatively consistent with the I-measure. In non-education countries, the monthly five-factor α is 5.5 bps for the non-top portfolio and 1.6 bps for the top portfolio, with no significant difference (p -value = 0.475). In education countries, the monthly α is -1.4 bps for the non-top portfolio and 34.7 bps for the top portfolio, with the differential significant at the 1% level (p -value = 0.008). The education-country top portfolio also significantly outperforms its non-education counterpart (p -value = 0.012).

5.3 Excluding the United Kingdom from the Analysis

As Table 5 shows, the United Kingdom (GBR) accounts for a disproportionately large share of non-U.S. education backgrounds in our sample: 657 funds have had at least one U.K.-educated manager, while the next most common foreign education countries, Canada (CAN) and India (IND), are associated with 198 and 186 funds respectively. Because U.K. ties are so much more common than any other, the U.K. is the one country that could plausibly drive the result on its own. To verify that it does not, we exclude U.K.-related investments from the analysis.

Specifically, we remove all U.K. equities from the portfolio construction and repeat the sorting exercise. Figure B3 (Appendix B.3) reports the results. In non-education countries, the monthly five-factor α is 9.0 bps for the non-top portfolio and 6.0 bps for the top portfolio (p -value = 0.634). In education countries, the non-top portfolio earns a monthly α of 2.4 bps and the top portfolio earns 62.3 bps, with the top–non-top differential significant at the 1% level (p -value = 0.003). The education-country top portfolio also significantly outperforms its non-education counterpart (p -value = 0.005). These conclusions do not depend on the factor model or on how performance is displayed: Appendix B.3 additionally reports the

ex-U.K. alphas adjusting for the benchmark only (a 42.0 bps monthly α for the education-country top portfolio) and the cumulative-return paths, in which the education-country top portfolio again pulls far above the non-top portfolio and the benchmarks.

5.4 Undergraduate Education Only

Our baseline treats a manager’s education country as any country in which the manager received post-secondary education, whether at the undergraduate or graduate level. As a further robustness check, we restrict the definition to undergraduate education only. Undergraduate study is more likely than graduate study to occur in a manager’s home country and formative environment, so this restriction captures a measure more closely related to formative or home-country familiarity. We estimate the undergraduate-only regressions on the same 1,663-fund panel used in Table 1; fund-country cells without an undergraduate tie are coded as zero, so the robustness check changes the education-link definition rather than the regression sample.

Table B3 re-estimates all eight specifications in Table 1 using undergraduate ties only. In the raw-weight specifications, the coefficient on $\mathbb{1}(EC)$ rises from 0.97 to 1.09 percentage points and the coefficient on $Frac_{EC}$ rises from 2.47 to 2.98. In the benchmark-scaled specifications, the undergraduate-only coefficients are somewhat smaller than the baseline estimates: 30.38% versus 39.39% for the indicator and 73.03% versus 80.72% for the team fraction. All eight undergraduate-only coefficients are significant at the 1% level, and the fund $\times$ time columns are nearly identical to their fund-fixed-effect counterparts (for example, 1.06 versus 1.09 percentage points for the raw-weight indicator). Thus, undergraduate ties robustly predict over-allocation, with larger effects in raw percentage-point terms but modestly smaller effects relative to benchmark weights (Table B3, Appendix B.9).

The stock-selection result is likewise robust and stronger. Figure B12 (Appendix B.9) repeats the baseline I-measure portfolio sort using undergraduate-only education countries. The education-country top-quartile portfolio earns a monthly five-factor α of 47.8 basis

points—compared with 41.0 in the full-education baseline (Section 4.3)—with the top–non-top differential significant at the 1% level, while the non-top and non-education portfolios remain indistinguishable from zero. The concentration of value in the most active education-country positions is thus not an artifact of pooling undergraduate and graduate ties; if anything, the undergraduate signal is sharper, consistent with undergraduate education proxying more directly for home-country familiarity.

5.5 Adding Momentum and Removing the Benchmark Control

Our baseline five-factor model contains no momentum factor, so a natural question is whether the education-country top-quartile α survives a momentum control. A second and distinct question concerns the benchmark: the baseline includes the fund’s assigned benchmark excess return on the right-hand side alongside the factors, so the reported α is measured relative to a benchmark-plus-factor span rather than a pure factor model. This subsection addresses both.

We re-estimate the baseline sort under two modifications to the return regression, introduced one at a time so that each can be attributed separately. The first adds the Fama-French momentum factor (WML) to each regional block,⁸ so that developed-market and emerging-market returns are each spanned by a six-factor model (market, size, value, profitability, investment, and momentum) rather than five, while retaining the benchmark excess return. The second additionally drops the benchmark excess return, leaving a pure factor-model α . The three specifications therefore form a ladder in which each step changes exactly one element:

⁸WML (“winners minus losers”) is the momentum factor, commonly denoted MOM in the Carhart four-factor model. Ken French labels it WML in his international factor data and Mom in his U.S. data; both share the identical winners-minus-losers construction, and we write WML throughout to match the source data. Momentum is taken from the same region as the other factors in each specification: the developed ex-U.S. and emerging WML series in the split model here, and the Global ex-U.S. WML in the un-split Global ex-U.S. four-factor model used in the footnote variant below and in the Jagannathan et al. (2022) comparison (Appendix D). These regional WML series are nearly perfectly correlated but not numerically identical, as they aggregate different underlying markets.

Specification	Right-hand side	Change
Baseline (Section 4.3)	$r^b - r^f + \text{FF5} \times 2 \text{ regions}$	—
+ momentum	$r^b - r^f + \text{FF5} + \text{WML} \times 2 \text{ regions}$	adds momentum
+ momentum, no benchmark	$\text{FF5} + \text{WML} \times 2 \text{ regions}$	then drops $r^b - r^f$

Portfolio construction is untouched throughout: stocks are ranked by the I-measure within fund-country-quarter cells, the top quartile forms the top portfolio, and portfolios are value-weighted and rebalanced quarterly exactly as in Section 4.3. All three specifications are estimated on the same portfolios; only the performance-attribution model changes.

Adding momentum alone reduces the education-country top-quartile α from 41.0 to 37.7 basis points per month (standard error 13.0). The top–non-top differential within education countries remains significant at the 1% level (p -value = 0.003), as does the difference between the education- and non-education-country top portfolios (p -value = 0.007); the other three portfolios stay indistinguishable from zero (4.4 and 0.7 basis points in non-education countries, p -value = 0.525, and -4.3 basis points for the education-country non-top portfolio). Dropping the benchmark term as well, as in Figure B13 (Appendix B.10), moves the estimate a further 1.4 basis points, to 36.3 (standard error 12.9), with the same pattern of significance (p -value = 0.005 for the within-education differential and 0.007 across country groups) and the remaining three portfolios again near zero (3.2 and -0.9 basis points, p -value = 0.499, and -3.9 basis points).

Of the 4.7-basis-point decline from the baseline, about 3.3 basis points are attributable to the momentum control and about 1.4 to the removal of the benchmark term. Both components are small relative to the premium itself, which retains close to 90% of its baseline magnitude under the most demanding of the three specifications. The large majority of the

effect, and every statistical conclusion drawn from it, survives both modifications.⁹

6 Discussion: Concentrated Value, Pervasive Bias

Taken together, the evidence points to an asymmetry between familiarity’s performance footprint and its allocation footprint. Managers appear to have a real advantage in familiar markets, but it pays off in only a narrow set of stocks; education ties are nevertheless associated with much broader portfolio tilts. We refer to this contrast as the allocation-to-performance wedge.

Across positions, value is concentrated while allocation is broad. Within education countries, the return premium is confined to the most active quartile of holdings. The top-quartile education-country portfolio earns an α of 41.0 basis points per month (5.03% annualized) with a Sharpe ratio of 0.44, while the remaining three quartiles earn alphas indistinguishable from zero (Section 4.3). The education-linked allocation tilt, however, does not stop at these positions: even after the top quartile is removed, funds continue to overweight education countries by economically meaningful amounts (Section 4.6, Table 15). Familiarity is therefore associated with economically large value in a concentrated subset of positions but with allocation changes across the broader education-country portfolio.

Across fund teams and over time, the advantage is heterogeneous and persistent. The top-quartile education-country portfolio is largely fund-specific: even in country-time cells with more than 50 unique high-I equities, 62.9% are held by only one fund team (Section 4.5.1, Figure 6). The premium therefore does not appear to reflect a single common signal extracted by all education-affiliated funds; its content varies across fund teams. The

⁹We also estimated a variant that replaces the split developed/emerging factor blocks with a single un-split Global ex-U.S. four-factor model (market, size, value, momentum). That specification yields a qualitatively identical pattern, with an education-country top-portfolio α of 38.9 basis points. We do not report it in the main text because Ken French’s Global ex-U.S. series—the three factors and momentum alike—have not been updated since June 2019, so this variant is estimated on 2000–2019 rather than on the full 2000–2024 sample used in the other checks. The truncation constrains nothing we report: the split developed- and emerging-market factors used throughout the main text run through 2024, and the Global ex-U.S. model enters elsewhere only in the Jagannathan et al. (2022) replication (Appendix D), whose sample period ends in 2014.

premium is also economically large in recent subsamples: the education-country top portfolio earns 21.9 basis points per month after the recession, 24.3 basis points per month over the most recent decade, and 26.0 basis points per month over the most recent five years (Section 4.3, Figure B11 in Appendix B.8). Although precision declines as the sample shortens, these estimates provide no evidence of a sharp recent decay.

These two dimensions reinforce a common conclusion: education-linked value is concentrated, heterogeneous across fund teams, and persistent in the available subsamples, whereas education-linked allocation is broad. The portion of the education-country tilt that generates α —the high-I subset—is narrow yet economically important. The remainder of the tilt does not outperform, even though the same fund teams display strong performance in their most active education-country positions. This contrast is difficult to reconcile with a country-wide informational advantage.

Interpretation of the concentrated premium. The return premium does not identify a unique mechanism. Education ties may proxy for knowledge of local institutions, language and cultural familiarity, professional or social networks, or persistent country-specific expertise. Each channel can generate an advantage that is security-specific and costly to acquire or deploy at scale, making it more likely to appear in selected high-conviction positions than across the country portfolio as a whole. Our evidence identifies this limited-scope feature: abnormal returns are concentrated in high-I positions, while the broader education-country portfolio does not outperform. We therefore interpret education ties as marking where country-specific information or skill is most valuable, rather than as establishing that education itself causes superior performance.

Against this background, our preferred interpretation of the allocation-to-performance wedge is that fund teams have a genuine but narrowly scoped advantage in familiar markets and that education-linked exposure extends beyond the positions in which that advantage pays. The return evidence alone cannot exclude every non-performance motive for the broader positions, including tracking considerations, mandate constraints, or client demand.

Nevertheless, the persistence of a substantial education-linked overweight outside the outperforming subset is consistent with a broad familiarity-based allocation bias coexisting with a concentrated information or skill channel. This coexistence is the paper’s central economic takeaway. It reconciles the long-standing tension between familiarity as information and familiarity as bias and shows why country-level averages can blend two distinct portfolio margins.

7 Conclusion

This paper combines detailed international mutual fund holdings with manager-level data on educational background to examine how education-based familiarity is associated with portfolio allocation and performance. Funds systematically overweight countries where their managers received post-secondary education, and an event study around manager turnover shows this tilt building after an education-linked manager arrives and receding after the last one departs—evidence that the allocation follows the managers rather than a fixed fund mandate. Yet these broad country tilts do not generate higher country-level returns. Familiarity’s allocation footprint is therefore wider than its performance footprint.

We then turn to the stock level. To capture the intensity of fund teams’ active positions, we develop the I-measure, which isolates within-country deviations from a passive proportional-tracking allocation over the same held-stock reference set. Sorting on the I-measure reveals that the top quartile of education-country positions generates a monthly α of 41.0 basis points (5.03% annualized) with a Sharpe ratio of 0.44, while all other positions earn returns indistinguishable from the benchmark. The premium survives estimated bid-ask spread costs and is not driven by the United Kingdom or by the choice of benchmark. These positions are also largely fund-specific: most high-I equities in a given country-quarter are held by only one fund team, so the premium does not reflect a single signal shared across education-linked funds.

Together, these results clarify a dual role of familiarity in international equity markets. Familiarity is associated with a concentrated performance advantage and a much broader allocation tilt. This reconciles seemingly conflicting views of home-linked investing: familiarity does carry information or skill, but its measured value is concentrated in a narrow set of high-I positions rather than realized across the education-country portfolio, which is why the country-level average shows no premium. Education-linked overweighting persists even after these outperforming positions are removed, consistent with familiarity-based allocation bias extending beyond the scope of the advantage. Our evidence locates where that advantage appears rather than identifying its source: education ties may proxy for knowledge of local institutions, language, professional networks, or country-specific expertise, and separating these channels is left to future work. These findings contribute to the literatures on familiarity bias, fund-manager skill, and international portfolio choice.

The findings have practical implications for evaluating active international funds. Portfolio-level statistics may understate education-linked stock-selection skill because the associated performance is concentrated in a small subset of positions, while broad familiar-country exposure may overstate the scope of that advantage. More generally, investors and fund boards should evaluate separately the positions in which familiarity is associated with performance and the broader allocations that familiarity appears to influence.

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Tables

Table 2: Summary Statistics of Fund Characteristics

	Mean	Median	Std	N
<i>Panel A: Portfolio and manager-education characteristics (fund-quarter)</i>				
Number of countries held	18.99	19.00	8.44	67,380
Number of manager education countries	1.58	1.00	0.94	63,364
Number of manager education countries (non-U.S.)	0.72	0.00	0.94	63,364
<i>Panel B: Fund characteristics (fund-year)</i>				
Fund Total Net Asset (billions \$)	1.59	0.23	6.66	21,084
Annual Net Expense Ratio (%)	1.22	1.17	0.50	19,695
Turnover Ratio (%)	64.31	48.00	60.96	18,999
Fund Return, annual (%)	8.68	10.86	22.59	18,773
Fund Volatility, annualized (%)	16.46	14.81	7.09	18,773
<i>Panel C: Manager characteristics (one observation per fund)</i>				
Average Manager Tenure (yrs)	5.57	4.93	3.27	1,663
No. of Managers (team size)	3.12	2.44	2.44	1,663

Notes: This table reports summary statistics for the sample funds over 2000–2024, using Morningstar data. A fund is a Morningstar FundID and a share class an ISIN; Panel B aggregates a fund’s share classes, summing net assets and net-asset-weighting the remaining characteristics. Panel A is measured at the fund-quarter level from quarter-end holdings, with the education counts covering fund-quarters in which the management team has at least one education link. Panel B is measured at the fund-year level, as expense and turnover ratios are reported annually; turnover is portfolio turnover, annual returns compound the year’s twelve monthly gross returns, and volatility scales their standard deviation by $\sqrt{12}$, both for complete years only. Panel C is measured once per fund from the manager panel: tenure averages the managers’ spells with the fund, and team size the monthly number of managers. All averages are equal-weighted across observations, and the reported N differs across rows with the unit of observation.

Table 3: Sample Fund TNA Breakdown by Benchmarks

Morningstar Index	Mean TNA (B)	Ave. Num Funds
Morningstar Gbl NR USD	4.672	51.55
Morningstar Gbl xUS Val TME NR USD	2.853	86.63
Morningstar Gbl Growth TME NR USD	2.790	51.36
Morningstar Gbl xUS Growth TME NR USD	1.787	86.45
Morningstar Gbl xUS NR USD	1.260	164.27
Morningstar EM Americas NR USD	1.091	2.57
Morningstar DM APAC xJpn NR USD	0.800	14.87
Morningstar EM NR USD	0.757	125.59
Morningstar Gbl Val TME NR USD	0.606	38.64
Morningstar Gbl xUS SMID NR USD	0.550	56.65
Morningstar DM Eur NR USD	0.528	10.16
Morningstar Gbl SMID NR USD	0.460	18.04
Morningstar Jpn NR USD	0.367	6.99
Morningstar DM APAC NR USD	0.257	3.15
Morningstar China NR USD	0.180	13.05
Morningstar India NR USD	0.121	2.34
Morningstar Gbl Lrg NR USD	0.093	28.97

Notes: This table reports the mean fund-level total net assets and the mean number of funds in each Morningstar benchmark category, computed at quarterly frequency over the 1,663-fund sample. Because the counts are per-quarter averages, the “Ave. Num Funds” column reflects funds active in a given quarter and sums to about 760—the average number of funds active per quarter—rather than the 1,663 distinct funds observed over the full sample, since funds enter and exit over time.

Table 4: Country Distribution of Manager Educations

Notes: The last column reports the share of managers educated in each country, computed as the number of managers educated in that country divided by the total number of managers (2,976). The shares sum to more than one because a manager can have education ties to multiple countries.

Country	# Mgrs	Share	Country	# Mgrs	Share
USA	2237	0.7517	BEL	5	0.0017
GBR	577	0.1939	CHL	4	0.0013
CAN	82	0.0276	PAK	4	0.0013
IND	68	0.0228	POL	4	0.0013
CHN	63	0.0212	ARG	3	0.0010
AUS	44	0.0148	NGA	3	0.0010
IRL	43	0.0144	NOR	3	0.0010
DEU	35	0.0118	TUR	3	0.0010
JPN	35	0.0118	CZE	2	0.0007
HKG	19	0.0064	EGY	2	0.0007
FRA	17	0.0057	ESP	2	0.0007
ZAF	16	0.0054	GRC	2	0.0007
SGP	13	0.0044	VEN	2	0.0007
KOR	12	0.0040	BGR	1	0.0003
ITA	11	0.0037	COL	1	0.0003
DNK	10	0.0034	CUB	1	0.0003
NLD	10	0.0034	EST	1	0.0003
CHE	9	0.0030	GTM	1	0.0003
SWE	8	0.0027	HUN	1	0.0003
ISR	7	0.0024	IDN	1	0.0003
TWN	7	0.0024	LBN	1	0.0003
BRA	6	0.0020	PRT	1	0.0003
MEX	6	0.0020	ROU	1	0.0003
NZL	6	0.0020	SAU	1	0.0003
RUS	6	0.0020	THA	1	0.0003
AUT	6	0.0020	UKR	1	0.0003
Total # of Managers = 2,976					

Table 5: Country Distribution of Manager Education at Fund Level

Notes: The last column reports the share of funds with at least one manager educated in each country, computed as the number of such funds divided by the total number of funds (1,663). The shares sum to more than one because a fund can have education ties to multiple countries through its managers.

Country	# Funds	Share	Country	# Funds	Share
USA	1525	0.917	MEX	12	0.007
GBR	657	0.395	POL	12	0.007
CAN	198	0.119	BRA	11	0.007
IND	186	0.112	CZE	8	0.005
CHN	137	0.082	NGA	8	0.005
IRL	133	0.080	CUB	7	0.004
AUS	86	0.052	EGY	7	0.004
JPN	78	0.047	ARG	6	0.004
DEU	52	0.031	CHL	6	0.004
ITA	48	0.029	GRC	6	0.004
HKG	47	0.028	NOR	6	0.004
SGP	43	0.026	PAK	6	0.004
FRA	39	0.023	ROU	6	0.004
ZAF	37	0.022	TUR	6	0.004
DNK	34	0.020	VEN	5	0.003
NLD	33	0.020	IDN	3	0.002
KOR	33	0.020	GTM	2	0.001
ISR	32	0.019	LBN	2	0.001
NZL	28	0.017	THA	2	0.001
CHE	24	0.014	UKR	2	0.001
RUS	19	0.011	BGR	1	0.001
SWE	19	0.011	COL	1	0.001
TWN	17	0.010	EST	1	0.001
ESP	15	0.009	HUN	1	0.001
AUT	15	0.009	PRT	1	0.001
BEL	13	0.008	SAU	1	0.001
Total # of Funds = 1,663					

Table 6: Portfolio Holdings Patterns

	Time-series Average			
	Mean	Median	Std	N
Sample Funds				
# of countries held	18.99	19	8.44	67,380
Country-level portfolio share(%)	4.83	2.17	9.47	1,279,345
# of stocks held	210.26	81	589.15	71,742
Stock-level portfolio share(%)	0.45	0.09	0.88	15,084,718
Morningstar Gbl xUS NR USD				
# of countries held	49.47	51	3.46	88
Country-level portfolio share(%)	1.97	0.58	3.35	4,353
# of stocks held	5,136.20	5,104	982.78	88
Stock-level portfolio share(%)	0.02	0.00	0.06	451,986
All Benchmarks (equal weights)				
# of countries held	27.55	26	20.28	1,947
Country-level portfolio share(%)	3.43	0.57	10.15	53,639
# of stocks held	1,887.58	1,344	1,841.74	1,947
Stock-level portfolio share(%)	0.05	0.01	0.23	3,675,116

Notes: This table reports country-level and stock-level holding counts and portfolio shares for the sample funds, the Morningstar Global ex-US benchmark, and the average across all Morningstar benchmarks assigned to sample funds. The sample-fund rows are computed on the same 1,663-fund sample and 2000–2024 window as Table 2. The “# held” rows are computed at the fund-quarter (or benchmark-quarter) level; the “portfolio share” rows are averaged across individual held positions (fund-country or fund-stock), so that the mean number held times the mean share equals the average total portfolio weight ($\approx 100\%$, below 100 because funds also hold some U.S. equity and cash). Positions with zero weight are excluded.

Table 7: Excess Weight over Benchmark (%) – Robustness

VARIABLES	(1) Excess ω	(2) Excess ω	(3) Excess ω	(4) Excess ω	(5) Excess ω	(6) Excess ω
1(EC)	0.974*** (0.219)	0.975*** (0.214)	1.051*** (0.208)	1.072*** (0.206)	0.955*** (0.218)	1.035*** (0.204)
Constant	-0.0462*** (0.00339)	-0.0462*** (0.0120)	-0.0146*** (0.00302)	-0.0149 (0.0117)	-0.0474*** (0.00335)	-0.0153*** (0.00296)
Observations	5,613,030	5,613,030	4,704,292	4,704,292	5,611,449	4,704,206
R-squared	0.145	0.129	0.188	0.177	0.191	0.237
Fund FE	Y	N	Y	N	–	–
Fund×Time FE	N	N	N	N	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y
Sub-Sample	N	N	Y	Y	N	Y
Funds Number	1663	1663	1185	1185	1663	1185

Notes: This table reports robustness checks for the raw excess-weight specification in Table 1. The dependent variable is the fund’s portfolio weight in a country minus the benchmark country weight (in percentage points). Columns (1)–(2) use the full sample; columns (3)–(4) restrict to the sub-sample of global funds—those assigned to a global Morningstar benchmark rather than a regional or single-country benchmark. Columns (1) and (3) include both fund and country×time fixed effects; columns (2) and (4) drop the fund fixed effect to gauge how much of the effect is identified within fund. Columns (5)–(6) instead replace the fund fixed effect with a fund×time fixed effect—absorbing all time-varying fund-level heterogeneity—on the full sample and the global sub-sample, respectively. Robust standard errors clustered at the fund level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Scaled Excess Weight over Benchmark (%) – Robustness

VARIABLES	(1) Scaled Excess ω	(2) Scaled Excess ω	(3) Scaled Excess ω	(4) Scaled Excess ω	(5) Scaled Excess ω	(6) Scaled Excess ω
1(EC)	39.39*** (5.857)	43.82*** (5.810)	28.25*** (5.177)	30.15*** (5.019)	40.22*** (5.917)	28.63*** (5.227)
Constant	-19.42*** (0.0842)	-19.48*** (1.015)	-23.31*** (0.0730)	-23.34*** (0.971)	-19.43*** (0.0844)	-23.32*** (0.0737)
Observations	5,244,238	5,244,238	4,508,382	4,508,382	5,242,603	4,508,382
R-squared	0.119	0.086	0.118	0.095	0.139	0.136
Fund FE	Y	N	Y	N	–	–
Fund×Time FE	N	N	N	N	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y
Sub-Sample	N	N	Y	Y	N	Y
Funds Number	1663	1663	1185	1185	1663	1185

Notes: This table reports robustness checks for the benchmark-scaled excess-weight specification in Table 1. The dependent variable is $(country_weight - benchmark_country_weight)/benchmark_country_weight$ multiplied by 100. Columns (1)–(2) use the full sample; columns (3)–(4) restrict to the sub-sample of global funds—those assigned to a global Morningstar benchmark rather than a regional or single-country benchmark. Columns (1) and (3) include both fund and country×time fixed effects; columns (2) and (4) drop the fund fixed effect. Columns (5)–(6) instead replace the fund fixed effect with a fund×time fixed effect—absorbing all time-varying fund-level heterogeneity—on the full sample and the global sub-sample, respectively. Robust standard errors clustered at the fund level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9: Excess Weight over Benchmark (%)

VARIABLES	(1) Excess ω	(2) Excess ω	(3) Excess ω	(4) Excess ω	(5) Excess ω	(6) Excess ω
Frac _{EC}	2.465*** (0.435)	2.429*** (0.422)	2.737*** (0.433)	2.774*** (0.431)	2.417*** (0.427)	2.662*** (0.420)
Constant	-0.0488*** (0.00312)	-0.0485*** (0.0119)	-0.0170*** (0.00279)	-0.0172 (0.0117)	-0.0498*** (0.00303)	-0.0175*** (0.00271)
Observations	5,613,030	5,613,030	4,704,292	4,704,292	5,611,449	4,704,206
R-squared	0.146	0.130	0.190	0.179	0.192	0.238
Fund FE	Y	N	Y	N	—	—
Fund×Time FE	N	N	N	N	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y
Sub-Sample	N	N	Y	Y	N	Y
Funds Number	1663	1663	1185	1185	1663	1185

Notes: This table reports robustness checks for the raw excess-weight specification using the continuous fraction-of-education-country exposure (Frac_{EC}) in place of the binary indicator. The dependent variable is the country-level excess portfolio weight (in percent) of fund i in country c at quarter t , defined as the fund's reported weight minus the benchmark weight. Frac_{EC} is the share of the fund management team whose education country corresponds to country c . Columns (3)–(4) restrict to the sub-sample of global funds—those assigned to a global Morningstar benchmark rather than a regional or single-country benchmark. All specifications include country×time fixed effects; columns (1) and (3) further include fund fixed effects, and columns (5)–(6) replace the fund fixed effect with a fund×time fixed effect—absorbing all time-varying fund-level heterogeneity—on the full sample and the global sub-sample, respectively. Standard errors clustered at the fund level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10: Scaled Excess Weight over Benchmark (%)

VARIABLES	(1) Scaled Excess ω	(2) Scaled Excess ω	(3) Scaled Excess ω	(4) Scaled Excess ω	(5) Scaled Excess ω	(6) Scaled Excess ω
Frac _{EC}	80.72*** (11.29)	83.01*** (11.32)	62.54*** (9.245)	61.48*** (9.239)	81.61*** (11.41)	62.41*** (9.277)
Constant	-19.38*** (0.0737)	-19.40*** (1.013)	-23.31*** (0.0578)	-23.30*** (0.969)	-19.38*** (0.0735)	-23.30*** (0.0580)
Observations	5,244,238	5,244,238	4,508,382	4,508,382	5,242,603	4,508,382
R-squared	0.120	0.086	0.118	0.095	0.139	0.136
Fund FE	Y	N	Y	N	—	—
Fund×Time FE	N	N	N	N	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y
Sub-Sample	N	N	Y	Y	N	Y
Funds Number	1663	1663	1185	1185	1663	1185

Notes: This table reports robustness checks for the benchmark-scaled excess-weight specification using the continuous fraction-of-education-country exposure (Frac_{EC}). The dependent variable is the scaled excess weight, defined as (fund weight – benchmark weight)/benchmark weight, expressed in percent. Frac_{EC} is the share of the fund management team whose education country corresponds to country c . Columns (3)–(4) restrict to the sub-sample of global funds—those assigned to a global Morningstar benchmark rather than a regional or single-country benchmark. All specifications include country×time fixed effects; columns (1) and (3) further include fund fixed effects, and columns (5)–(6) replace the fund fixed effect with a fund×time fixed effect—absorbing all time-varying fund-level heterogeneity—on the full sample and the global sub-sample, respectively. Standard errors clustered at the fund level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Education-Country Return Predictability at One- and Three-Month Horizons (bps)

VARIABLES	(1) Return	(2) Return	(3) Return	(4) Return
<i>Panel A: One-month forward return</i>				
1(EC)	2.839 (2.694)	0.714 (2.385)		
Frac _{EC}			-1.624 (3.983)	-2.746 (3.651)
Constant	89.40*** (0.0744)	89.44*** (0.0659)	89.50*** (0.0513)	89.50*** (0.0470)
Observations	2,227,181	2,227,030	2,227,181	2,227,030
R-squared	0.139	0.236	0.139	0.236
<i>Panel B: Cumulative three-month forward return</i>				
1(EC)	12.81* (6.810)	8.969 (6.567)		
Frac _{EC}			-0.119 (11.21)	-1.344 (10.98)
Constant	220.2*** (0.203)	220.3*** (0.196)	220.6*** (0.153)	220.6*** (0.150)
Observations	1,086,990	1,086,955	1,086,990	1,086,955
R-squared	0.363	0.564	0.363	0.564
Time FE	Y		Y	
Country FE	Y		Y	
Fund FE	Y	Y	Y	Y
Country×Time FE		Y		Y

Notes: This table reports panel regressions of country-level forward returns, in basis points, on manager-country education ties. Panel A uses the value-weighted return over the next month. Panel B uses the cumulative return compounded over the next three months, $\prod_{h=1}^3 (1 + r_{t+h}) - 1$, at quarter-end formation dates so the horizons do not overlap. 1(EC) indicates that at least one current manager was educated in the country; Frac_{EC} is the corresponding management-team share. Columns (1) and (3) include separate time and country fixed effects; columns (2) and (4) include country×time fixed effects. All specifications include fund fixed effects and cluster standard errors by fund. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Portfolio	RF-Adj. Mean	Mean Ret.	RF-Adj. SD	Sharpe	IR
Glb. ex-US Benchmark	0.358	0.507	4.879	0.253	—
Edu Top	0.643	0.792	5.085	0.438	0.546
Edu Non-Top	0.396	0.545	5.073	0.271	0.098
Non-Edu Top	0.441	0.590	4.986	0.307	0.287
Non-Edu Non-Top	0.373	0.522	4.979	0.260	0.055

Table 12: Portfolio performance, 2000–2024 (monthly frequency)

Notes: Columns report (i) risk-free-adjusted mean monthly return (in %/month), (ii) unadjusted mean monthly return (in %/month), (iii) risk-free-adjusted return standard deviation at monthly frequency (in %/month), (iv) the annualized Sharpe ratio (monthly $\text{mean}(R_p - R_f)/\text{sd}(R_p - R_f)$ multiplied by $\sqrt{12}$), and (v) the annualized information ratio (monthly $\text{mean}(R_p - R_b)/\text{sd}(R_p - R_b)$ multiplied by $\sqrt{12}$, with R_b the Glb. ex-US benchmark return). The information ratio is undefined for the baseline benchmark.

Table 13: Portfolio Returns, 2000–2024 (Yearly, in %)

Portfolio	# Years	Mean Return	Std. Dev.	Min Return	Max Return
Glb. ex-US Benchmark	25	6.828	20.32	-45.37	42.82
Edu Top	25	11.18	21.11	-37.41	55.23
Edu Non-Top	25	7.534	21.21	-43.97	51.11
Non-Edu Top	25	8.034	21.34	-45.78	45.92
Non-Edu Non-Top	25	7.061	21.26	-45.73	43.13
US Benchmark	25	9.615	18.81	-36.99	35.25
Glb. ex-US Benchmark (RF Adj.)	25	4.983	20.25	-46.06	42.62
Edu Top (RF Adj.)	25	9.219	20.80	-38.20	53.71
Edu Non-Top (RF Adj.)	25	5.671	21.15	-44.67	49.63
Non-Edu Top (RF Adj.)	25	6.173	21.28	-46.47	45.71
Non-Edu Non-Top (RF Adj.)	25	5.214	21.19	-46.42	42.92
US Benchmark (RF Adj.)	25	7.752	18.94	-37.78	35.18

8

Notes: This table reports yearly summary statistics for four sub-portfolios formed by interacting the manager education-country indicator (Edu vs. Non-Edu) with the within-country I-measure quartile (Top vs. Non-Top), alongside the Global ex-US and US benchmark portfolios. The Edu Top portfolio holds top-quartile (highest I-measure) stocks in countries where at least one fund manager was educated; analogous definitions apply to the other three Edu/Non-Edu Top/Non-Top portfolios. Returns are in percent and computed at annual frequency over 2000–2024 by compounding monthly returns within each year. “RF Adj.” rows subtract the monthly risk-free rate—the 3-month U.S. Treasury rate used throughout the paper, de-annualized to a monthly frequency—from each monthly return before compounding, so the adjusted annual figure is $\prod_m (1 + R_m - r_m) - 1$ rather than the annual return less an annual rate. Because the adjustment is applied inside the compounding, the gap between a raw row and its adjusted counterpart varies slightly across portfolios.

VARIABLES	(1) Excess Weight	(2) Excess Weight	(3) Excess Weight	(4) Excess Weight
$\mathbb{1}(\text{EC})$	0.193*** (0.0616)	0.192*** (0.0615)	0.110** (0.0531)	0.107** (0.0530)
$\mathbb{1}(\text{Top})$	0.100*** (0.0165)	0.101*** (0.0164)	0.128*** (0.0176)	0.130*** (0.0175)
$\mathbb{1}(\text{EC}) \times \mathbb{1}(\text{Top})$	0.300*** (0.0700)	0.300*** (0.0703)	0.316*** (0.0685)	0.314*** (0.0687)
Number of Equities Held			-0.000350*** (0.000109)	-0.000373*** (0.000120)
Mkt Cap (In Billion USD)			0.00306*** (0.000515)	0.00302*** (0.000515)
Constant	0.252*** (0.00400)	0.252*** (0.00398)	0.287*** (0.0310)	0.293*** (0.0337)
Observations	31,148,436	31,147,961	31,148,436	31,147,961
R-squared	0.049	0.056	0.105	0.112
Time FE	Y		Y	
Country FE	Y		Y	
Country Cluster	Y	Y	Y	Y
Country $\times$ Time FE		Y		Y

Table 14: Excess Weight Comparison

Notes: This table reports stock-level regressions of excess portfolio weight on the manager education-country indicator interacted with a top-pick dummy. The unit of observation is a fund-stock-month. The dependent variable is the excess weight (in percent) of stock i in country c within fund j at month t , $\omega_j^i(t) - B_j^i(t)$. $\mathbb{1}(\text{EC})$ equals one if any current manager of fund j was educated in country c ; $\mathbb{1}(\text{Top})$ equals one for stocks in the top quartile of fund j 's I-measure within country c at month t . Columns (3)–(4) add controls for the number of equities held by the fund within the country and the stock's market capitalization (in billion USD). Columns (1) and (3) include separate time and country fixed effects; columns (2) and (4) absorb country $\times$ time fixed effects. Standard errors clustered at the country level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

VARIABLES	(1) Excess NT ω	(2) Excess NT ω	(3) Scaled Excess NT ω	(4) Scaled Excess NT ω	(5) Excess NT ω	(6) Excess NT ω	(7) Scaled Excess NT ω	(8) Scaled Excess NT ω
1 (EC)	0.577*** (0.182)		32.06*** (5.272)		0.593*** (0.183)		33.05*** (5.312)	
Frac _{EC}		1.658*** (0.353)		67.90*** (10.05)		1.697*** (0.348)		68.65*** (10.15)
Constant	-0.545*** (0.00283)	-0.548*** (0.00254)	-31.58*** (0.0758)	-31.56*** (0.0656)	-0.537*** (0.00281)	-0.540*** (0.00247)	-31.59*** (0.0758)	-31.56*** (0.0653)
Observations	5,613,030	5,613,030	5,244,238	5,244,238	5,611,449	5,611,449	5,242,603	5,242,603
R-squared	0.253	0.253	0.108	0.108	0.280	0.281	0.128	0.128
Fund FE	Y	Y	Y	Y	—	—	—	—
Fund×Time FE	N	N	N	N	Y	Y	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y	Y	Y
Sub-Sample	N	N	N	N	N	N	N	N
Funds Number	1663	1663	1663	1663	1663	1663	1663	1663

Table 15: Education Links and Excess Country Weight Outside the Top Quartile

Notes: This table re-estimates the over-allocation regressions of Table 1 using country weights reconstructed from non-top holdings only. For each fund-country-quarter, equities are ranked by the I-measure within the cell and the top quartile is excluded; the remaining bottom three quartiles' weights are summed to obtain the non-top country weight $\omega_{jc}^{NT}(t)$. Columns (1)–(2) and (5)–(6) use the resulting raw excess weight $\omega_{jc}^{NT}(t) - B_{jc}(t)$ (in percentage points), and columns (3)–(4) and (7)–(8) scale the same difference by the benchmark country weight. In both cases $B_{jc}(t)$ is the fund's full benchmark country weight, which is not restricted to the bottom three quartiles: the I-measure is defined only over the fund's own holdings, so the benchmark has no corresponding quartile to drop. Coefficients here are therefore mechanically smaller than in Table 1 and should be compared to it after rescaling by the share of country weight the bottom three quartiles represent (about 73.6%). Columns (1)–(4) include fund and country×time fixed effects; columns (5)–(8) repeat the same specifications but replace the fund fixed effect with a fund×time fixed effect, absorbing all time-varying fund-level heterogeneity. Robust standard errors clustered at the fund level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figures

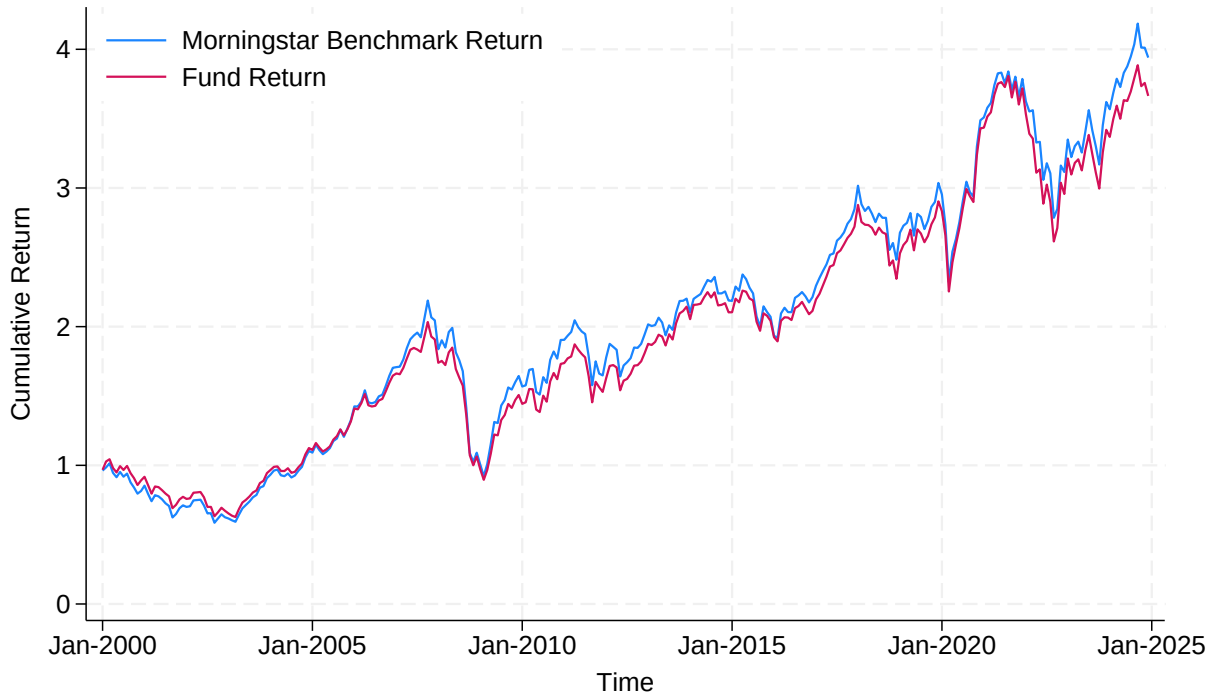


Figure 4: Weighted Benchmark Return VS Fund Sample Return

Notes: This figure plots the monthly cumulative total return of sample funds and the weighted total return of their corresponding benchmarks. The two lines align well with each other, with correlation of cumulative index 99.81%, and correlation of monthly return 98.84%.

Alt text: Line chart of cumulative monthly total returns over 2000–2024 for the sample funds and their assignment-weighted Morningstar benchmarks, both normalized to one at the start; the two lines rise to roughly four times their initial value and track each other closely, with the fund line ending slightly below the benchmark line.

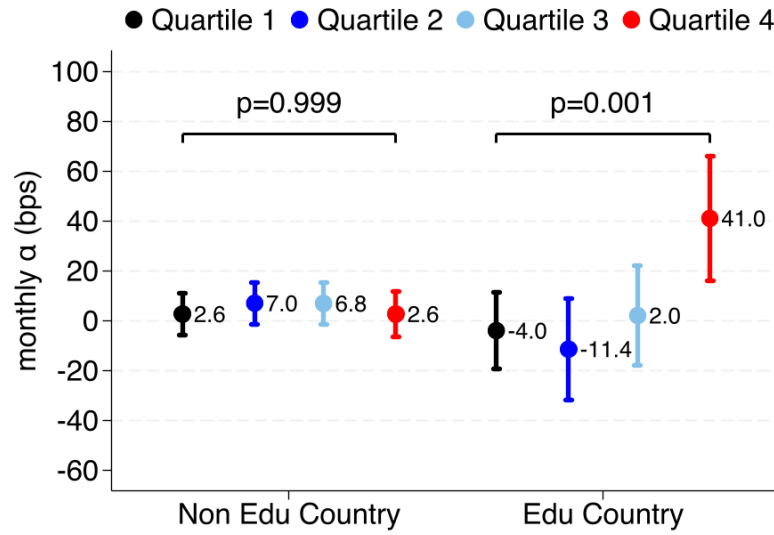


Figure 5: Portfolio α Across Four I-Measure Quartiles

Notes: This figure provides portfolio α for all 4 quartiles using the Fama-French five-factor model in education and non-education countries separately. The bars provide 95% confidence intervals. The p -values are for the joint test of the difference between the top quartile (quartile 4) and bottom quartile (quartile 1).

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals for the four I-measure quartiles, shown separately for non-education and education countries. Alphas are near zero in all quartiles except quartile four in education countries, which is about 41 basis points and significantly above that group's bottom quartile.

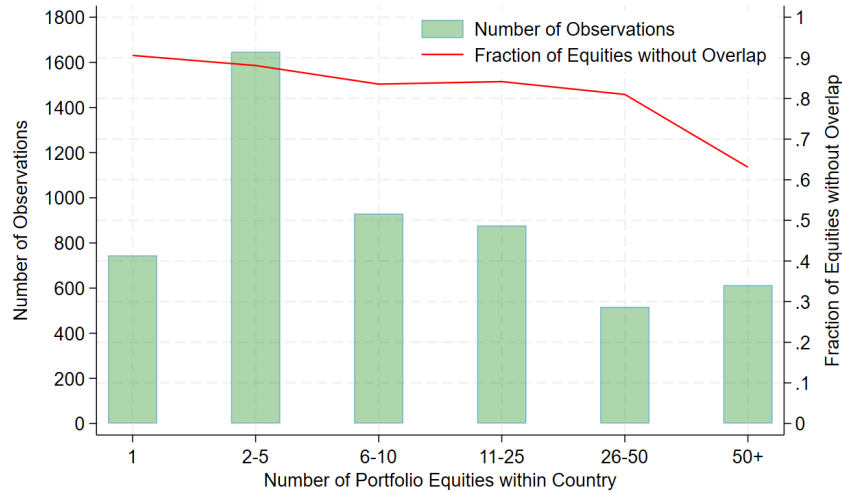


Figure 6: Top Pick Overlap Pattern

Notes: This figure shows the share of education-country high-I equities held by only one fund team (i.e., not shared with another fund team's high-I positions) within each country-quarter cell. The horizontal axis groups country-quarter observations by the number of unique equities in the education-country top portfolio, which varies with the number of education-affiliated funds represented in that country. The vertical axis reports the average non-overlap rate within each bin. Higher non-overlap rates indicate that high-I positions are more fund-specific rather than convergent on common targets.

Alt text: Combination chart in which green bars show the number of country-quarter cells in each unique-equity-count group (1, 2-5, 6-10, 11-25, 26-50, and more than 50) and a red line shows the average share of equities held by only one fund team; the share stays above 80% in most groups and is 62.9% for cells with more than 50 equities.

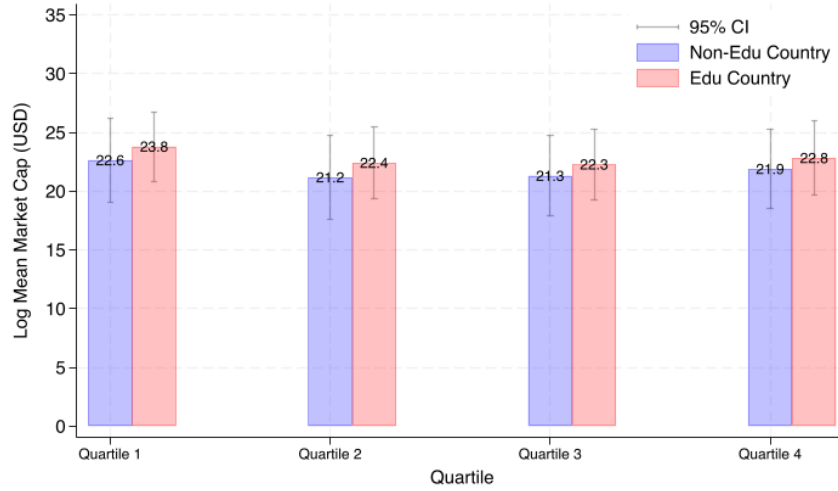


Figure 7: Market Cap Comparison

Notes: This figure plots the average log market capitalization of equities held by sample funds across the four I-measure quartiles, separately for education and non-education countries. Quartile 1 contains the least active (lowest I-measure) holdings and Quartile 4 the most active (top picks). Capped bars denote 95% confidence intervals for the within-bin mean. The figure addresses whether differences in stock-picking behavior between education and non-education countries are mechanically driven by differences in the size of firms targeted.

Alt text: Dot plot of average log market capitalization with 95% confidence intervals across the four I-measure quartiles for education and non-education countries; averages lie between about 21 and 23 in every quartile, education-country holdings are larger throughout by a roughly constant amount, and the intervals for the two groups overlap in every quartile.

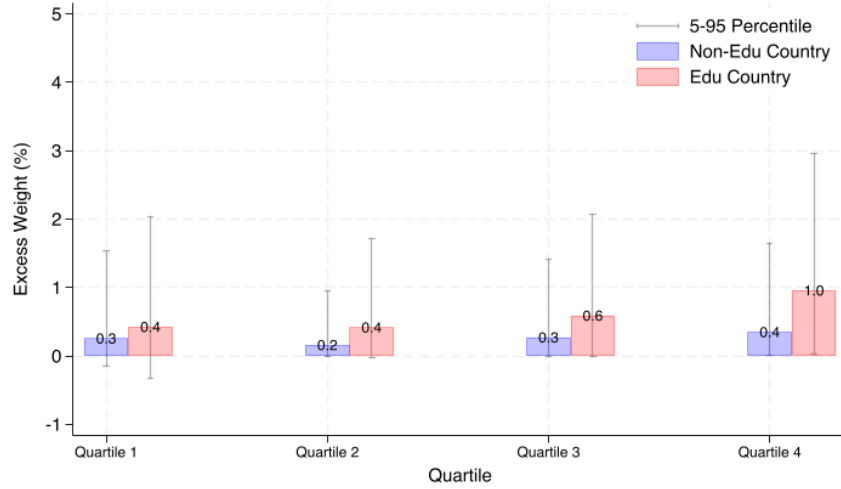


Figure 8: Excess Weight Comparison

Notes: This figure plots the average raw excess portfolio weight, $\omega_j^i(t) - B_j^i(t)$, across the four I-measure quartiles, separately for education and non-education countries. Excess weight measures how much a fund deviates from the benchmark in a given equity. Quartile 4 corresponds to top picks. Capped bars show the 5th–95th percentile range across funds within each bin, illustrating the dispersion in position sizing.

Alt text: Chart of average raw excess portfolio weight by I-measure quartile for education and non-education countries, with whiskers marking the 5th–95th percentile range; averages are similar across groups in the bottom three quartiles, while in the top quartile education-country positions average about 1.0% versus 0.4% for non-education countries.

Appendix A. Equity Return Construction

Appendix A.1. Return Index Construction

As mentioned in the description of the data, we use the Datastream return index to calculate the monthly gross return on each equity. The return index includes both the price return and the dividend return. However, due to the data limitation on international equity, around 20% of the equities in the sample have a missing return index, but almost no equities have the price data missing. Thus, we back out the gross return as the sum of the price return and the country’s average dividend of the month. We use the change in the market cap of the stock to discipline the price return. This is because price changes might be caused by stock splits and reverse splits. We use only the price returns that are within 0.1 percentage point of the percentage change in market cap in the same month. Concretely, for those equity-month observations with a missing return index, we define the monthly gross return as $price_return_{i,t} + dividend_{c,t}$ if $|price_return_{i,t} - \% \Delta market_cap| < 0.1\%$, where $dividend_{c,t}$ is the country average dividend in country c to which equity i belongs in month t . This conservative rule fills a missing return only when the price move is corroborated by the change in market capitalization—that is, when no stock split or share issuance drives a wedge between the two—and leaves the month missing otherwise; it applies only to the roughly 20% of equity-months for which the return index is unavailable.

VARIABLES	(1) Constructed Return	(2) Constructed Return
Reported Return	1.025*** (0.00472)	0.982*** (0.0102)
Constant	0.000180 (0.000230)	-0.000146 (0.000497)
Observations	63,108	300
R-squared	0.974	0.969
Correlation	0.9871	0.9844
Sample	All Sample Funds	Global xUS Benchmark

Table A1: Comparison of Reported Return and Constructed Return

Notes: This table reports OLS regressions of the constructed monthly gross return on the reported monthly gross return at the fund-month level. The constructed return is computed as the weighted average of equity-level monthly returns using lagged portfolio weights from holdings data, with weights re-normalized to sum to one within the equity universe. Column (1) uses all sample funds and column (2) uses the Morningstar Global ex-US (Gbl xUS USD) benchmark. The reported R-squared and the correlation between constructed and reported returns are both close to one, indicating that the constructed return closely tracks the reported return. Robust standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix A.2. Testing the Reliability of the Return-Index Construction and Reported Holdings

As a joint test of the reliability of the detailed holdings data and our constructed equity-level returns, we compute each fund’s return implied by its disclosed holdings and constructed equity returns, and compare this holdings-based return with the fund’s reported return. This comparison is the analogue, in our setting, of the *return gap* of Kacperczyk et al. (2008): the difference between a fund’s reported return and the return on a portfolio of its previously disclosed holdings, which captures interim trades, trading costs, and other actions that holdings snapshots do not reveal. They find the average return gap close to zero but with substantial and persistent cross-sectional variation, which is why a holdings-based analysis such as ours warrants this check. We perform this exercise both for all funds in our sample and for the Morningstar Global ex-US benchmark (Morningstar Gbl xUS USD).

We construct the fund return as the weighted return of individual equities,

$$R_{f,t} = \frac{\sum_{i \in \Omega_{f,t-1}} \omega_{f,i,t-1} r_{i,t}}{\sum_{i \in \Omega_{f,t-1}} \omega_{f,i,t-1}},$$

where $\Omega_{f,t-1}$ is the set of equities held by fund f at the end of month $t - 1$ and $\omega_{f,i,t-1}$ is the weight of equity i in that fund’s disclosed portfolio. Note that this captures only the return on equity investments. The benchmark holds nothing other than equities, but funds may hold a small fraction of liquid fixed income or cash, which the reported return reflects and our constructed return does not.

We run a simple OLS regression of the constructed return on the reported return with observations at the month-fund level, and the results are reported in Table A1. The R-squared is 0.97 for the sample of all funds, and the correlation of the two series is 0.99. The regression coefficient is 1.02, which is 2.5% larger than 1. This might come from the fact that the funds in our sample hold on average 2% – 3% cash or very liquid assets as buffers

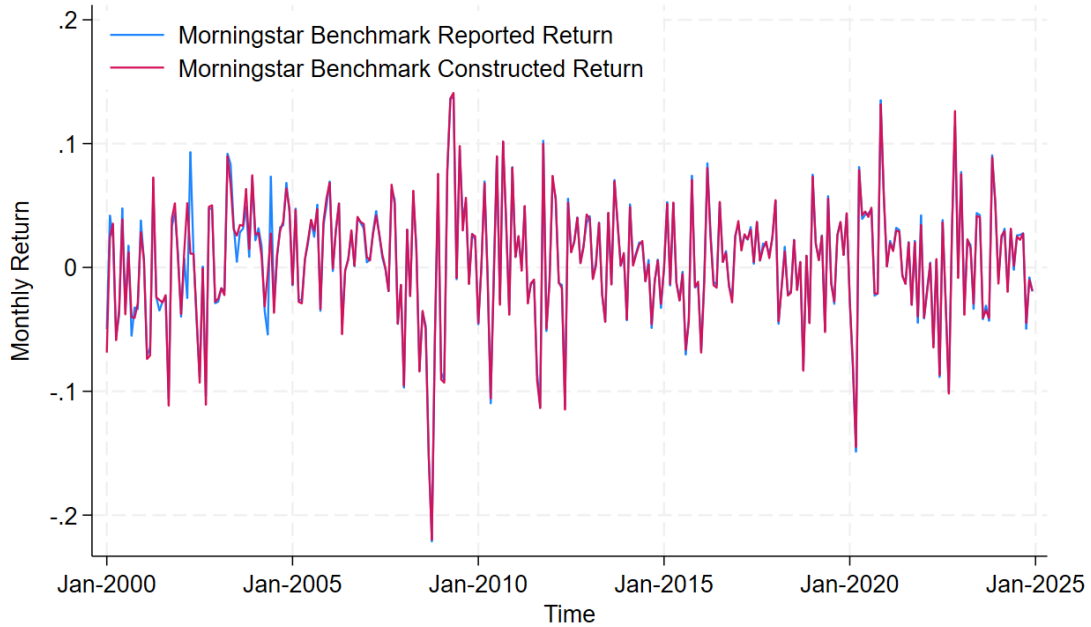


Figure A1: Reported Benchmark Return VS Constructed Benchmark Return

Notes: This figure plots the monthly gross return reported by Morningstar for benchmark Morningstar Gbl xUS USD against the constructed monthly gross return for the same index from holding and price data. The two lines align well with each other, with a correlation of monthly returns of 98.44%.

Alt text: Line chart comparing the reported and constructed monthly gross returns of the Morningstar Global ex-US benchmark over 2000–2024; the two series overlap almost completely, with a monthly return correlation of 98.44%.

for redemption. We also do the same for the Morningstar Gbl xUS USD benchmark, with the R-squared and correlation at 0.97 and 0.98, which are very close to the values for all sample funds.

To visualize the alignment, Figure A1 plots the monthly constructed return series against the reported series; the two lines almost overlap, especially from 2005 onward.

The comparison of constructed and reported returns verifies the reliability of both the reported holdings and the constructed equity-level returns. The results in the main text are therefore unlikely to be driven by errors in either the equity-return construction or the holdings reported by Morningstar.

Appendix B. Robustness Check Results

Appendix B.1 Sorting with the Assigned-Benchmark B-Measure (Assigned Morningstar Benchmark)

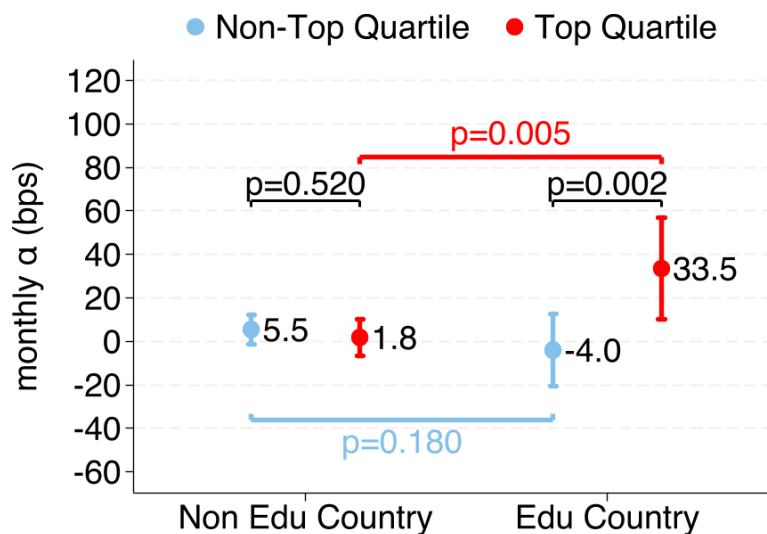


Figure B1: Portfolio α with assigned-benchmark B-measure sorting (5 Factors)

Notes: This figure provides portfolio α with assigned-benchmark B-measure sorting (the proportional-tracking reference uses the fund's assigned Morningstar benchmark rather than market capitalization) using the Fama-French five-factor model. The two black brackets near the markers report the p -value of the top vs non-top quartile difference within each fund type; the wider red bracket on top reports the p -value of the education vs non-education country difference within the top quartile, and the wider blue bracket on the bottom reports the corresponding difference within the non-top quartile. The bars provide 95% confidence intervals.

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals for portfolios sorted by the B-measure. Non-education-country non-top and top portfolios earn 5.5 and 1.8 basis points; education-country non-top and top portfolios earn -4.0 and 33.5 basis points, and the education-country top versus non-top difference is significant at the 1% level.

Appendix B.2 Expanding the Fund's Reference Set to Equities Held During the Prior Two Years

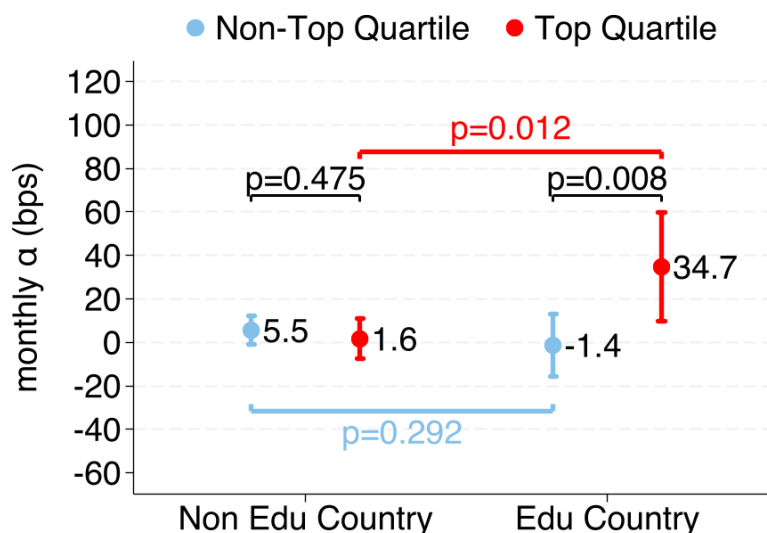


Figure B2: Portfolio α with I-measure sorting with Expanded Universe (5 Factors)

Notes: This figure provides portfolio α with I-measure sorting with expanded funds' investment universe, using the Fama-French five-factor model. The two black brackets near the markers report the p -value of the top vs non-top quartile difference within each fund type; the wider red bracket on top reports the p -value of the education vs non-education country difference within the top quartile, and the wider blue bracket on the bottom reports the corresponding difference within the non-top quartile. The bars provide 95% confidence intervals.

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals for portfolios sorted by the U-measure. Non-education-country non-top and top portfolios earn 5.5 and 1.6 basis points; education-country non-top and top portfolios earn -1.4 and 34.7 basis points, and the education-country top versus non-top difference is significant at the 1% level.

Appendix B.3 Excluding U.K. from Analysis

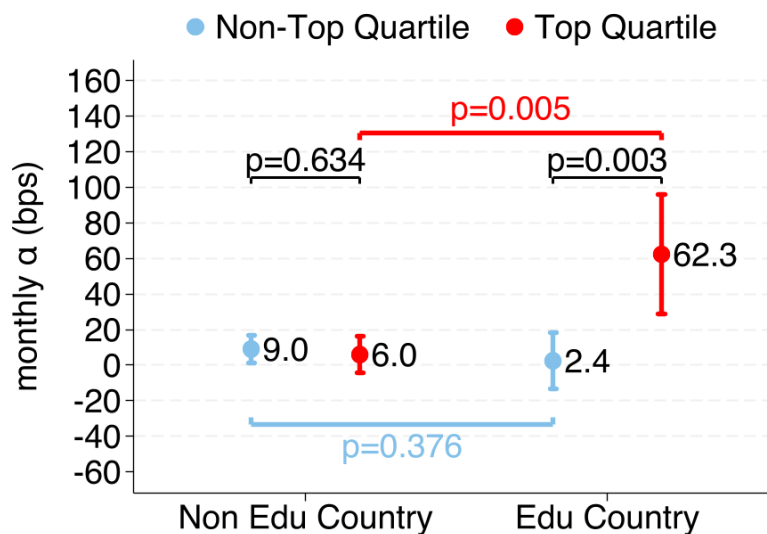


Figure B3: Portfolio α with I-measure sorting Excluding U.K. (5 Factors)

Notes: This figure provides portfolio α from I-measure sorting, excluding U.K. investments, using the Fama-French five-factor model. The two black brackets near the markers report the p -value of the top vs non-top quartile difference within each fund type; the wider red bracket on top reports the p -value of the education vs non-education country difference within the top quartile, and the wider blue bracket on the bottom reports the corresponding difference within the non-top quartile. The bars provide 95% confidence intervals.

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals for the I-measure sort excluding U.K. investments. Non-education-country non-top and top portfolios earn 9.0 and 6.0 basis points; education-country non-top and top portfolios earn 2.4 and 62.3 basis points, and the education-country top versus non-top difference is significant at the 1% level.

The ex-U.K. result does not depend on the factor adjustment, and the outperformance is equally visible in cumulative returns. Figure B4 reports the ex-U.K. portfolio alphas adjusting for the assigned benchmark only (dropping the Fama-French factors): the education-country top portfolio earns a monthly benchmark-adjusted α of 42.0 basis points, significantly above the education-country non-top portfolio (p -value = 0.044) and above the non-education-country top portfolio (p -value = 0.041). Figure B5 plots the corresponding cumulative returns: even after removing all U.K. holdings, the education-country top portfolio grows to roughly eleven times its initial value while the non-top portfolio and both

benchmarks grow about fourfold.

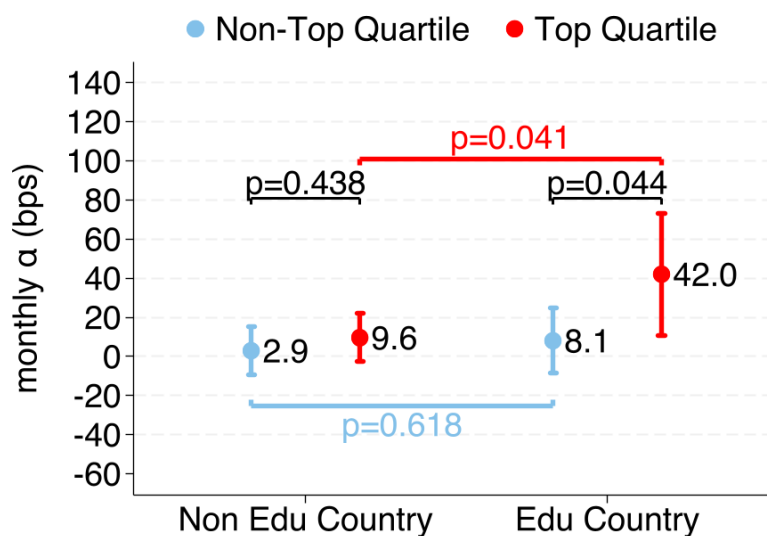
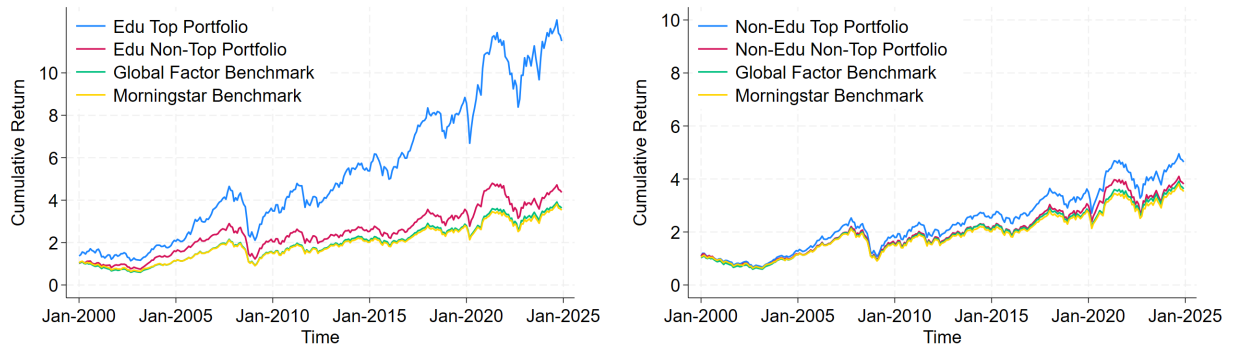


Figure B4: Portfolio α Excluding U.K., benchmark adjustment only (no factors)

Notes: This figure repeats the ex-U.K. I-measure portfolio sort of Figure B3, but adjusts returns using only the assigned Morningstar Global ex-US benchmark excess return, omitting the ten Fama-French developed- and emerging-market factors. The brackets and p -values are defined as in Figure B3, and the bars provide 95% confidence intervals.

Alt text: Dot plot of monthly benchmark-adjusted alphas, with no factor controls, for the ex-U.K. sort; the education-country top portfolio earns 42.0 basis points, significantly above both the education-country non-top portfolio and the non-education-country top portfolio.



Education-Country Portfolios

Non-Education-Country Portfolios

Figure B5: Cumulative Returns Excluding U.K.

Notes: This figure plots cumulative returns of the ex-U.K. I-measure sorted portfolios, normalized to one at January 2000. The left panel reports the education-country top and non-top portfolios together with the global factor and Morningstar benchmark series; the right panel reports the analogous non-education-country portfolios. All U.K. holdings are excluded from portfolio construction.

Alt text: Two line charts of cumulative returns excluding U.K. holdings, normalized to one in January 2000; the education-country top portfolio grows to roughly eleven times its initial value while the non-top portfolio and the benchmarks grow about fourfold, and the non-education-country portfolios track the benchmarks.

Appendix B.4 Manager Entry and Exit Event Study

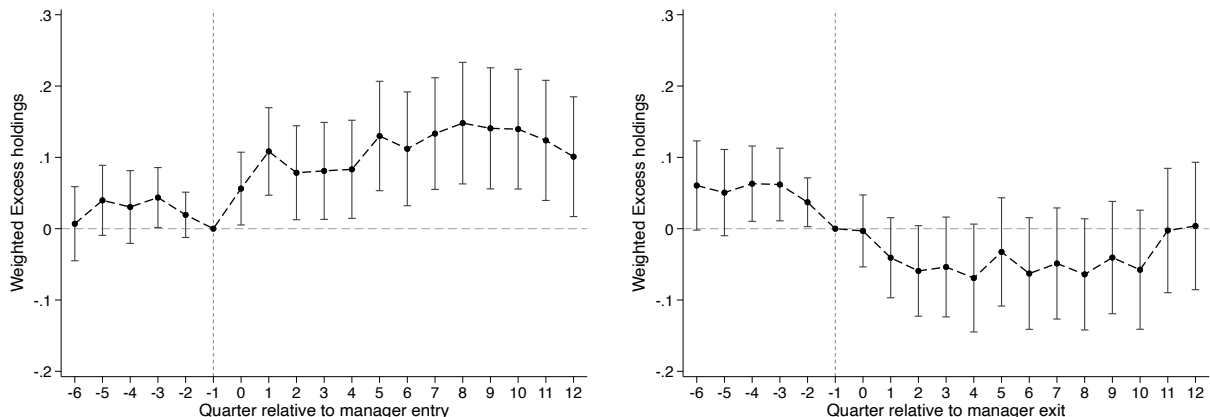
The event-study design and headline results appear in the main text (Section 3.2.1 and Section 4.1, Figure 1). This appendix records the implementation details and reports a robustness variant that scales the outcome by the benchmark country weight. Because the analysis focuses on foreign familiarity among U.S. mutual funds, U.S. education ties are excluded throughout.

The analysis runs on a quarterly fund-country panel. An entry event occurs when $Fund_EC_{jq}$ switches from zero to one, excluding the first observed quarter for a given fund-country pair; an exit event occurs when it switches from one to zero. Each event is followed for up to twelve quarters, but only for as long as the treatment status does not reverse: after entry, the post-event window ends once $Fund_EC_{jq}$ returns to zero, and after exit, it ends once $Fund_EC_{jq}$ returns to one. To guarantee that the pre-event window is genuinely untreated (for entry) or treated (for exit), we impose a clean-lead washout: an entry event is dropped if the fund had an education tie to country c at any point in the six quarters before the event, and an exit event is dropped if the fund lacked such a tie at any point in the six quarters before the event. This filter removes events adjacent to a neighboring spell, whose residual tilts would otherwise contaminate the pre-event baseline. Remaining overlapping event windows are rare, but we remove them so that each window is cleanly identified. After these filters, the sample contains 974 entry events and 777 exit events in the raw excess-holding specification, and 872 entry and 695 exit events in the benchmark-weighted specification, whose panel excludes fund-country cells with zero benchmark weight.

The main text reports results for raw excess holding, the fund’s portfolio weight in a country minus the benchmark weight. As a robustness variant, we rescale the outcome by the importance of the country in the benchmark, defining benchmark-weighted excess holding as

$$\frac{country_weight - benchmark_country_weight}{benchmark_country_weight},$$

winsorized at the 5th and 95th percentiles to limit the influence of observations with very small benchmark weights. Figure B6 re-estimates Equation 7 with this outcome. The pattern matches the raw-outcome results in the main text: benchmark-relative exposure builds gradually after entry and unwinds after exit, so the two-sided pattern that identifies manager-specific familiarity is not an artifact of how the tilt is scaled.



Entry: Benchmark-Weighted Excess Holding Exit: Benchmark-Weighted Excess Holding

Figure B6: Benchmark-Weighted Excess Holdings Around Manager Entry and Exit

Notes: This figure re-estimates the event study of Equation 7 using benchmark-weighted excess holding, defined as $(country_weight - benchmark_country_weight)/benchmark_country_weight$ and winsorized at the 5th and 95th percentiles, as the outcome. Entry and exit events, event windows, the six-quarter clean-lead washout, and the removal of overlapping windows follow the description in the text; the estimation sample contains 872 entry and 695 exit events, as the panel excludes fund-country cells with zero benchmark weight. The two panels share the same vertical scale. Regressions include fund-country, country-quarter, and fund-quarter fixed effects, with quarter -1 omitted as the reference period. Capped bars denote 95% confidence intervals, and standard errors are clustered at the fund level.

Alt text: Two event-study panels plotting coefficients with 95% confidence intervals for benchmark-weighted excess holdings around manager entry and exit; the tilt builds gradually after the first education-linked manager arrives and unwinds after the last one departs, matching the raw-outcome pattern in the main text.

Appendix B.5 Non-Education Familiarity Countries

The main analysis uses managers' education histories to identify countries in which they are likely to possess deeper familiarity or informational advantages. A natural question is whether similar results arise when familiarity is defined more broadly using non-education background information. To examine this, we construct an alternative country-familiarity measure based on managers' other non-U.S. location ties reported in biography text, such as birthplace, past work location, or other persistent location-related background information, while excluding countries linked through education. We then re-estimate the portfolio-sorting exercise using this non-education familiarity definition in place of the education-country indicator. Unlike education countries, which are systematically reported in manager biographies across funds, non-education foreign familiarity ties are observed much less consistently. Among our 2,976 sample managers, this information yields 377 distinct manager-country pairs spread across 44 countries.

To summarize the composition of this non-education familiarity measure, Table B1 reports the full country distribution after collapsing the non-education country fields into distinct manager-country pairs. The distribution is highly concentrated in a small number of countries. The United Kingdom and Hong Kong are the most common non-education familiarity countries, accounting for 21.2% and 20.4% of all (manager, country) pairs, followed by Japan and Singapore (8.2% each) and France (6.1%). The top five countries jointly cover roughly 64% of all non-education familiarity pairs. This concentration indicates that the non-education familiarity measure is both limited in coverage and skewed toward a narrow set of locations, making it a noisier and less systematic proxy for country familiarity than the education-based measure. One likely reason is that much of the non-education information reflects prior work experience, which is disproportionately concentrated in financial centers such as Hong Kong, the United Kingdom, and Singapore.

The portfolio construction follows the baseline procedure in Section 4.3. Within each

Table B1: Country Distribution of Non-Education Familiarity

Country	# Mgrs	Share	Country	# Mgrs	Share
GBR	80	0.2122	SLV	3	0.0080
HKG	77	0.2042	THA	3	0.0080
JPN	31	0.0822	ZAF	3	0.0080
SGP	31	0.0822	IND	2	0.0053
FRA	23	0.0610	KOR	2	0.0053
CAN	11	0.0292	PNG	2	0.0053
BRA	10	0.0265	SSD	2	0.0053
DEU	10	0.0265	ARE	1	0.0027
CHN	8	0.0212	AUT	1	0.0027
IDN	8	0.0212	BEL	1	0.0027
MEX	7	0.0186	CHE	1	0.0027
NZL	7	0.0186	DOM	1	0.0027
SWE	7	0.0186	ECU	1	0.0027
ESP	6	0.0159	EGY	1	0.0027
NOR	6	0.0159	IRN	1	0.0027
AUS	4	0.0106	NGA	1	0.0027
ITA	4	0.0106	NLD	1	0.0027
ARG	3	0.0080	POL	1	0.0027
HUN	3	0.0080	RUS	1	0.0027
MCO	3	0.0080	SAU	1	0.0027
MYS	3	0.0080	TWN	1	0.0027
NPL	3	0.0080	UKR	1	0.0027
Total Pairs	377	1.000	Total Countries	44	

Notes: This table reports the country distribution of non-education familiarity for the sample managers, listing every country observed (no top-N truncation). The “# Mgrs” column counts the number of distinct sample managers with at least one non-education familiarity tie to that country, and the “Share” column expresses each count as a fraction of the 377 total non-missing (manager, country) pairs (so the column sums to 1). Non-education familiarity is constructed from the other-country-location fields in the manager-level biography file (sources include birthplace, prior work location, and other persistent location ties), excluding any tie that already corresponds to the manager’s education country. These 377 pairs are drawn from the same universe of 2,976 distinct sample managers used in Table 4.

fund-country-quarter cell, we rank the stocks held by the fund using the I-measure and define the top quartile as the top portfolio, while pooling the remaining three quartiles into the non-top portfolio. As in the main specification, we rebalance portfolios quarterly, carry weights forward within the quarter using realized returns, and reweight the stock positions at each quarterly formation date so that each constructed portfolio matches the benchmark’s country allocation, allowing the weights to drift within the quarter. We then estimate monthly portfolio α using the same five-factor specification as in the main text.

Figure B7 reports the results. In countries with no such non-education familiarity match, the estimated monthly α is 3.5 basis points for the non-top portfolio and 4.9 basis points for the top portfolio, with no significant difference between them (p -value = 0.797). In countries classified as familiar under the non-education measure, the top portfolio earns -13.0 basis points per month and the non-top portfolio -18.8 basis points. The resulting 5.8-basis-point spread is not statistically distinguishable from zero (p -value = 0.726), and—unlike the education-based sort—the familiar-country top portfolio does not outperform its non-familiar counterpart: its point estimate is, if anything, lower (-13.0 versus 4.9 basis points). The non-education familiarity measure therefore yields no reliable evidence of superior stock selection. The contrast is sharp: re-running this same exercise with familiarity defined by education ties instead—identical panel, construction, and specification, differing only in how a country is classified as familiar—recovers the baseline top-portfolio α of 41.0 basis points (p -value = 0.001), the same figure reported in Section 4.3.

The contrast with the baseline education-based results is informative. Education ties appear to be a sharper and more systematic proxy for the type of country-specific familiarity that translates into superior stock selection. By comparison, non-education biography-based ties are likely more heterogeneous and noisier, which attenuates precision. The appendix evidence therefore suggests that not all familiarity proxies are equally informative: the strongest and cleanest return patterns continue to be concentrated in countries linked to managers through education.

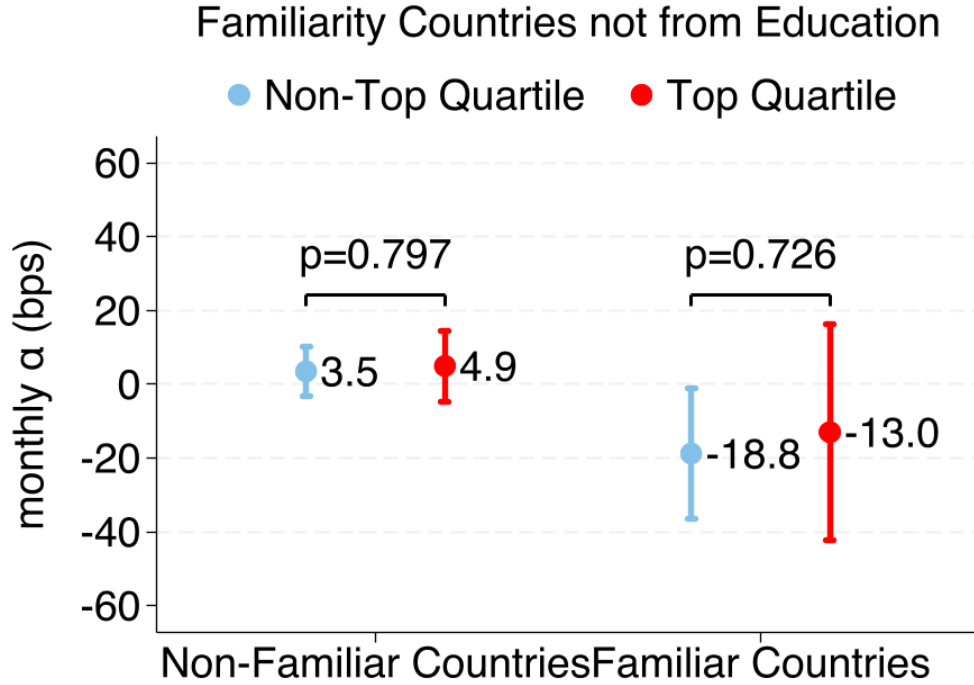


Figure B7: Portfolio α Using Non-Education Familiarity Countries

Notes: This figure repeats the baseline I-measure portfolio sort using an alternative familiarity definition based on managers' non-education country ties. A country is classified as familiar if at least one current manager of the fund has a non-U.S. non-education location tie to that country. Within each fund-country-quarter cell, stocks are ranked by the I-measure, with the top quartile forming the top portfolio and the remaining three quartiles forming the non-top portfolio. Portfolios are value-weighted within country, reweighted to match benchmark country allocations, and rebalanced quarterly. Reported coefficients are monthly alphas in basis points from the same five-factor specification used in the main text. Capped bars denote 95% confidence intervals, and reported p -values test the equality of the top and non-top portfolio alphas within each country group.

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals for portfolios sorted on non-education familiarity ties. Non-familiar-country non-top and top portfolios earn 3.5 and 4.9 basis points; familiar-country non-top and top portfolios earn -18.8 and -13.0 basis points, and no difference is statistically significant.

Appendix B.6 Portfolio Sorting Without Factor Adjustment

The main portfolio result risk-adjusts returns with the Fama-French five-factor model. To show that the concentration of value in education-country top picks does not depend on this factor adjustment, we re-estimate the four sorted-portfolio alphas controlling only for the assigned benchmark—that is, we drop the ten developed- and emerging-market factors and retain only the benchmark excess return. Figure B8 reports the resulting benchmark-adjusted alphas.

The pattern is unchanged. In education countries, the top portfolio earns a monthly benchmark-adjusted α of 33.9 basis points while the non-top portfolio earns 3.4 basis points, with the top–non-top difference statistically significant (p -value = 0.025). In non-education countries the top and non-top portfolios remain indistinguishable (9.1 vs. 2.0 basis points, p -value = 0.375). If anything, the benchmark-only alpha of the education-country top portfolio (33.9 bps) is *smaller* than its five-factor counterpart (41.0 bps in Section 4.3): the top-pick portfolio carries a growth tilt—a statistically significant negative loading on the value (HML) factor—and value earned a positive premium over the sample, so netting out factor exposures *raises* the estimated selection alpha rather than lowering it. The headline result of large, concentrated abnormal returns in education-country top picks therefore does not rely on the factor model, and is if anything understated by a benchmark-only adjustment.

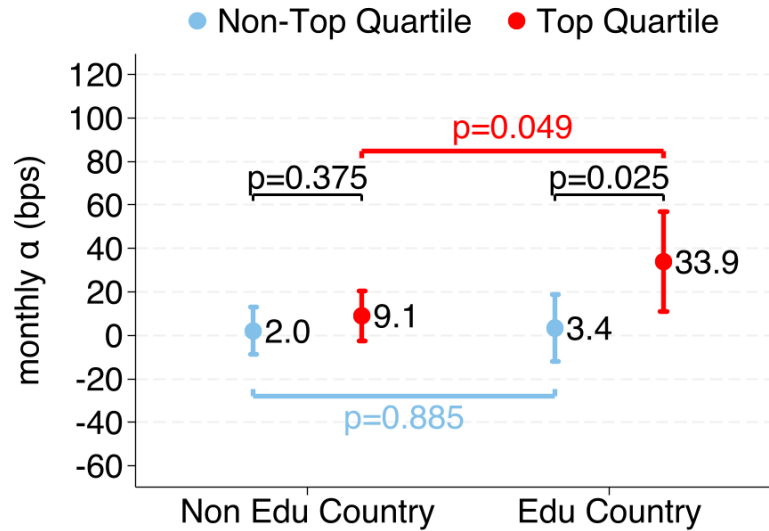


Figure B8: Portfolio α with I-measure sorting, benchmark adjustment only (no factors)

Notes: This figure repeats the main I-measure portfolio sort but adjusts returns using only the assigned Morningstar Global ex-US benchmark excess return, omitting the ten Fama-French developed- and emerging-market factors used in the main specification. The two black brackets near the markers report the p -value of the top vs non-top quartile difference within each country group; the wider red bracket on top reports the p -value of the education vs non-education difference within the top quartile, and the wider blue bracket on the bottom reports the corresponding difference within the non-top quartile. The bars provide 95% confidence intervals.

Alt text: Dot plot of monthly benchmark-adjusted alphas without factor controls; the education-country top portfolio earns 33.9 basis points versus 3.4 for the education-country non-top portfolio, a difference significant at the 5% level, while the non-education-country top and non-top portfolios earn 9.1 and 2.0 basis points and are statistically indistinguishable.

Appendix B.7 Portfolio Performance After Bid-Ask Spread Adjustment

A potential concern with the portfolio-sorting results is that the top portfolios may require substantial trading, so that their gross outperformance may not survive plausible implementation costs. To address this concern, we re-estimate the baseline portfolio analysis after incorporating security-level bid-ask spread costs at quarterly rebalancing dates.

The construction begins from the same quarterly target portfolios used in the main text. We merge a quarterly bid-ask spread measure at the security level via the $\text{ISIN} \times \text{year} \times \text{month}$ key. At each quarterly rebalance, we compare the new target weight $w_{i,q}^{\text{target}}$ with the preceding target weight carried forward to the rebalance date using realized returns, $w_{i,q}^{\text{drift}}$. We define reported one-way turnover as one-half of the sum of absolute weight changes, $\frac{1}{2} \sum_i |w_{i,q}^{\text{target}} - w_{i,q}^{\text{drift}}|$. For transaction costs, we define the proportional half-spread as $s_{i,q} = (\text{Ask}_{i,q} - \text{Bid}_{i,q}) / (\text{Ask}_{i,q} + \text{Bid}_{i,q})$ and compute the quarterly cost as $\sum_i s_{i,q} |w_{i,q}^{\text{target}} - w_{i,q}^{\text{drift}}|$. This charges the half-spread on both purchases and sales and therefore does not divide the cost measure by two. We subtract the resulting quarterly cost from gross portfolio returns evenly across the three holding months and then recompute net monthly returns, cumulative returns, and factor-adjusted alphas.

Figure B9 reports the spread-adjusted α estimates. The main conclusion is robust to estimated bid-ask spread costs: under the factor specification, the education-country top portfolio earns a monthly α of 33.7 basis points and the education-country non-top portfolio earns -11.4 basis points, with the top–non-top differential significant at the 1% level ($p\text{-value} < 0.001$). The non-education-country portfolios remain statistically indistinguishable, with monthly alphas of 0.6 (non-top) and -2.4 (top) basis points ($p\text{-value} = 0.590$).

Figure B10 translates these alphas into cumulative returns. Even net of estimated bid-ask spread costs, the education-country top portfolio pulls away sharply from its non-top counterpart and from the benchmark, while the non-education-country top and non-top portfolios remain close to each other and to the benchmark. The performance gap is therefore

not driven by unrealistic gross-return assumptions.

Table B2 reports average annual turnover and estimated annual bid-ask spread costs for the four cost-adjusted portfolios. As expected, top portfolios are more actively traded than non-top portfolios. The education-country top portfolio has the highest average one-way annual turnover at 133.22% of portfolio value, followed by the non-education-country top portfolio at 86.43%; the corresponding average annual spread costs are 0.85% and 0.61%, respectively. Despite these higher estimated costs, the education-country top portfolio is the only portfolio that delivers large positive alpha after the adjustment, reinforcing the interpretation that the result is not compensation for mechanically higher turnover.

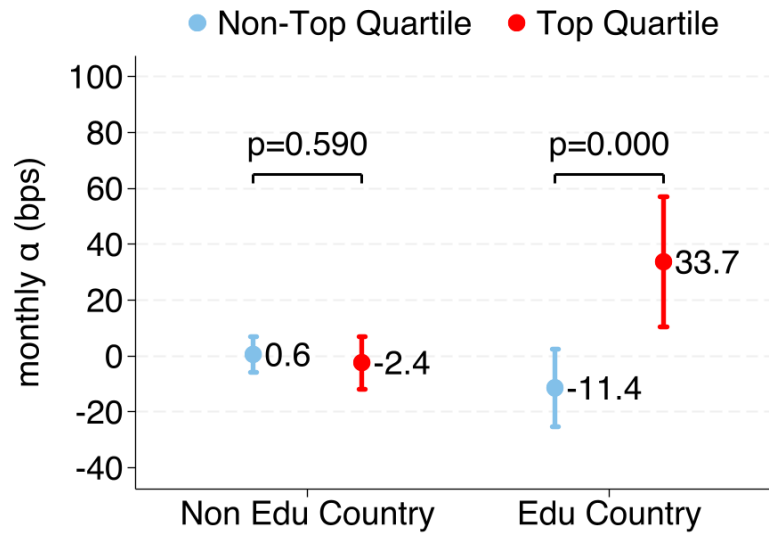
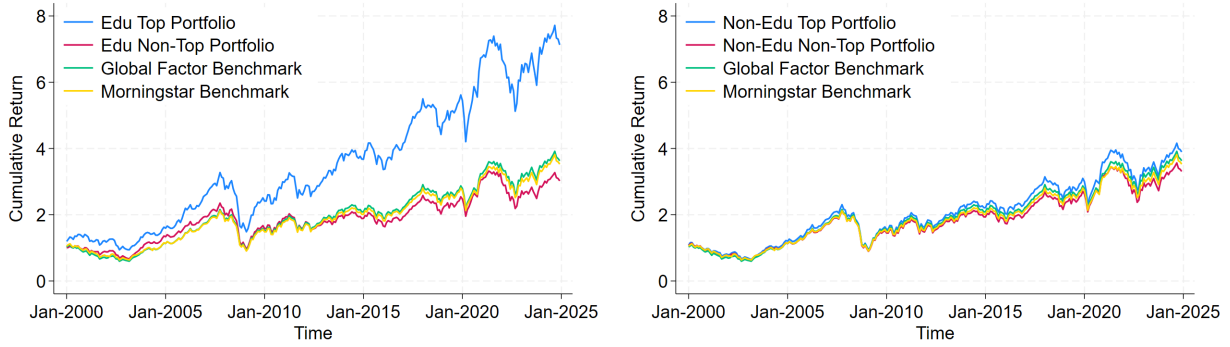


Figure B9: Portfolio α After Bid-Ask Spread Adjustment

Notes: This figure reports portfolio alphas after incorporating quarterly bid-ask spread transaction costs, using the Fama-French five-factor specification. Capped bars denote 95% confidence intervals, and reported p -values test the equality of the two alphas within each country group.

Alt text: Dot plot of monthly five-factor alphas net of estimated bid-ask spread costs; the education-country top portfolio earns 33.7 basis points versus -11.4 for the education-country non-top portfolio, a difference significant at the 1% level, while the non-education-country non-top and top portfolios earn 0.6 and -2.4 basis points and are statistically indistinguishable.



Education-Country Portfolios

Non-Education-Country Portfolios

Figure B10: Cumulative Returns After Bid-Ask Spread Adjustment

Notes: This figure plots cumulative net returns after incorporating quarterly bid-ask spread transaction costs. The left panel reports the education-country top and non-top portfolios together with the benchmark series, and the right panel reports the analogous non-education-country portfolios. All series are normalized near the start of the sample, and returns are net of estimated transaction costs incurred at quarterly rebalancing dates.

Alt text: Two line charts of cumulative returns net of estimated bid-ask spread costs; the education-country top portfolio pulls far above the non-top portfolio and the benchmark, while the non-education-country top and non-top portfolios remain close to each other and to the benchmark.

Portfolio	Avg. annual turnover	Avg. annual spread cost
Edu non-top	102.73%	0.73%
Edu top	133.22%	0.85%
Non-edu non-top	52.23%	0.34%
Non-edu top	86.43%	0.61%

Table B2: Turnover and Transaction Costs for Cost-Adjusted Portfolios

Notes: This table reports turnover and transaction-cost statistics for the four portfolios in the cost-adjusted analysis. At each quarterly rebalance, absolute changes are measured relative to the preceding target weights carried forward using realized returns; entries and exits are included. Average annual turnover is the one-way measure, computed by summing these absolute weight changes across securities and quarterly rebalances within a year and dividing by two. Estimated spread costs instead apply the proportional half-spread, $(Ask - Bid)/(Ask + Bid)$, to each absolute security-level weight change, sum across securities and rebalances within the year, and are not divided by two. Initial portfolio formation is measured relative to zero holdings and is therefore included in the first year's turnover and cost. The table reports time-series averages of these annual quantities.

Appendix B.8 Portfolio Performance Across Subperiods

Figure B11 reports the I-measure portfolio sort over three subsamples, supporting the persistence discussion in Section 4.3.

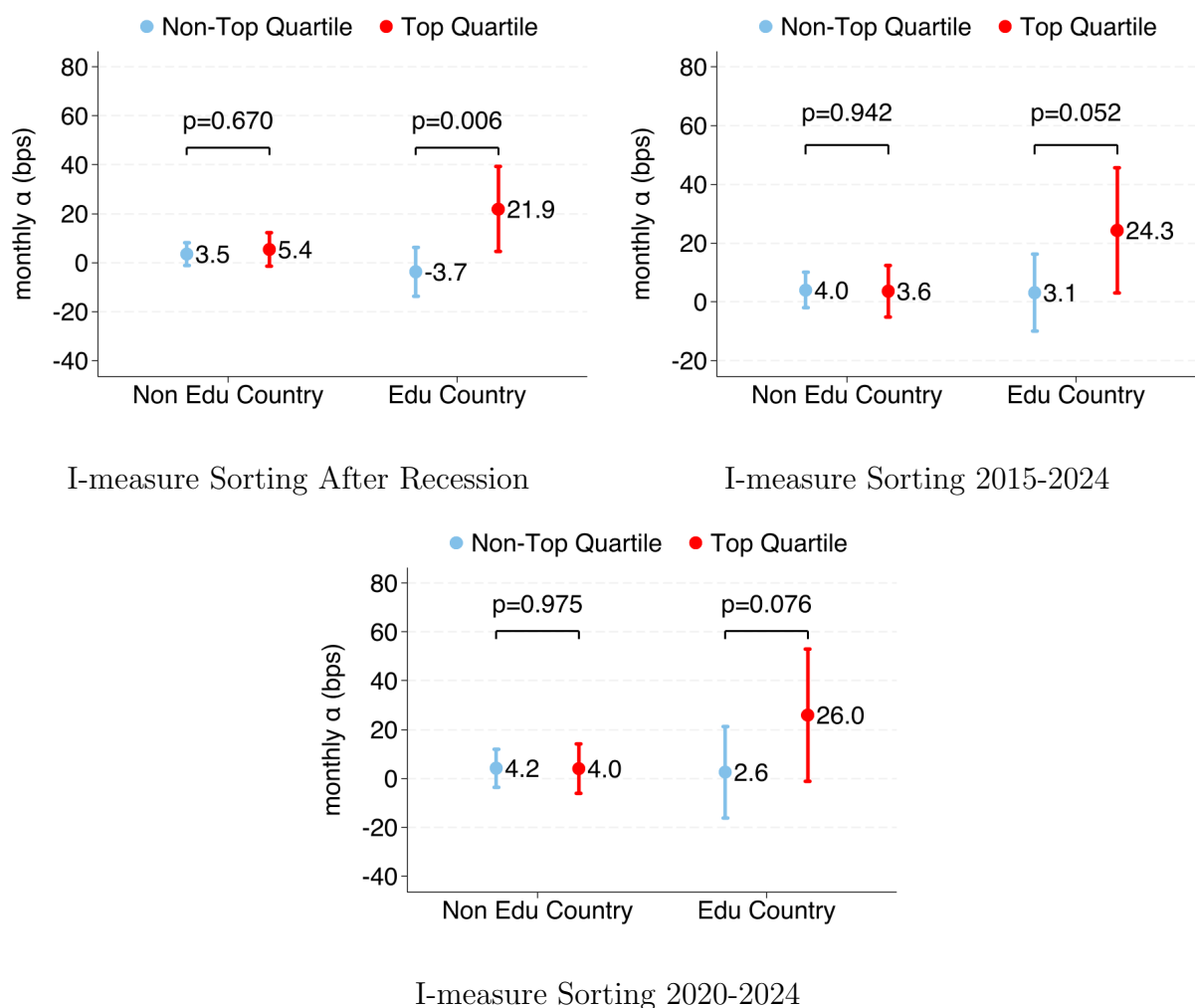


Figure B11: Portfolio α Across Subperiods

Notes: This figure reports portfolio α from I-measure sorting under the Fama-French five-factor model for three sub-samples. The top-left panel uses the post-recession sample (July 2009 to December 2024); the top-right panel uses the most recent 10-year sample (January 2015 to December 2024); the bottom panel uses the most recent 5-year sample (January 2020 to December 2024). Capped bars denote 95% confidence intervals.

Alt text: Three dot plots of monthly five-factor alphas with 95% confidence intervals for subsamples beginning July 2009, January 2015, and January 2020; the education-country top portfolio earns 21.9, 24.3, and 26.0 basis points across the three windows and remains above the other three portfolios, whose alphas stay near zero.

Appendix B.9 Undergraduate-Only Education

Table B3 and Figure B12 report the undergraduate-only allocation regressions and portfolio sort discussed in Section 5.

VARIABLES	(1) Excess Weights	(2) Excess Weights	(3) Scaled Excess Weights	(4) Scaled Excess Weights	(5) Excess Weights	(6) Excess Weights	(7) Scaled Excess Weights	(8) Scaled Excess Weights
$\mathbb{1}(EC_{UG})$	1.093*** (0.256)		30.38*** (6.761)		1.057*** (0.253)		30.85*** (6.797)	
$Frac_{EC,UG}$		2.981*** (0.588)		73.03*** (14.99)		2.865*** (0.571)		72.94*** (15.16)
Constant	-0.0429*** (0.00277)	-0.0436*** (0.00247)	-19.15*** (0.0667)	-19.13*** (0.0564)	-0.0441*** (0.00272)	-0.0446*** (0.00237)	-19.16*** (0.0666)	-19.13*** (0.0561)
Observations	5,613,030	5,613,030	5,244,238	5,244,238	5,611,449	5,611,449	5,242,603	5,242,603
R-squared	0.145	0.146	0.119	0.119	0.191	0.192	0.139	0.139
Fund FE	Y	Y	Y	Y	–	–	–	–
Fund×Time FE	N	N	N	N	Y	Y	Y	Y
Country×Time FE	Y	Y	Y	Y	Y	Y	Y	Y
Funds Number	1663	1663	1663	1663	1663	1663	1663	1663

Table B3: Education Links and Excess Country Weight: Undergraduate Education Only

Notes: This table re-estimates Table 1 on the same regression panel while defining education-country links using undergraduate degrees only. Columns (1)–(2) and (5)–(6) use raw excess weight (fund country weight minus benchmark country weight, in percentage points); columns (3)–(4) and (7)–(8) divide the same difference by the benchmark country weight and multiply by 100. $\mathbb{1}(EC_{UG})$ equals one if any current manager received an undergraduate degree in the country, and $Frac_{EC,UG}$ is the share of the current management team with an undergraduate degree in that country. Columns (1)–(4) include fund and country×time fixed effects; columns (5)–(8) repeat the same specifications but replace the fund fixed effect with a fund×time fixed effect, which subsumes it and absorbs all time-varying fund-level heterogeneity. Robust standard errors clustered at the fund level are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

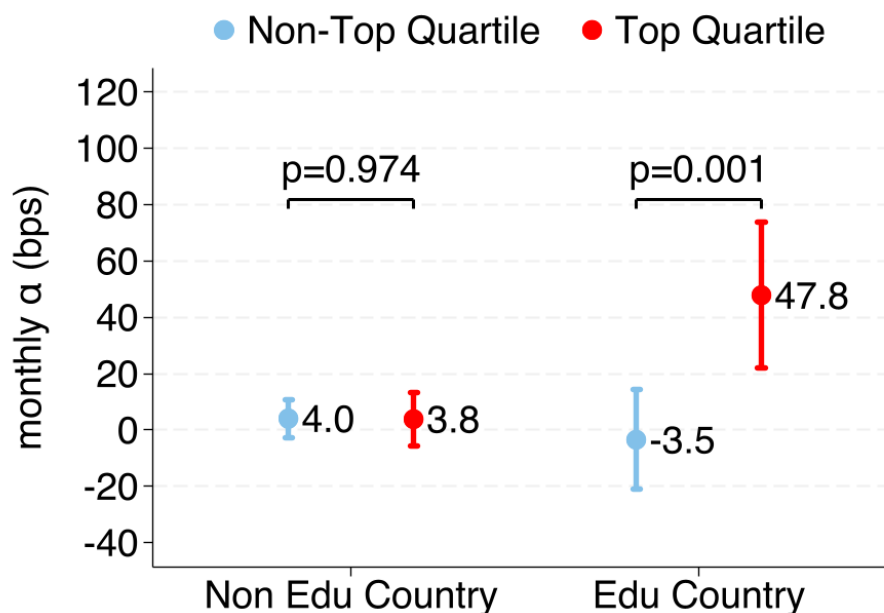


Figure B12: Portfolio α Using Undergraduate-Only Education Countries

Notes: This figure repeats the baseline I-measure portfolio sort, but defines a fund's education countries using only its managers' undergraduate degrees. Within each fund-country-quarter cell, stocks are ranked by the I-measure, with the top quartile forming the top portfolio and the remaining three quartiles the non-top portfolio; portfolios are value-weighted within country, reweighted to match benchmark country allocations, and rebalanced quarterly. Reported coefficients are monthly five-factor alphas in basis points; capped bars denote 95% confidence intervals, and p -values test the equality of the top and non-top alphas within each country group.

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals using undergraduate-only education ties; the education-country top portfolio earns 47.8 basis points, significantly above the education-country non-top portfolio, while the non-education-country portfolios remain near zero.

Appendix B.10 Adding Momentum and Removing the Benchmark Control

Figure B13 reports the momentum-augmented, no-benchmark portfolio sort discussed in Section 5.5.

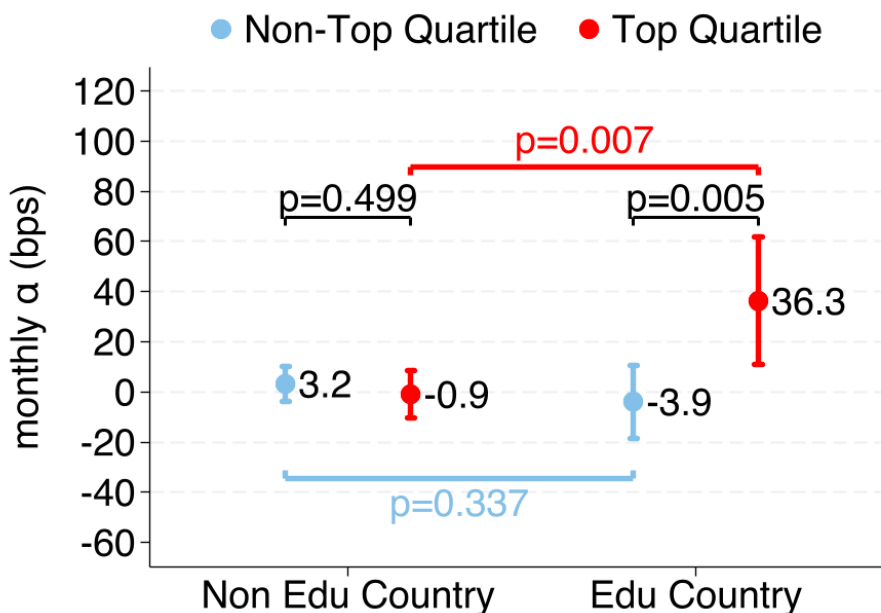


Figure B13: Portfolio α with Momentum and No Benchmark Control

Notes: This figure repeats the baseline I-measure portfolio sort of Section 4.3 while changing only the return-attribution model. The fund's assigned benchmark excess return is dropped from the right-hand side, and the Fama-French momentum factor (WML) is added to each of the developed ex-U.S. and emerging-market factor blocks, giving a six-factor model per region. Portfolio construction, weighting, and rebalancing are unchanged. Reported coefficients are monthly alphas in basis points; capped bars denote 95% confidence intervals. Black p -values test the equality of the top and non-top alphas within a country group; the red p -value tests the equality of the two top-quartile alphas across country groups, and the blue p -value the corresponding test for the non-top portfolios.

Alt text: Dot plot of monthly alphas from the six-factor model that adds momentum and drops the benchmark control; the education-country top portfolio earns 36.3 basis points, significantly above the education-country non-top portfolio at -3.9 , while the non-education-country non-top and top portfolios earn 3.2 and -0.9 basis points.

Appendix B.11 Sorting Without the Universe Restriction

Figure B14 reports the portfolio sort based on the country-scaled excess weight $A_{jc}^i(t)$ of Equation 1, which omits the universe-restriction step of the I-measure; Sections 3 and 4.3 discuss the comparison.

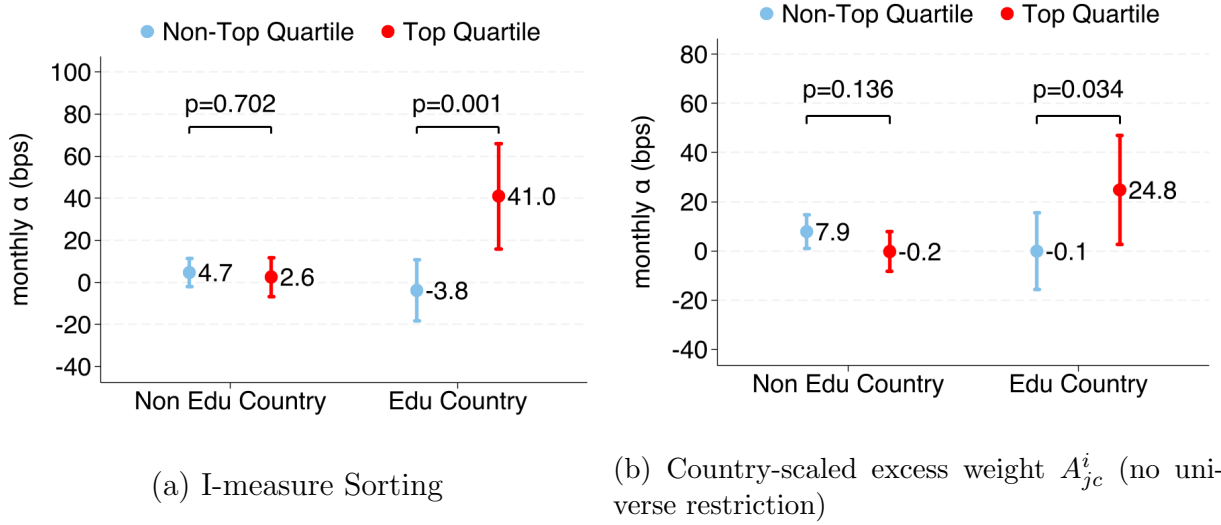


Figure B14: Portfolio α from I-measure and country-scaled excess-weight sorting (5 factors)
Notes: This figure provides portfolio α under the Fama-French five-factor model for two sorting variables: (a) the I-measure of Equation 3, whose passive reference is a market-cap tracker restricted to the stocks the fund holds in the country, and (b) the country-scaled excess weight $A_{jc}^i(t)$ of Equation 1, which compares the fund against the full benchmark and so omits the universe-restriction step. The two dots on the left of each graph are portfolio α for the sub-portfolio of equities with activeness (as measured in the two ways above) in the bottom three quartiles and the top quartile separately, in countries where the fund manager was not educated. The two dots on the right of each graph are the portfolio α for sub-portfolios in managers' education countries sorted by activeness. The bars provide 95% confidence intervals.

Alt text: Two dot plots of monthly five-factor alphas with 95% confidence intervals comparing the I-measure sort with a sort on the country-scaled excess weight that omits the universe restriction; the education-country top-portfolio premium is pronounced under the I-measure and much weaker and noisier under the unrestricted measure.

Appendix C. Comparison to the Antón et al. Activeness Measure

Appendix C.1 Measure Construction

In Antón et al. (2021), the authors set up a simple portfolio choice problem as follows:

$$\max_{\lambda} \lambda'_t(E_t \mathbf{R}_{t+1} - R_{f,t+1} \mathbb{1}) - \frac{k}{2} \lambda'_t \Omega_t \lambda_t \quad (15)$$

Here the managers choose vector of portfolio weight λ_t at time t to maximize risk-adjusted portfolio return. $E_t \mathbf{R}_{t+1}$ is the expected return of the next period, $R_{f,t+1}$ is the risk-free rate, k is the risk-aversion parameter, and Ω_t is the return variance-covariance matrix.

The solution to this maximization problem is:

$$\lambda_t = \frac{1}{k} \Omega_t^{-1} (E_t \mathbf{R}_{t+1} - R_{f,t+1} \mathbb{1}) \quad (16)$$

Defining $\mu_t \equiv (E_t \mathbf{R}_{t+1} - R_{f,t+1} \mathbb{1})$, the excess return, then $\mu_t = k \Omega_t \lambda_t$. Then for the portfolio manager, the subjective CAPM α is $\mu_t - \mu_{Vt}$, where μ_{Vt} is the excess return on the market (value-weighted) portfolio of the equities in the fund's investment universe, and we denote that portfolio's weight vector by λ_{Vt} . Assuming that the variance-covariance matrix Ω is diagonal, the subjective Sharpe ratio of equity i is $\frac{k \sigma_{i,t}^2 (\omega_{i,t} - \omega_{V,i,t})}{\sigma_{i,t}}$, where $\omega_{i,t}$ is the i -th element of the fund's weight vector λ_t , $\omega_{V,i,t}$ is the i -th element of the market portfolio's weight vector λ_{Vt} , and $\sigma_{i,t}^2$ is the i -th diagonal element of Ω_t . The equity with the largest Sharpe ratio is therefore the one with the largest $\sigma_{i,t}(\omega_{i,t} - \omega_{V,i,t})$. Antón et al. (2021) refer to this measure as the information ratio (IR).

σ_{it} is the predicted volatility using the following predictive regression:

$$\sigma_{i,t} = \beta \sigma_{i,t-1} + \gamma \log(\text{Market_Cap}_{i,t-1}) + F E_{c,t} + \epsilon_{i,t} \quad (17)$$

In our specification, we use last year's volatility and the log of market cap to predict next year's volatility, and absorb for the variation in country-time level using country-time fixed

VARIABLES	(1)
	Next Year Volatility
log(Market Cap)	-0.0131*** (0.00191)
Previous Year Volatility	0.343*** (0.0256)
Constant	0.348*** (0.0387)
Observations	4,306,131
R-squared	0.391
Country×Time FE	Y
Country Cluster	Y

Table C1: Volatility Regression

Notes: This table reports the predictive regression used to construct the equity-level expected volatility input ($\sigma_{i,t}$) entering the Antón et al. (2021) information-ratio measure. The dependent variable is realized next-year (12-month) return volatility for stock i . Regressors are the previous-year realized volatility and the log of market capitalization (in USD), measured at $t - 1$. The specification absorbs country-time fixed effects so that $\sigma_{i,t}$ is identified from within-country-time variation in size and lagged volatility, and standard errors are clustered at the country level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

effect. The regression results are reported in Table C1.

Appendix C.2 Portfolio Sorting Result

Following Antón et al. (2021), we sort each fund’s portfolio at each point in time using the activeness measure $\sigma_{i,t}(\omega_{i,t} - \omega_{V,i,t})$, consistent with their methodology. We make one minor modification: instead of selecting the top X picks as in their original approach, we select the top 2% of holdings, which is approximately equivalent to the top two picks in Antón et al. (2021), given that a typical sample fund holds around 80 equities (the median in Table 6). This adjustment helps mitigate the high variance associated with constructing top portfolios from a very small number of equities.

It is important to note two key differences between this sorting method and the I-measure-based sorting used in our main analysis. First, this sorting is applied at the overall fund

level and does not condition on country, whereas we use the I-measure to sort within country sub-portfolios. Second, the activeness measure used here does not account for differences in the investment universe between the fund and its benchmark. In Appendix C.3, we adjust for these two differences and present the corresponding sorting results.

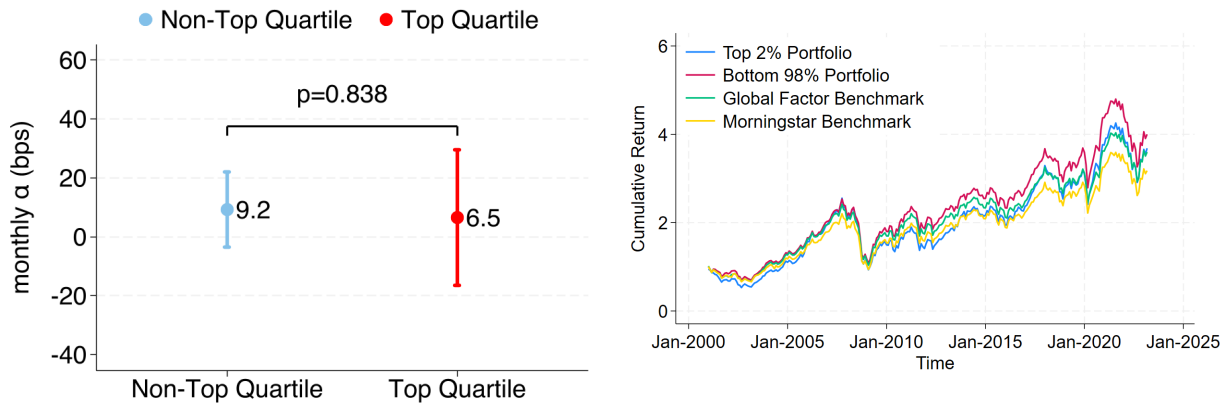


Figure C1: Antón et al. (2021) Sorted Return

Notes: This figure reports results using the activeness measure of Antón et al. (2021). The left panel shows the monthly benchmark-adjusted α of the “best ideas” portfolio (top 2%) and the remaining portfolio (bottom 98%), along with the p -value for the alpha difference between the two. The right panel plots their cumulative returns alongside two global benchmarks.

Alt text: Left panel: dot plot of monthly benchmark-adjusted alphas for the top 2% and bottom 98% portfolios under the original best-ideas information-ratio measure, earning 6.5 and 9.2 basis points with a p -value of 0.838 for the difference. Right panel: line chart of cumulative returns in which the top 2% portfolio grows roughly fourfold, similar to the benchmarks.

Figure C1 presents the results of applying the Antón et al. (2021) methodology directly to our international sample. The left panel reports the monthly benchmark-adjusted alphas for portfolios sorted by their activeness measure. The top 2% portfolio—identified by their information-ratio measure—earns a monthly alpha of 6.5 basis points, while the remaining 98% earns 9.2 basis points; the difference is not statistically distinguishable from zero (p -value = 0.838). The original measure was developed for U.S. domestic mutual funds, where conditioning on country and adjusting for investment-universe differences is unnecessary; in the international setting, those adjustments matter for the reasons described in Section 3.

The right panel of Figure C1 plots the cumulative returns of the same top 2% and bottom

98% portfolios alongside the global benchmark. Consistent with the alpha results, the top 2% portfolio grows roughly fourfold over the sample period, compared with approximately tenfold growth for the top portfolio identified using our I-measure. These results indicate that the direct application of the original Antón et al. (2021) measure produces weak and statistically insignificant sorting in the international setting—the top-2% portfolio is no better than the rest—which we view as evidence that the within-country, universe-adjusted construction in our I-measure is important in this context.

Appendix C.3 Sorting using Modified IR from Antón et al. (2021)

In this section, we adjust the sorting above using IR by sorting within each country sub-portfolio, and take into account the differential investment universe when constructing the measure.

In our international equity context, we suppose that the portfolio choice problem happens within the country due to friction in international equity investment across countries. Then we define the relevant market portfolio as the benchmark sub-portfolio of equities in the fund’s universe in the respective country. Then, following the same derivation as above, the within-country portfolio choice problem implies that the equity with the largest $\sigma_{it} \left(\frac{\omega_{jc}^i(t)}{\sum_{i' \in \Omega_{jc}(t)} \omega_{jc}^{i'}(t)} - \frac{B_{jc}^i(t)}{\sum_{i' \in \Omega_{jc}(t)} B_{jc}^{i'}(t)} \right)$ is the one with the largest subjective information ratio relative to the defined benchmark, exactly as in the derivation above. This equals σ_{it} times the B-measure of Equation 2; because the assigned benchmarks are predominantly market-capitalization-weighted (Section 3.1.2), it is numerically close to $\sigma_{it} I_{jc}^i(t)$. We refer to it as the modified IR measure.

Using the modified IR measure, we find that in non-education countries, the monthly 5-factor α is 4.7 basis points for the non-top portfolio and 1.4 basis points for the top portfolio (Figure C2). The difference between the two is not statistically significant, with a p -value of 0.567. It is worth noting that, since this measure tends to over-select high-volatility stocks into the top portfolio, the estimated α for the non-education country top portfolio exhibits

substantially larger variance relative to results based on other sorting measures.

In contrast, for education countries, the monthly 5-factor α is -10.2 basis points for the non-top portfolio and 29.0 basis points for the top portfolio. The top-vs-non-top difference is sizable and statistically significant at the 1% level, with a p -value of 0.005 . The α of the education country top portfolio is also significantly higher than that of the non-education country top portfolio (p -value = 0.034).

While the modified IR sorting is not as sharp as the I-measure sorting—largely due to the tendency to select volatile small-cap equities, as discussed in Antón et al. (2021)—it nonetheless reveals a consistent pattern. The contrast between the results in Appendix C.2 and Appendix C.3 highlights two key points in the international context: the importance of conducting portfolio sorting within countries, and the importance of the interaction between education and activeness, rather than activeness alone.

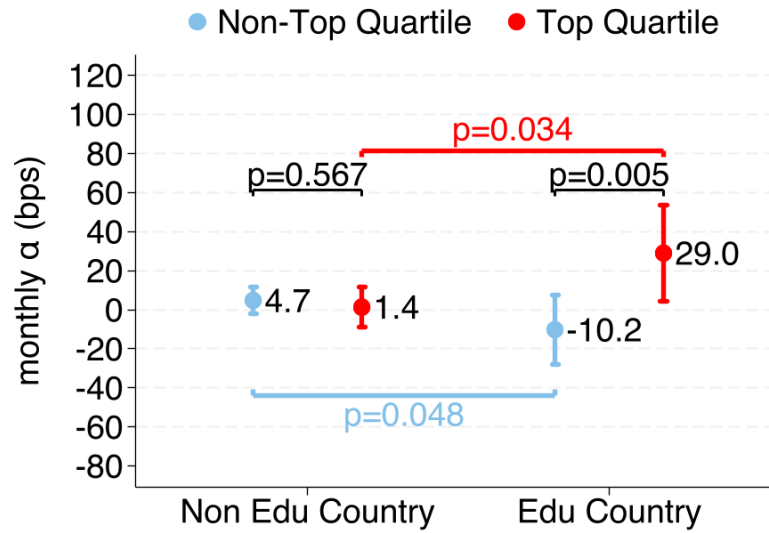


Figure C2: IR Measure Sorted Return (5 Factors)

Notes: This figure provides portfolio α with modified IR measure sorting using the Fama-French five-factor model. The two black brackets near the markers report the p -value of the top vs non-top quartile difference within each fund type; the wider red bracket on top reports the p -value of the education vs non-education country difference within the top quartile, and the wider blue bracket on the bottom reports the corresponding difference within the non-top quartile. The bars provide 95% confidence intervals.

Alt text: Dot plot of monthly five-factor alphas with 95% confidence intervals for portfolios sorted by the modified information-ratio measure; education-country non-top and top portfolios earn -10.2 and 29.0 basis points, a difference significant at the 1% level, while non-education-country non-top and top portfolios earn 4.7 and 1.4 basis points and are statistically indistinguishable.

Appendix D. Comparison to Jagannathan et al. (2022)

This appendix compares our analysis with Jagannathan et al. (2022) (JJK). The introduction summarizes the conceptual distinction between the two papers; here we document the implementation details behind the country-level comparison: fund coverage, manager education coverage, education-tie definitions, specification, and portfolio construction.

D.1. Overview of the comparison

JJK study U.S. international mutual funds over 1991–2014 and identify managers’ “home countries” using foreign undergraduate degrees. Our baseline sample covers 2000–2024 and uses all post-secondary education locations to measure managers’ familiar markets. To make the comparison as direct as possible, we reproduce the country-level overholding and performance exercises as closely as possible using JJK’s exact fund list, their 1991–2014 sample period, and their undergraduate-based definition of education links, while relying on the current Morningstar holdings and manager biography data available to us. Importantly, in this section, we also use their exact definitions and empirical methods applied to our full fund sample from 2000 to 2024, so the headline results on this full sample are not the same as the results in the main text.

The comparison yields two main conclusions. First, we qualitatively confirm JJK’s central allocation result: funds overweight the countries in which their managers were educated. The estimated tilt is economically meaningful, although roughly one-fifth the magnitude reported in JJK. This gap is driven largely by the regression specification rather than by a difference in the data: the home tilt is isolated cleanly only when the specification absorbs the fund’s mandate (a category fixed effect) and each country’s baseline weight (a country fixed effect) together. With either fixed effect alone the coefficient remains far above ours—on JJK’s fund list, 9.5 pp under the mandate fixed effect and 6.5 pp under the country fixed effect—and only when both are included simultaneously does it fall to 3.2 pp, close to our headline

estimate of 2.5 pp on that sample. Our full sample behaves the same way: 11.0 pp under the mandate fixed effect alone, falling to 4.3 pp once the country fixed effect joins it and 3.0 pp under our headline specification. Crucially, JJK’s specifications never absorb these two fixed effects jointly: their Table 2 applies at most one fixed effect at a time (none, category, fund, or country, each on its own), and their Figure 2 breakdowns likewise carry a single fixed effect throughout. Because the mandate and country-size confounds are therefore left in across all of their specifications, their reported coefficient of roughly 14 pp is mechanically large—the gap is not that our data show a weaker tilt, but that our headline specification nets out both confounds at once while theirs never does. We argue the joint specification is the correct one, because each fixed effect removes a distinct confound—the fund’s mandate, and a country’s overall size in the international market—that would otherwise be misread as the manager’s personal home bias; this is also what separates the two panels of JJK’s own Figure 2 (Section D.5).

Second, in our data, the average education-country portfolio does not earn robust abnormal returns relative to the non-education-country portfolio—under JJK’s own portfolio construction and factor model on their own fund list, as well as under our baseline construction (Section D.6). Both conclusions are similar when we use the JJK fund list over 1991–2014 and when we use our broader 2000–2024 sample.

The remaining subsections document the implementation details behind this comparison. Section D.2 summarizes the headline country-level estimates. Sections D.3 and D.4 compare fund coverage and manager education coverage. Section D.5 reconciles the country-level over-allocation estimate with JJK, covering their Table 2, their Figure 2, and the role of fixed effects; Section D.6 compares portfolio performance results; and Section D.7 summarizes. Table D2 reports the year-by-year coverage comparison, and Table D1 summarizes the headline country-level results.

D.2. JJK’s headline results and a summary comparison

We begin by stating JJK’s headline findings, which this exercise is designed to reproduce. Studying 1,090 actively managed U.S. international equity funds over 1991–2014, JJK document two central results. First, home-linked managers—those holding a foreign undergraduate degree from a country within the fund’s mandate—tilt their portfolios heavily toward those countries. The magnitude is the coefficient on the home-linked team share: a fund whose management team is entirely home-linked to a country holds, on average, about 14 percentage points more of its assets in that country than an otherwise-similar fund with no home-linked managers, and roughly 22 and 30 percentage points more for regional and country funds, respectively (their Table 2). These tilts are economically very large: because the average international fund allocates less than 5% of its assets to any single country, a 14–30-percentage-point increase amounts to a several-fold overweight in the manager’s home country. Second, these home-country holdings earn positive abnormal returns: the education-country (home-country) portfolio earns a four-factor alpha of about 54 basis points per month, and a long–short portfolio that buys home-country holdings and shorts the same countries’ stocks held by unlinked managers earns 31 basis points per month, significant at the 10% level (their Table 4). JJK read the first result as home bias and the second as an informational advantage of home-linked managers.

Table D1 places these numbers next to our estimates, both on JJK’s exact fund list over their sample period and on our full sample. Three features of this exercise are worth noting. The replication runs on JJK’s exact fund sample rather than on a universe reconstructed from filters, and the implementation follows their published design. And their holdings and manager-education data, like ours, come from Morningstar, so the two studies draw on the same underlying source. Differences between our replication estimates and theirs therefore reflect specification and data vintage rather than a different sample or a different data provider. Our results echo JJK’s central allocation finding—funds do overweight the countries where their managers were educated—while the magnitude we estimate is smaller:

about 2.5 percentage points for a fully home-linked team on their fund list, and a comparable 3.0 percentage points on our full sample, next to their $\approx 14\%$. On performance, our education-country portfolios earn abnormal returns close to zero on average, rather than the significantly positive alpha JJK document. Reassuringly, our two samples, their exact fund list and our broader universe, deliver the same picture despite differing in coverage, time period, and education definition. As the remaining subsections discuss, differences in Morningstar data vintage and coverage—especially in sparsely covered early years—may contribute to the divergence. Because their fund holdings and manager education data are not publicly available, we cannot identify the exact source of the remaining divergence.

D.3. Fund coverage

We first compare the mutual fund universe. JJK identify international equity funds using the following Morningstar categories: World Stock, Foreign Large Blend, Foreign Large Growth, Foreign Large Value, Foreign Small/Mid Blend, Foreign Small/Mid Growth, Foreign Small/Mid Value, Diversified Emerging Markets, Diversified Pacific/Asia, Pacific/Asia ex-Japan Stock, China Region, India Equity, Japan Stock, Europe Stock, and Latin America Stock. JJK identify actively managed funds using the Morningstar filter *Index Fund = No*. After merging with Morningstar manager education data, JJK’s final sample contains 1,090 unique funds with manager education information over 1991–2014.

Our baseline sample begins in 2000 for two reasons. First, the U.S. international mutual fund universe is substantially smaller in the early 1990s, and historical holdings and manager-biography coverage are more limited. Second, the Morningstar benchmark data we use to compute excess holdings and excess returns are not consistently available until 1999.

For the direct JJK comparison, we do not reconstruct their fund universe using filters. We obtain the fund identifiers for the 1,090 funds in JJK’s sample and retrieve holdings and fund characteristics directly for those funds from current Morningstar Direct. The country-

Table D1: Headline results: JJK versus our replication and full sample

	Sample	Funds	Education	Over-holding (pp)			Long-short
	period	(N)	tie	cat $\times$ qtr	+country	full FE	α (bps/mo)
JJK (reported)	1991–2014	1,090	UG only	≈ 14	—	—	+31*
JJK replication	1991–2014	1,090	UG only	9.5	3.2	2.5	+4.7
Full sample	2000–2024	1,663	all post-sec.	10.8	4.3	3.0	+5.2

Notes: The three “Over-holding (pp)” columns report the coefficient on the home-linked team share (JJK’s $HLMgr / our Frac_{EC}$) in a fund-country-quarter panel of raw country weights (held countries, ex-U.S., quarterly; matching JJK Table 2), under three fixed-effect specifications: *cat $\times$ qtr* = category $\times$ quarter fixed effect only; *+country* = category plus (time-invariant) country fixed effects; *full FE* = country $\times$ time plus fund fixed effects (our main specification). All three columns are estimated without fund-level controls, so they are directly comparable to one another across the two rows; the cat $\times$ qtr entries reported here are therefore estimated without JJK’s five fund controls, which Table D4 adds; on the fund list the two differ only slightly (9.5 here versus 9.8 with controls). The estimate depends critically on the fixed effects: it is largest under the category-only design and shrinks by roughly two-thirds once a country fixed effect is added alongside the category, and further with the fund fixed effect (Section D.5, Table D5). JJK apply one fixed effect at a time throughout, so only the cat $\times$ qtr column is comparable to their reported magnitude (which rises further, to about 22% for regional and 30% for country funds); our main-text benchmark-excess measure (Table 1) gives a comparable full-FE tilt, 0.97 pp in its indicator specification. “Long-short α ” is the monthly abnormal return of a self-financing portfolio that is long familiar-country holdings and short comparable unfamiliar-country holdings, so all three rows are on the same footing. JJK report +31 bps for their home versus unlinked-manager long-short, significant at the 10% level (*); their long-only home portfolio earns a four-factor α of +54 bps. Our faithful country $\times$ category-matched replication on JJK’s fund list (Section D.6, Table D7) gives +4.7 bps for the matched long-short ($t = 0.34$), against +16.7 bps for the home-linked long-only leg ($t = 0.78$); both are statistically insignificant. In the full sample, using our benchmark-scaled construction over the 2000–2024 period (no activeness split), the long-only education-country portfolio earns +7.6 bps and the non-education-country portfolio +2.4 bps, so the education-minus-non-education long-short earns +5.2 bps—again statistically insignificant ($t = 0.72$). The full sample—the sample used throughout the paper’s main analysis—includes JJK’s funds together with additional international equity funds; “UG” denotes undergraduate-only education ties, “all post-sec.” all post-secondary education locations.

level overholding and performance exercises below are therefore conducted on JJK’s exact fund list, and we reproduce the implementation as closely as possible from their published design. One remaining source of difference is data vintage: Morningstar’s historical holdings coverage has improved over time, so the holdings currently recorded for a given fund-quarter may differ from those available in the Morningstar data vintage used in JJK, even for the same fund. This shows up directly in the size of the estimation panel. Holding the fund list and the sample window fixed—the same 1,090 funds over 1991–2014, the same ex-U.S., held-only, quarterly fund-country structure—our panel contains 588,779 fund $\times$ quarter $\times$ country observations, against the 464,578 reported by JJK. Because the fund list and window are

identical by construction, the roughly 27% additional observations cannot reflect a broader universe or a longer sample; they most plausibly reflect the improved retrospective holdings coverage of the current Morningstar vintage, which fills in fund-quarters that were missing when JJK drew their extract. Our replication of their design therefore rests on somewhat denser data than the original, which is worth bearing in mind when comparing magnitudes: the specifications are matched, but the underlying panels are not identical.

Our full sample, used in the paper’s main analysis, is not restricted to the JJK fund list. We construct it in current Morningstar Direct using *U.S. Category Group = International Equity* together with *Index Fund = No*, which is the closest current equivalent to JJK’s category-based fund definition. In Table D2, the “Full” column counts funds in this broader universe that enter our analysis sample, requiring that the fund holds at least one non-U.S. equity and has at least one manager matched to the education data. The “Fundlist” column applies the same requirements but restricts the universe to the JJK fund list. The broader universe contains substantially more international equity funds than the JJK list during 1991–2014, and the gap widens in recent years as the international mutual fund universe expands and manager-biography coverage becomes more complete.

D.4. Manager education coverage

We next compare manager education coverage. Both JJK and our paper obtain manager education information from Morningstar manager biographies, but the underlying data vintage differs. Current Morningstar Direct provides broader manager biography coverage in recent years, but less complete historical coverage for some managers who left the fund industry before the current data vintage. Because JJK’s manager-level education records are not available to us, we cannot compare coverage at the manager level. Instead, we benchmark our coverage against their published aggregate series by applying their definition of home-linked managers: non-U.S. managers with an undergraduate degree from a country in which their fund holds stocks.

Table D2 reports the year-by-year comparison. The “Home-linked managers” block reports, for each year, the quarterly average number of non-U.S. managers whose fund holds a stock in a country where the manager was educated. The “Any deg.” column uses any education country, while the “UG” and “Non-UG” columns split these into undergraduate and above-undergraduate education countries. The “UG (FL)” column is the undergraduate count computed on the JJK fund list, and is therefore the series directly comparable to JJK’s undergraduate-based definition. The final “JJK Fig. A1” column reproduces the “No. of Home-linked Managers” series read from Figure A1 of JJK, which spans 1991–2014.

The table shows that our coverage differs from JJK’s in a systematic way over time. On the same fund list, we observe fewer home-linked managers than JJK before 2004, but more thereafter, with the gap widening in later years. This pattern appears in the last two columns of Table D2: our “UG (FL)” count falls below the “JJK Fig. A1” series through the early 2000s (2.8 against 8.5 in 1991, and 40.5 against 52.3 in 2000), crosses it in 2003–2004, and exceeds it in every year thereafter (83.0 against 81.3 in 2004, and 156.0 against 148.3 in 2014). The first two columns also show that our broader sample covers more funds than the JJK fund list, especially in more recent years as the international mutual fund universe expands and manager-biography coverage becomes more complete.

The consequences of these coverage differences for the country-level comparison are largest early in the sample. In the nine years before 2000, the number of foreign-undergraduate home-linked managers is very small and heavily concentrated in a small number of countries (predominantly the United Kingdom), so even a modest difference in which managers are covered can materially shift the constructed education-country portfolios and the country-level overweight estimates. This sensitivity, together with our broader fund coverage throughout the sample, is a possible source of the divergence between our early-sample estimates and JJK’s, even when we use their exact fund list.

Outside those early years, however, the comparison favors our manager data on every margin. In 2014, the last year the two samples overlap, we identify 347 home-linked man-

agers per quarter across our full universe against the 148 in JJK’s published series—more than twice as many—and 156 on their own fund list under their own undergraduate-only definition. Three differences account for this. Our fund universe is broader (927 funds in 2014 against 589 on their list); our education definition admits graduate as well as undergraduate degrees, which adds a further 139 managers in 2014 (the “Non-UG” column); and our sample continues for another decade, through 2024, over which the number of home-linked managers roughly doubles again. The sparsity that complicates the early-sample comparison is therefore a feature of the 1990s rather than of our data as a whole: from 2004 onward our coverage exceeds JJK’s in every year, including under the undergraduate-only definition they use.

Table D2: Sample Coverage and Home-Linked Managers: Our Data vs. Jagannathan et al. (2022)

Year	Funds ($\cap$ bio/edu)		Home-linked managers (quarterly avg.)				JJK
	Full	Fundlist	Any deg.	UG	Non-UG	UG (FL)	Fig. A1
1991	43	31	7.75	3.00	4.50	2.75	8.50
1992	61	44	10.25	6.00	3.50	5.00	11.50
1993	92	73	12.50	7.00	4.50	6.50	16.50
1994	129	105	15.00	8.50	5.50	7.75	20.75
1995	167	134	22.75	14.25	7.00	14.00	26.00
1996	217	173	26.75	15.75	9.00	15.25	34.25
1997	258	202	39.75	22.75	14.75	21.00	45.25
1998	333	259	62.75	38.00	24.50	34.50	53.00
1999	377	294	70.00	41.25	28.75	35.50	49.25
2000	420	319	80.50	49.50	31.00	40.50	52.25
2001	499	373	96.75	57.25	39.25	47.00	54.25
2002	552	415	115.00	73.50	40.75	57.75	66.50
2003	582	440	145.25	94.25	51.50	74.75	75.25
2004	566	429	160.50	103.75	54.00	83.00	81.25
2005	591	450	184.75	120.00	64.75	95.50	80.25
2006	630	472	214.00	135.25	78.50	104.50	94.50
2007	690	506	243.75	151.50	93.75	114.25	103.25
2008	760	536	268.75	166.00	111.00	123.00	112.00
2009	775	544	276.00	166.75	118.25	126.00	115.25
2010	780	545	287.75	178.25	121.50	128.50	116.25
2011	830	565	302.25	192.75	129.25	141.75	124.50
2012	851	574	320.25	210.00	136.50	151.00	130.00
2013	882	586	327.50	221.50	133.75	156.50	141.75
2014	927	589	347.25	237.50	138.75	156.00	148.25
2015	966	567	385.50	262.25	152.75	163.25	—
2016	988	547	401.50	274.75	162.00	169.50	—
2017	1000	528	420.25	287.25	171.50	171.75	—
2018	1002	501	450.50	301.25	186.25	177.25	—
2019	1009	471	466.50	296.75	199.75	173.25	—
2020	993	446	439.50	273.50	185.25	165.25	—
2021	978	423	454.00	284.00	194.00	169.00	—
2022	973	407	446.25	273.75	192.25	160.25	—
2023	955	395	410.50	245.00	181.25	136.00	—
2024	900	372	368.50	214.25	167.25	112.25	—

Notes: “Funds” columns report the number of distinct international equity funds each year that both hold a non-U.S. equity and have at least one manager matched to the education (bio) data—the analysis sample gate. “Full” is our full extended (broader) universe; “Fundlist” restricts to the exact fund list that Jagannathan et al. (2022) (JJK) shared with us (1,090 funds). “Home-linked managers” count non-U.S. managers whose fund holds a stock in a country where the manager was educated, computed *per quarter* and averaged over the four quarters of the year (hence non-integer). “Any deg.” uses any education country; “UG”/“Non-UG” restrict to undergraduate / graduate (above-undergraduate) education countries. The “Any deg.” column need not equal “UG”+“Non-UG”: a manager whose foreign degree level is unknown enters “Any deg.” but neither split, while a manager holding foreign degrees at both the undergraduate and graduate levels is counted once in “Any deg.” but in both splits. “UG (FL)” is the undergraduate count on the JJK fundlist, directly comparable to JJK’s published series. “JJK Fig. A1” is the “No. of Home-linked Managers” read from Figure A1 of Jagannathan et al. (2022) (time-averaged within year); their series spans 1991–2014 only (“—” afterward).

D.5. Comparing Country-Level Over-Allocation Results

This subsection reconciles the country-level over-allocation that JJK document with the smaller magnitude we report in the main text. JJK find that funds with home-linked managers overweight their managers’ education countries by roughly twelve to seventeen percentage points—in the pooled coefficient of their Table 2 and in the by-year breakdown of their Figure 2—whereas the corresponding raw-weight estimate under our preferred specification is about 3.0 percentage points (pp) in the full sample and 2.5 pp on JJK’s fund list. (The main text’s benchmark-excess Frac_{EC} coefficient, 2.47 pp in Table 1, is a different object and is not directly comparable to these raw-weight figures.)

We show that this difference is one of specification rather than of data. Reproducing JJK’s own design recovers their magnitudes; the estimate falls to about 4.3 pp in the full sample and 3.2 pp on JJK’s fund list once the regression controls simultaneously for the fund’s mandate and for each country’s baseline weight—that is, once a country fixed effect is included alongside the mandate (category) fixed effect. We first replicate JJK’s Table 2 and trace this compression through a fixed-effects ladder, and then replicate the by-country and by-year breakdowns of their Figure 2, which reproduce JJK’s pattern and display the same compression year by year.

Replicating JJK’s Table 2

JJK’s Table 2 regresses a fund’s country weight (a fraction of assets) on $\text{HLMgr}_{i,c,t}$, the share of the management team home-linked to country c , pooling all funds and applying one fixed effect at a time: none, category $\times$ quarter, fund, and country $\times$ quarter, with standard errors clustered by country and year. Table D3 reports this specification on JJK’s fund list in their original format, with each control’s coefficient: column (1) is the raw HLMgr regression, and columns (2)–(4) add JJK’s five fund-level controls (log size, expense ratio, log turnover, log age, and log number of managers) together with one fixed effect at a time. The controls carry sensible signs but are economically negligible and leave the HLMgr coefficient close to

its no-control value.

Table D3: JJK Table 2, literal replication on the **JJK fund list** (1991–2014). Dependent variable: fund country weight (fraction). HLMgr = undergraduate home-linked team share. Column (1) is the raw regression; columns (2)–(4) add JJK’s five fund controls and one fixed effect. SEs (in parentheses) two-way clustered by country and year. */**/***: 10/5/1%.

	(1)	(2)	(3)	(4)
HLMgr	0.125*** (0.021)	0.098*** (0.023)	0.090*** (0.027)	0.071** (0.031)
Fund size (log)		−0.0008** (0.0003)	−0.0005*** (0.0001)	−0.0003 (0.0004)
Expense ratio		0.0012 (0.0019)	−0.0023** (0.0008)	0.0087** (0.0039)
Turnover (log)		−0.0014*** (0.0003)	−0.0006*** (0.0001)	−0.0029*** (0.0007)
Fund age (log)		0.0007 (0.0008)	0.0023*** (0.0002)	0.0000 (0.0011)
No. managers (log)		−0.0025*** (0.0007)	−0.0008 (0.0005)	−0.0033*** (0.0010)
Controls	No	Yes	Yes	Yes
Category×quarter FE	No	Yes	No	No
Fund FE	No	No	Yes	No
Country×quarter FE	No	No	No	Yes
Observations	588,779	427,096	427,109	426,677
R^2	0.020	0.210	0.301	0.372

Table D4 summarizes the same four specifications on both samples, reporting the HLMgr coefficient (with controls in rows (2)–(4)) alongside JJK’s reported values.

The pattern replicates: the coefficient declines monotonically across the four designs and averages about 0.11 in the full sample against JJK’s 0.14. Because Table 2 applies only one fixed effect at a time, its magnitudes remain near 0.10–0.17; our main-text coefficient of about 0.030 (3.0 pp) is smaller because that specification simultaneously absorbs country×time and fund fixed effects (shown next). Reproducing JJK’s own design thus recovers JJK-like magnitudes, indicating that the gap between the two studies reflects the choice of fixed effects rather than a difference in the data.

The role of fixed effects

The pooled Table 2 magnitude shrinks toward our headline estimate once the specification controls for two distinct confounds. The first is the fund’s *mandate*: home-linked managers concentrate in funds already tilted toward their country—a manager educated in Japan tends to work at a Japan or Asia-Pacific fund, one educated in Brazil at a Latin America fund—so part of the measured tilt belongs to the mandate rather than to the manager. A Morningstar Category fixed effect absorbs it. The second is the *country’s baseline weight*: large markets such as Japan and the United Kingdom carry heavy weights in every international portfolio and also supply many home-linked managers, so HLMgr is correlated with a country’s sheer size in the international market. A country fixed effect absorbs that. JJK’s specifications include one of these at a time and never both—their Table 2 applies at most one fixed effect at a time (none, category, fund, or country, each on its own), and their Figure 2 carries a single fixed effect throughout—so at least one confound survives in every estimate they report. Table D6 adds fixed effects one at a time on each sample, and Table D5 reports the full menu on the fund-list subsample.

Two housekeeping notes before interpreting these tables. Standard errors are two-way clustered by country and year throughout, following JJK, so that the comparison isolates the specification rather than the inference method; in the two most saturated cells the two-way variance matrix is degenerate, and the notes to Tables D4 and D6 report well-defined country-clustered standard errors instead ($t = 6.8$ and $t = 3.31$), so no conclusion turns on the clustering choice. And Tables D5 and D6 omit JJK’s five fund controls while Tables D3 and D4 include them; the controls barely move the coefficient (0.095 without against 0.098 with, under the category $\times$ quarter fixed effect), so the compression documented next is a property of the fixed effects, not of the covariates.

Neither of these two fixed effects alone is sufficient. On JJK’s fund list, where the full menu of specifications is reported in Table D5, absorbing the mandate alone leaves 9.5 pp

and absorbing the country alone leaves 6.5 pp, but the two together reduce the coefficient to 3.2 pp. The full sample behaves the same way: the category fixed effect alone leaves 11.0 pp, and adding the country fixed effect brings the estimate to 4.3 pp (Table D6). Adding the fund and country $\times$ time fixed effects moves the estimate only the remaining short distance, to 3.0 pp in the full sample and 2.5 pp on the fund list. Almost all of the compression therefore occurs at the point where the country fixed effect joins the category fixed effect: on the fund list that single step accounts for 6.3 of the 7.0 pp total decline, and the further saturation contributes less than one percentage point. These raw-weight estimates are not directly comparable to the Frac_{EC} coefficient of 2.47 pp in Table 1, which uses the benchmark-excess weight, zero-fills each fund’s held-and-benchmark country universe, and defines education ties at all post-secondary levels; the specifications here instead follow JJK in using the raw country weight, held countries only, and undergraduate ties, so that the magnitudes line up with theirs. We regard the specification with both a category and a country fixed effect as the preferred one. Only with both does the coefficient compare a home-linked against a non-home-linked manager running the same type of fund and holding the same country—the contrast the home-bias hypothesis is actually about. Dropping either fixed effect re-introduces overweighting that the preferred specification removes, and inflates the coefficient back toward JJK’s magnitude. This is also what separates the two panels of JJK’s Figure 2: Panel A estimates the tilt one country at a time, so each regression is already within a single country—its intercept nets out that country’s baseline weight just as a country fixed effect would, on top of the category fixed effect—and its pooled common-slope analogue is the compressed ~ 3 pp figure; Panel B, pooling across countries under a category fixed effect only, leaves each country’s baseline in and so reports the inflated 9–11 pp level (falling to 2–5 pp once a country fixed effect is added). Our estimate matches the country-controlled Panel A magnitude, which we take to be the preferred specification.

Our main analysis also differs from JJK in two constructions of the dependent variable: how non-held country-quarters are treated, and whether the country weight is measured raw

or in excess of the fund’s benchmark. JJK use the raw country weight and set it to missing whenever a fund does not hold a country (held countries only). Our main text instead uses the benchmark-excess weight and zero-fills non-held countries within each fund’s held-and-benchmark investment universe. In unreported tests, the home-tilt coefficient is insensitive to both choices.

Replication of JJK’s Figure 2

Table D4 and the fixed-effects ladder report the home tilt as a single pooled coefficient. JJK’s Figure 2 displays the same relationship disaggregated in two ways—Panel A estimates it country by country, and Panel B year by year—and we are able to reproduce both breakdowns. This serves two purposes. First, it confirms that our replication recovers JJK’s qualitative pattern within each slice: funds overweight the countries where their managers were educated, concentrated in a familiar set of countries and present throughout the sample period. Second, the by-year panel lets us make the preceding subsection’s point visible year by year, because a by-year regression can carry a country fixed effect alongside the category fixed effect; comparing the two versions of Panel B shows how far the estimate falls once the mandate and each country’s baseline weight are absorbed simultaneously. Panel A, with a single country per regression, cannot add an explicit country fixed effect, but the single-country restriction itself nets out that country’s baseline weight exactly as a country fixed effect would—so Panel A is already country-controlled (on top of the category fixed effect), and its pooled common-slope analogue is the compressed category-plus-country level rather than the uncompressed category-only one. Its per-country slopes are large for a handful of strong-home-bias countries (Brazil, Japan) but small for most, and their obs-weighted average sits near the compressed ~ 3 pp estimate.

The dependent variable is the fund’s raw portfolio weight in a country, $w_{i,c,t}$ (in %, held countries only, ex-U.S., quarter-end), and the regressor is $\text{HLMgr}_{i,c,t}$, the fraction of fund i ’s managers home-linked to country c through an undergraduate degree. Matching JJK, the

baseline for both panels uses a category (Morningstar Category) fixed effect only:

Panel A (by country c): $w_{i,t} = \beta_c \text{HLMgr}_{i,t} + \alpha_{\text{cat}} + \varepsilon$, one regression per country;

Panel B (by year t): $w_{i,c} = \beta_t \text{HLMgr}_{i,c} + \alpha_{\text{cat}} + \varepsilon$, one regression per year.

Standard errors are clustered by fund. Panel A carries the category fixed effect only; Panel B is shown both with the category fixed effect alone and with a country fixed effect added.

Figures D1 and D2 report the two panels on JJK's fund list.

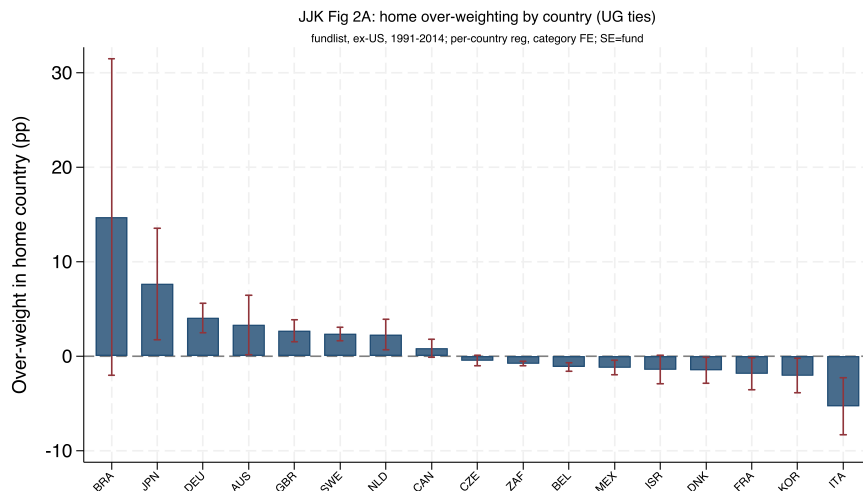
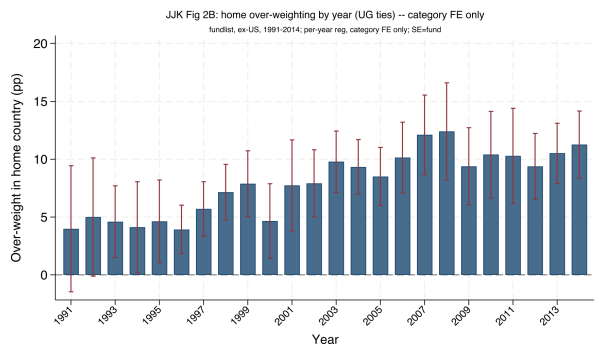


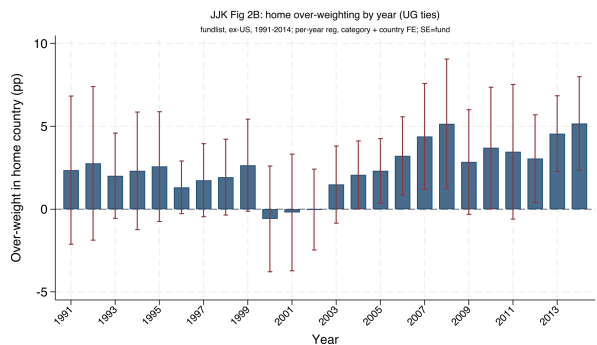
Figure D1: **Panel A — by country (fund list, 1991–2014), category FE only.** Bars are the estimated HLMgr coefficient β_c (extra pp of country weight for a fully home-linked fund), for countries significant at $p < 0.10$, with 95% confidence intervals. Home bias is strongest for Brazil (+14.7), Japan (+7.6), Germany (+4.0), Australia (+3.3), the U.K. (+2.7), Sweden (+2.4) and the Netherlands (+2.3). Panel A is a set of per-country regressions, so it admits no explicit country fixed effect, but each within-country regression already nets out that country's baseline weight—so these slopes are country-controlled, and their obs-weighted average sits near the compressed category-plus-country level rather than the category-only one.

Alt text: Bar chart of per-country home-linked team-share coefficients with 95% confidence intervals for countries significant at the 10% level; Brazil is largest at about 14.7 percentage points, followed by Japan at 7.6, with most other significant countries between about 2 and 4 percentage points.

These panels should be read in light of their fixed-effect structure. Panel A carries an



(a) category FE only



(b) category + country FE

Figure D2: **Panel B — by year (fund list, 1991–2014).** The per-year coefficient β_t under (a) a category fixed effect only and (b) an added country fixed effect. Under (a) the coefficient rises from about 4 pp in the thin early years to 9–11 pp in recent years (2014: 11.2); adding the country fixed effect in (b) reduces it to about 2–5 pp (2014: 5.1), consistent with the compression in Table D6.

Alt text: Two panels of year-by-year home-linked team-share coefficients over 1991–2014; with a category fixed effect only, coefficients rise from about 4 percentage points in the early years to 9–11 in recent years, and adding a country fixed effect lowers them to about 2–5 percentage points.

explicit mandate fixed effect only, but because it estimates one country at a time it also nets out each country’s baseline weight, so its slopes are already country-controlled rather than sitting at the uncompressed category-only level; the large individual bars are genuine per-country home tilts (strongest for Brazil and Japan), while their obs-weighted average sits near the fully-controlled ~ 3 pp level. Each bar is a per-unit-of-fraction slope, so a fund with $\text{HLMgr} = 0.3$ in a country with a +6 pp coefficient tilts by $0.3 \times 6 \approx 1.8$ pp toward that country. Panel B estimates a separate slope in each year; the category-only coefficients rise to 9–11 pp in recent years, and adding the country fixed effect reduces them to about 2–5 pp (Figure D2). Pooling the by-year regression with a constant slope and $\text{country} \times \text{time}$ fixed effects returns a coefficient in line with the fixed-effects ladder in Table D6, and sub-period estimates are flat-to-rising rather than trending—so the year-by-year pattern reflects thin annual cross-sections, not a time trend in the home bias.

JJK’s own published Figure 2 displays the same asymmetry. In their by-country panel, the typical estimate is on the order of 2–3 pp and only Brazil exceeds 10 pp, while their by-year panel runs above 10 pp—the two panels differ by roughly the factor the country

control accounts for. The compression our fixed-effects ladder documents is therefore visible inside JJK's own exhibits: their country-by-country estimates, which net out each country's baseline weight by construction, already sit near our preferred magnitude, and the large headline numbers come from the designs that leave each country's baseline weight in.

Table D4: JJK Table 2 replication: country weight (fraction) on HLMgr, whole sample, one fixed effect at a time. Row (1) is the raw regression with no controls; rows (2)–(4) add JJK’s five fund controls (log size, expense ratio, log turnover, log age, log #managers). JJK “Global” column from their Table 2.

Model (fixed effect)	JJK Reported (Global funds)	Our replication 1991–2014 (JJK fund list)	Our sample 2000–24 (full sample)
(1) none (raw)	0.174***	0.125*** (0.021)	0.146*** (0.029)
(2) category $\times$ quarter	0.139***	0.098*** (0.023)	0.108*** (0.011)
(3) fund	0.138***	0.090*** (0.027)	0.099*** [0.015]
(4) country $\times$ quarter	0.121***	0.071** (0.031)	0.088* (0.045)
average	0.143	0.096	0.110
Fund controls	None in row (1); JJK’s five in rows (2)–(4)		

Notes: Standard errors in parentheses, on the same (fraction) scale as the coefficients. */**/** denote significance at the 10/5/1% level. The JJK column shows no standard errors because JJK do not report them—their Table 2 gives coefficients and t -statistics only—so we leave those cells blank rather than back out a figure they never published. JJK’s significance stars are as printed in their Table 2. Standard errors are two-way clustered by country and year because that is JJK’s own choice (their Table 2 note states that “standard errors are clustered at the country and year level”); we retain it throughout this appendix so that our estimates and theirs are computed the same way and the comparison isolates the specification rather than the inference method. Because the second cluster dimension is small—24 years on the fund list and 25 on the full sample—inference uses the corresponding degrees of freedom, which is why row (4) of the full sample ($t = 1.96$) is significant only at the 10% level. The JJK and replication columns are not drawn from identical samples—JJK’s figures are for their Global funds, whereas our replication pools fund types—so the two are comparable in design rather than estimated on the same observations. For the row-(3) full-sample cell, two-way clustering forms the variance as $V_{country} + V_{year} - V_{country \times year}$, which is not guaranteed positive semi-definite; in that regression it is not, and after the Cameron–Gelbach–Miller adjustment the matrix remains singular, so no two-way standard error is defined for any regressor. We therefore report in square brackets the standard error clustered by country alone (86 clusters; $t = 6.8$), which is well defined. The coefficient itself is unaffected—clustering changes only the variance estimate, never the point estimate. No significance stars are shown for that cell because stars elsewhere are based on two-way clustering. No conclusion rests on this cell: the coefficient is unaffected by the clustering choice, and the country-clustered inference is significant at the 1% level.

Table D5: Full fixed-effects menu on the **JJK fund-list subsample** (1991–2014). Dependent variable is the country weight as a fraction; category = Morningstar Category. **No fund controls**: the regressor is HLMgr alone.

Fixed effects	coef (fraction)	pp	se (pp)	<i>N</i>
no FE	0.125***	12.53	(2.06)	588,779
category (Mstar Cat)	0.095***	9.49	(1.96)	588,779
country (time-invariant)	0.065**	6.50	(2.91)	588,779
category + country	0.032***	3.17	(0.81)	588,779
category $\times$ quarter	0.095***	9.46	(2.09)	588,746
country $\times$ quarter	0.064**	6.45	(2.95)	588,419
category + country $\times$ qtr	0.031***	3.07	(0.84)	588,419
+ fund (cat + country + fund)	0.025***	2.55	(0.54)	588,777
MAIN: country $\times$ time + fund	0.025***	2.46	(0.54)	588,417

Notes: Standard errors, in percentage points and shown in parentheses, are two-way clustered by country and year, following JJK’s own choice so that every estimate in this appendix is computed the same way as theirs. */**/** denote significance at the 10/5/1% level, evaluated with the degrees of freedom implied by the smaller cluster dimension (24 years). **This table includes no fund controls**—the regressor is HLMgr alone—which is what distinguishes it from Table D4, whose rows (2)–(4) add JJK’s five fund controls. Comparing the two isolates the role of the fixed effects from the role of the controls: the category $\times$ quarter coefficient, for instance, is 0.095 here and 0.098 with controls. Every coefficient is significant at the 5% level or better; the two rows that absorb country alone are the weakest ($t = 2.24$ and $t = 2.18$), and every row combining a category with a country fixed effect is significant at the 1% level.

Table D6: Fixed-effects ladder: $w_{i,c,t}$ on $\text{HLMgr}_{i,c,t}$ (UG fraction), ex-U.S., quarterly, held-only, raw weight, Morningstar Category as the category FE. SEs two-way clustered by country and year. Row (4') replaces the UG team share with an indicator for any home-linked manager, estimated on the same panel as row (4).

Specification (fixed effects)	HLMgr coefficient (pp)	
	Full sample 2000–2024	Fund list 1991–2014
(1) Category FE only	11.05*** (1.13)	9.49*** (1.96)
(2) + Country FE	4.31** (1.62)	3.17*** (0.81)
(3) + Fund FE	3.05 [0.92]	2.55*** (0.54)
(4) Main: Country×Time FE + Fund FE	3.02*** (0.90)	2.46*** (0.54)
(4') Main spec, indicator $\mathbb{1}(\text{home-linked})$	1.29*** (0.24)	1.45*** (0.40)

Notes: Standard errors in percentage points are in parentheses. */**/** denote significance at the 10/5/1% level, evaluated with the degrees of freedom implied by the smaller cluster dimension (24 years on the fund list, 25 on the full sample). Standard errors are two-way clustered by country and year, following JJK's own choice, so that these estimates are computed exactly as theirs are. **No fund controls are included in any row:** the regressor is HLMgr alone, as in Table D5 and unlike rows (2)–(4) of Table D4. For the row-(3) full-sample cell, two-way clustering forms the variance as $V_{\text{country}} + V_{\text{year}} - V_{\text{country} \times \text{year}}$, which is not guaranteed positive semi-definite; in that regression it is not, and after the Cameron–Gelbach–Miller adjustment the matrix remains singular, so no two-way standard error is defined for any regressor—it is undefined, not merely large. We therefore report in square brackets the standard error clustered by country alone (86 clusters; $t = 3.31$), which is well defined. The coefficient itself is unaffected, since clustering changes only the variance estimate. No significance stars are shown for that cell because stars elsewhere are based on two-way clustering. No conclusion rests on this cell: row (4), which absorbs country×time and fund fixed effects together, returns a near-identical coefficient with a well-defined standard error and 1% significance.

D.6. Comparing Education-Linked Portfolio Performance Results

We also compare education-country performance estimates. JJK report that the education-country portfolio—home-linked managers’ holdings of stocks from their education countries—earns a large, significant abnormal return, with a four-factor alpha of about 54 basis points per month (their Table 4).¹⁰ In our data, comparable education-country portfolios earn close to zero abnormal returns on average.

To make the performance comparison direct, we replicate JJK’s security-level construction on their exact 1,090-fund list over 1991–2014, following their design as closely as our data permit. Within each Morningstar category and quarter, we form a long portfolio of the home-country stocks held by home-linked managers and a short portfolio of those same countries’ stocks held by unlinked managers in the same category; the resulting long–short isolates whether a home-linked manager selects better home-country stocks than an unlinked peer with the same mandate holding the same country. Portfolios are rebalanced quarterly and held for three months, value-weighted by dollar holdings, with the long legs measured in excess of the U.S. Treasury-bill rate, and we evaluate them with the Fama-French Global ex-U.S. four-factor model (market, size, value, and momentum) from Ken French’s data library—the same model JJK employ. The home-linked long-only portfolio earns a four-factor alpha of +16.7 basis points per month ($t = 0.78$), and the country- and category-matched long–short earns +4.7 basis points ($t = 0.34$); both are statistically indistinguishable from zero (Table D7). These estimates carry the same sign as JJK’s +54 and +31 but are considerably smaller and insignificant in our data.¹¹ The country- and category-matching is essential: it holds fixed both the country and the mandate, so the estimate cannot

¹⁰JJK also report a long-short portfolio that goes long these home-country holdings and short the same countries’ stocks held by unlinked managers in the same category, with an alpha of 31 basis points per month, significant at the 10% level.

¹¹Two features of JJK’s construction that we cannot reproduce account for much of the level difference: they weight funds by total net assets, which we do not observe (so we weight by dollar holdings), and they report separately for Global, Regional, and Country funds—their headline +54/+31 are for Global funds—whereas we pool fund types. The momentum factor is important here: home-linked managers tilt away from recent-winner stocks, so a factor model without momentum drives these alphas negative, and matching JJK’s four-factor model is what recovers the positive sign.

be driven mechanically by education ties being concentrated in a few high-return countries. Even under JJK’s own construction and factor model, then, the home-country stock-selection premium in our data is positive but economically small and statistically insignificant, rather than the large, reliable alpha JJK report.

Table D7: Faithful JJK country×category-matched performance (monthly four-factor α , bps; fund list, undergraduate ties, 1991–2014, 282–285 months). The long–short compares linked vs unlinked managers of the same country in the same Morningstar category. Factor model is JJK’s exact Fama-French **Global ex-U.S. four-factor** (Mkt-RF, SMB, HML, WML); long legs excess of the U.S. T-bill. t -statistics from the portfolio return series; the JJK α and t columns report JJK’s Global-fund figures from their Table 4.

Portfolio	α	t	p	JJK α	JJK t
Long-only: linked home-country holdings (excess of Rf)	+16.7	0.78	0.44	+54	2.67
Long–short (linked – unlinked)	+4.7	0.34	0.73	+31	1.69

Portfolio construction plays an important role in this comparison. Education ties are unevenly distributed across countries, especially in the early part of the sample, so an aggregate education-country portfolio that is not matched or reweighted can be influenced by realized returns in a small number of countries with high education coverage. The country- and category-matched long–short above addresses this directly by holding the country fixed; our country-level return tests in the main text address it in the same spirit, benchmarking education-country and non-education-country portfolios against comparable country exposures so that the estimated alpha is not driven mechanically by country composition.

This distinction is particularly important in the pre-2000 period, when the number of funds and manager education coverage are limited. In the 2000–2024 sample, the broader fund universe and richer manager coverage make education-country portfolio returns far less sensitive to a small number of high-coverage countries. Our main country-level performance results are similar in earlier and more recent subperiods.

D.7. Summary

The comparison with JJK yields four takeaways.

Allocation: the direction replicates, the magnitude does not. Managers do overweight the countries where they were educated, on both JJK’s fund list (1991–2014) and our broader 2000–2024 sample. But the tilt we estimate is several times smaller than the one they report.

The magnitude gap is largely a specification difference. JJK’s designs never absorb the fund’s mandate and each country’s baseline weight at the same time—their Table 2 uses one fixed effect at a time and their Figure 2 a single fixed effect throughout—so at least one confound remains in every estimate they report, and their coefficient (≈ 14 pp) is correspondingly overstated. Absorbing a category and a country fixed effect jointly—the specification we argue is the correct one—brings the estimate to about 4.3 pp in the full sample and 3.2 pp on JJK’s fund list, and this step, rather than the further saturation in our baseline, does nearly all of the work. Reproducing JJK’s own one-at-a-time designs returns coefficients in their range and in the same declining pattern, though about a third below their reported values; that residual we attribute to Morningstar vintage and coverage, which we cannot verify directly.

Performance: we do not reproduce their result. Under JJK’s own construction, matching, and four-factor model, applied to their own fund list, the long-only home portfolio earns 16.7 bps ($t = 0.78$) and the country- and category-matched long-short 4.7 bps ($t = 0.34$), against the 54 and 31 bps they report. Both are economically small and statistically indistinguishable from zero, and the same null holds in our broader sample. The country-level informational advantage JJK infer from this comparison does not survive in our data.

Interpretation. We cannot pin down the residual differences, since JJK’s code, holdings, and manager-education records are not available to us, and the sparse early-1990s data make country-level portfolios sensitive to coverage. But a near-zero average performance does not mean familiarity is uninformative—it means the country-level average is too coarse to reveal where that information lies. The main text separates broad familiar-country exposure from

managers' most active positions and shows the return premium concentrates in the latter. What we take from JJK is therefore the existence of a familiarity-driven allocation tilt, but neither its reported magnitude nor the country-level outperformance they attribute to it; our contribution is to locate where within the portfolio familiarity is, and is not, associated with value.